

Large Language Models for Generative Recommendation and User Behavior Modeling: A Comprehensive Survey

1 Introduction and the Paradigm Shift to Generative Recommendation

1.1 From Discriminative Ranking to Generative Text-to-Text Paradigms

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The evolution of recommender systems has undergone a profound transformation, shifting from rigid, discriminative ranking architectures to flexible, generative text-to-text paradigms. Historically, the foundation of modern recommendation engines was built upon collaborative filtering and matrix factorization techniques, which primarily relied on unique user and item identifiers (IDs) to model interactions. In this traditional discriminative framework, the core objective was to calculate a relevance score for every candidate item within a predefined corpus given a specific user context. This process typically involved embedding users and items into a unified latent space and performing approximate nearest neighbor searches to retrieve top candidates. While effective for static environments, this approach treats recommendation as a classification or regression problem, necessitating a distinct scoring function for each candidate and often struggling with the scalability required for massive, dynamic item catalogs. Crucially, the limitations of this ID-based paradigm become particularly apparent when dealing with cold-start items lacking interaction history or when attempting to incorporate rich semantic content beyond simple identifiers, a gap that purely interaction-based models fail to bridge.

The paradigm shift began with the recognition that user behaviors and item attributes could be modeled as sequences, drawing inspiration from natural language processing (NLP). Early attempts to bridge this gap involved adapting recurrent neural networks and attention mechanisms to capture temporal dynamics in user interaction logs. For instance, researchers explored causal masked multi-head attention to optimize both short and long-term user interests, treating user history as a sequence of tokens similar to sentences in a document [1]. Furthermore, the integration of temporal learning with sequence modeling allowed systems to capture complex user-item activity patterns, moving beyond static profiles to understand the evolving nature of user preferences [2]. These developments laid the essential groundwork for viewing recommendation not merely as a matching task but as a sequential prediction problem, where the goal is to anticipate the next element in a user’s journey based on historical context, setting the stage for the eventual adoption of Large Language Models.

However, the true revolution emerged with the advent of Large Language Models (LLMs) and the formulation of recommendation as a sequence-to-sequence generation task. In this emerging generative paradigm, the recommendation process is redefined: instead of scoring a fixed set of candidates, the model autoregressively generates the identifiers of the target items directly. This approach fundamentally alters the retrieval mechanism. As demonstrated in pioneering work, modern recommender systems can now perform large-scale retrieval by training a Transformer-based sequence-to-sequence model to predict the Semantic ID of the next item a user will interact with, effectively turning retrieval into a generation problem [3]. This method eliminates the need for a separate retrieval stage followed by ranking, unifying the pipeline into a single, end-to-end generative process that inherently supports more complex reasoning capabilities.

A critical component of this transition is the design of item identifiers that are compatible with language models. Traditional random IDs are semantically void and difficult for LLMs to reason about, creating a disconnect between the model’s linguistic knowledge and the recommendation task. Consequently, new strategies have been developed to create semantically meaningful tuples of codewords, known as Semantic IDs, which serve as unique identifiers for each item. By mapping items to these structured tokens, generative models can leverage their inherent linguistic reasoning capabilities to generalize to items with no prior interaction history, significantly enhancing retrieval performance in cold-start scenarios [3]. Further advancements have introduced hierarchical quantization methods, such as Graph RQ-VAE, to assign Rec-IDs derived from both multimodal content and collaborative filtering information, ensuring that the generated tokens carry rich semantic and structural meaning [4]. This innovation directly addresses the challenge of aligning semantic content with collaborative signals, a theme that will be explored in greater depth in the subsequent section.

The generative text-to-text paradigm also enables a more natural integration of diverse data modalities. Unlike traditional models that struggle to fuse textual reviews, item descriptions, and interaction logs, generative frameworks can convert all data sources into natural language sequences. The “Pretrain, Personalized Prompt, and Predict Paradigm” (P5) exemplifies this by unifying various recommendation tasks—such as sequential recommendation, review generation, and explanation—into a shared text-to-text framework [5]. In this setup, user-item interactions, metadata, and reviews are all formatted as natural language, allowing the model to learn deep semantics for personalization through a universal language modeling objective. This flexibility extends to handling infinite item spaces, where the system can generate personalized prompts to create new items rather than just retrieving existing ones, a concept explored in generative model recommendation tasks [6].

Moreover, the shift to generative models addresses the inefficiencies of auto-regressive generation in real-time industrial systems through innovative architectural designs. While early generative models relied heavily on autoregressive strategies, recent innovations have introduced non-autoregressive generative models for reranking to enhance efficiency without sacrificing the ability to model intra-list correlations [7]. Additionally, lightweight Transformer architectures specifically tailored for recommendation, such as LightLM, have been proposed to handle the direct generation of items with deep and narrow structures, challenging the notion that massive NLP-oriented models are strictly necessary for high-performance generative recommendation [8].

Ultimately, the transition from discriminative ranking to generative text-to-text paradigms represents more than just an architectural change; it signifies a conceptual leap towards foundational models for recommendation. By leveraging the universal reasoning capabilities of LLMs and formulating recommendation as a language generation task, these systems can achieve zero-shot generalization across diverse domains and platforms [9]. This evolution paves the way for “Artificial General Recommenders” that possess the universality and conversationality to understand complex user intents, reason over vast knowledge bases, and generate personalized responses in a manner that closely mimics human cognitive processes. Having established this generative foundation, the field now faces the pivotal challenge of effectively unifying the rich semantic understanding provided by these models with the implicit collaborative signals derived from user interactions, a topic addressed in the following section.

1.2 Unifying Semantic Content and Collaborative Signals

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Building upon the generative text-to-text foundation established in the preceding section, the integration of Large Language Models (LLMs) into recommender systems necessitates a critical synthesis: bridging the substantial gap between the rich, explicit semantic knowledge inherent in pre-trained language models and the implicit, often sparse collaborative filtering (CF) signals derived from user-item interactions. While the previous section highlighted how generative paradigms reframe recommendation as a sequence prediction task using Semantic IDs, a fundamental challenge persists in ensuring these generated identifiers accurately reflect both content attributes and collective behavior patterns. Traditional recommender systems have long relied on identity-based (ID) representations, where users and items are modeled as discrete indices learned solely from interaction matrices. While effective in warm-start scenarios with abundant data, these ID-based approaches struggle significantly when facing data sparsity or cold-start conditions, as they lack the ability to generalize to unseen entities based on content attributes [10]. Conversely, while LLMs possess vast world knowledge and can interpret item descriptions, reviews, and user profiles with high fidelity, they often overlook the vital collaborative signals that capture collective user behavior patterns, leading to a misalignment between semantic reasoning and task-specific interaction dynamics [11].

The core difficulty lies in the heterogeneity of the representation spaces. Collaborative signals are typically encoded in dense, low-dimensional vector spaces optimized for predicting interaction probabilities, whereas semantic content resides in the high-dimensional, linguistically structured space of LLMs. Simply concatenating these features or using them as independent inputs often fails to capture the complex interdependencies required for accurate recommendation. For instance, pure text-based approaches may excel in cold-start scenarios by leveraging item metadata but frequently underperform in warm scenarios where precise collaborative knowledge is paramount [12]. This dichotomy necessitates sophisticated alignment strategies that do not merely coexist but actively unify these two distinct information sources into a cohesive representation framework, a prerequisite for the robust identifier design discussed in the subsequent section.

To address this, recent research has focused on developing mechanisms to align content semantics with interaction-based representations. One prominent direction involves modifying the objective functions to maximize the mutual information between collaborative embeddings and semantic feature representations. By reformulating cold-start item representation learning from an information-theoretic standpoint, frameworks like Contrastive Learning-based Cold-start Recommendation (CLCRec) aim to preserve collaborative signals within content representations, ensuring that even items without historical interactions retain a footprint of collective behavior patterns [13]. Similarly, the CARec model introduces a collaborative alignment approach that integrates CF with semantic representations, ensuring that the alignment occurs directly within the semantic space while retaining key semantic meanings, thereby extending applicability to cold-start scenarios with notable accuracy enhancements [10].

Another critical strategy involves the use of contrastive learning and graph-based augmentation to fuse these signals. Approaches such as LLMRec leverage the reasoning capabilities of LLMs to augment interaction graphs by reinforcing user-item edges and enhancing node attributes through natural language profiling. This method addresses data sparsity by utilizing rich textual content to fill gaps in the interaction graph, while simultaneously employing denoising mechanisms to ensure the reliability of the augmented data [14]. Furthermore, knowledge graph-based semantic

contrastive learning has been proposed to jointly train on user-item and item-item interactions, utilizing descriptive text to better understand entity relationships. This hybrid multi-task learning not only improves ranking performance but also enhances recommendation diversity, benefiting cold-start users who have few interactions [15].

The challenge of unification is further complicated by the need to handle long-tail items and sparse data without succumbing to the limitations of either modality alone. Methods like CoLLM address this by seamlessly incorporating collaborative information into LLMs via external traditional models, mapping collaborative embeddings into the input token space of the LLM. This allows the model to exploit high-quality user/item embeddings trained by state-of-the-art CF systems while maintaining the LLM’s emergent reasoning abilities, effectively balancing performance across both cold and warm start scenarios [16]. Additionally, the concept of “Semantic Decoding” proposes viewing the collaboration between LLMs and tools as an optimization procedure in semantic space, where semantic tokens are exchanged to construct high-quality outputs, offering a fresh perspective on how to orchestrate these diverse information sources [17].

Moreover, the alignment process must account for the potential noise and bias introduced by implicit feedback data. Frameworks like RLMRec propose a model-agnostic approach that aligns the semantic space of LLMs with the representation space of collaborative relational signals through a cross-view alignment framework. This ensures that the intricate semantic aspects of user behaviors are captured while mitigating the noise inherent in implicit feedback [18]. The effectiveness of these unification strategies is evident in their ability to outperform traditional baselines in both sparse and dense data regimes, demonstrating that the synergy between semantic depth and collaborative breadth is essential for the next generation of recommender systems.

Ultimately, unifying semantic content and collaborative signals is not merely a technical exercise in feature fusion but a fundamental requirement for building robust, generalizable, and interpretable recommendation systems. As the field moves towards the generative paradigms detailed in Section 1.1, the ability to harmonize the explicit reasoning of LLMs with the implicit statistical strengths of collaborative filtering defines the efficacy of systems in overcoming data sparsity and solving cold-start problems. However, this unified representation must be effectively translated into a format that LLMs can generate and reason over. This leads directly to the pivotal challenge of item indexing: how to encode these fused semantic-collaborative signals into unique, LLM-compatible identifiers that enable efficient retrieval without hallucination, a topic explored in depth in the following section.

1.3 The Critical Role of Item Indexing and Identifier Design

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Building upon the unification of semantic content and collaborative signals discussed previously, the transition from discriminative ranking models to generative recommendation paradigms introduces a pivotal architectural shift: the interface between user intent and item representation. In traditional collaborative filtering systems, items are represented by opaque, high-dimensional ID embeddings learned jointly with user representations, or by dense vectors derived from content features. However, in the realm of Large Language Model (LLM)-based generative recommendation, the model is tasked with directly autoregressively generating the identifiers of recommended items as text tokens. This shift elevates the design of item indexing and identifier strategies from a mere implementation detail to a critical component that dictates the feasibility, efficiency, and reliability

of the entire system. The core challenge lies in constructing LLM-compatible item identifiers that are simultaneously concise, unique, semantically meaningful, and capable of supporting efficient retrieval without inducing hallucinations.

The necessity for robust indexing stems from the inherent limitations of LLMs when dealing with vast, dynamic item corpora. If items are indexed using trivial methods, such as assigning random numeric IDs or relying solely on raw titles, the generative process becomes prone to severe inefficiencies and errors. Random indexing, while ensuring uniqueness, fails to provide any semantic signal to the language model, forcing it to memorize arbitrary mappings between user contexts and meaningless token sequences. This approach often leads to excessive generation lengths and a high probability of the model hallucinating non-existent items, as the generated tokens lack grounding in the model’s pre-trained knowledge space. Similarly, using full item titles or descriptions as identifiers can result in excessively long output sequences, increasing computational latency and the likelihood of generation drift, where the model produces plausible-sounding but invalid item descriptions that do not correspond to any entity in the catalog. As highlighted in recent studies, creating LLM-compatible item IDs is essential to avoid generating excessively long text and hallucinated recommendations when deciding which items to recommend [19].

To address these challenges, the research community has begun to explore a spectrum of indexing strategies that balance distinctiveness with semantic richness, effectively operationalizing the alignment principles mentioned in the preceding section. A primary direction involves the development of semantic or content-based indexing, where identifiers are derived from the intrinsic attributes of the items. For instance, approaches like ACID propose using abstractive, content-based IDs composed of keyphrases generated by LLMs, rather than integer sequences. This method leverages the natural language understanding capabilities of the model, allowing it to generate identifiers that are human-readable and semantically grounded, thereby improving retrieval accuracy significantly compared to hierarchical clustering of embeddings [20]. Furthermore, the concept of semantic identifiers has been extended to learn hierarchical structures that preserve the semantics of documents or items within their IDs. Frameworks such as LMINDEXER introduce self-supervised methods to learn these semantic IDs using generative language models, tackling the challenge of sequential discrete representation through contrastive learning and progressive training [21]. These semantic approaches aim to bridge the gap between the continuous latent space of traditional recommenders and the discrete token space of LLMs.

Conversely, to fully leverage the collaborative signals unified in Section 1.2, collaborative indexing strategies seek to embed interaction patterns directly into the identifier structure. By analyzing user-item interaction graphs, these methods assign IDs that reflect collaborative signals, ensuring that items with similar user behaviors share structural similarities in their identifiers. This allows the LLM to leverage its reasoning capabilities to infer user preferences based on the structural relationships between IDs, rather than solely relying on content semantics. The study of indexing strategies systematically contrasts random, sequential, collaborative, and hybrid methods, demonstrating that the choice of indexing profoundly influences the performance of LLM-based recommenders. Specifically, hybrid indexing, which combines collaborative signals with semantic content, has shown to be particularly effective in real-world datasets, validating the need for multi-faceted identifier designs [19].

A significant advancement in this domain is the multi-facet paradigm, which addresses the limitations of single-view identifiers by synthesizing the diverse information sources discussed thus far. Traditional ID-based or description-based identifiers often compromise either semantic richness or uniqueness. To combat this, novel frameworks like TransRec employ multi-facet identifiers

that incorporate ID, title, and attribute information simultaneously. This approach achieves both distinctiveness and semantics, while specialized data structures guarantee that the generated identifiers remain within the corpus, effectively mitigating the risk of out-of-corpus generation [22]. Additionally, generative retrieval methods like GLEN utilize dynamic lexical identifiers and a two-phase index learning strategy to learn meaningful relevance signals, further enhancing the model’s ability to rank and retrieve items accurately without auxiliary index structures [23].

The design of these identifiers is also intrinsically linked to the mitigation of hallucination, a pervasive issue in generative systems that directly impacts the reliability of the user modeling discussed in the following section. Hallucination in recommendation manifests as the generation of item IDs that do not exist in the database or attributes that contradict factual item data. Effective indexing strategies act as a constraint mechanism, guiding the generative process toward valid token sequences. For example, hierarchical encoding-decoding methods like Hi-Gen utilize category-guided clustering to facilitate docID generation, ensuring that the positional information and inter-token relations are exploited to maintain generation validity in large-scale e-commerce settings [24]. Moreover, the integration of collaborative data refinement in frameworks like Persona-DB highlights how optimizing the representation of the database can improve generalization and reduce errors in cold-start scenarios, further emphasizing the role of structured indexing in personalized retrieval [25].

In conclusion, the paradigm shift to generative recommendation necessitates a re-evaluation of how items are indexed, serving as the foundational bridge between the unified semantic-collaborative representations and the dynamic user modeling capabilities explored next. The move away from opaque embeddings toward explicit, LLM-compatible identifiers requires a delicate balance between semantic expressiveness, collaborative signal integration, and structural constraints. Whether through abstractive keyphrases, hierarchical semantic codes, or multi-facet hybrid tokens, the ultimate goal is to create an indexing schema that enables the LLM to perform efficient, accurate, and hallucination-free generative retrieval. As the field evolves, the development of standardized, scalable, and adaptive indexing mechanisms will remain a foundational prerequisite for realizing the full potential of foundation models in recommendation systems, setting the stage for transforming how user contexts are contextualized and interpreted.

1.4 Transforming User Modeling via Natural Language Contextualization

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Building upon the robust item indexing strategies established in the preceding section, which provide the essential vocabulary for generative recommendation, the focus now shifts to the complementary challenge: constructing dynamic, language-grounded representations of users. The advent of Large Language Models (LLMs) has precipitated a fundamental paradigm shift in user modeling, moving away from static, dense vector representations derived from collaborative filtering or shallow neural encoders. While traditional methods offer computational efficiency, they often struggle to capture the nuanced, evolving, and multi-faceted nature of human preferences, particularly when confronted with long interaction histories or sparse data. In contrast, generative AI redefines the user not as a fixed point in a latent space, but as a dynamic entity whose context can be articulated, reasoned upon, and updated through natural language. This subsection explores the mechanisms by which LLMs convert complex, high-dimensional interaction logs into semantically rich natural language prompts or adaptable dense embeddings, thereby enabling the dynamic intent inference

and personalized reasoning required to effectively utilize the item identifiers discussed previously.

A primary challenge in leveraging LLMs for this transformed user modeling lies in the effective incorporation of potentially noisy and extensive user interaction data. Directly feeding raw historical logs into a model’s context window is often computationally prohibitive and can lead to performance degradation due to information overload. To address this scalability issue while maintaining the semantic richness enabled by the new indexing paradigms, recent frameworks have proposed leveraging user embeddings to contextualize LLMs dynamically. For instance, the [26] framework introduces a novel approach where user embeddings, distilled from diverse interactions via self-supervised pretraining, are integrated with LLMs through cross-attention and soft-prompting. This method allows the model to capture latent user preferences and their evolution over time without the inefficiency of processing raw text sequences, demonstrating significant performance gains on tasks requiring deep user understanding compared to traditional text-prompt-based contextualization. Similarly, the [27] study tackles the specific hurdle of modeling long user histories by introducing a User Embedding Module (UEM). This module efficiently compresses free-form text histories into embeddings that serve as soft prompts, biasing the language model with user signals and enabling the handling of significantly longer histories than conventional methods allow.

Beyond simple compression, the transformation of user data into natural language enables a deeper, more interpretable form of user understanding known as “interest journeys.” Traditional recommender systems encode activities into opaque representations that are difficult to examine or interpret, creating a disconnect between the user’s actual intent and the system’s internal state. In contrast, LLMs can reason through user activities to describe interests in nuanced, human-like narratives that align with the semantic structure of the item identifiers. The [28] paper argues that LLMs can identify persistent, non-transient interests—termed “interest journeys”—and summarize them using techniques like few-shot prompting and fine-tuning. This capability moves user modeling from mere prediction to assistive, journey-aware understanding, where the system can articulate *why* a user prefers certain items based on a coherent narrative of their evolving hobbies and routines. Furthermore, to handle the dynamic nature of user preferences in real-time applications, incremental modeling approaches have been developed. The [29] work proposes a paradigm where embeddings of recent interaction histories are dynamically integrated into accumulated history vectors via a transformer encoder. This allows for the creation of generalized user representations in a consecutive manner, effectively alleviating data management challenges while adapting to shifting user interests.

The contextualization of user models also extends to integrating external knowledge and specific domain contexts to enhance personalization, further bridging the gap between user intent and the item corpus. In scenarios such as web search, where understanding a user’s goals and existing knowledge is crucial, the [30] framework constructs an entity-centric knowledge store for each user. This store aggregates projections of user interests onto public knowledge graphs, providing contextually relevant augmentations to LLM prompts. This approach mitigates privacy concerns associated with deep profiling while significantly improving the relevance and utility of generated query suggestions. Moreover, the integration of summarization and retrieval has proven effective for personalization under practical constraints. The [31] study demonstrates that extending retrieval-augmented personalization with task-aware user summaries generated offline by LLMs can outperform pure retrieval methods. This hybrid strategy ensures that the model retains a profound understanding of the user even when runtime constraints limit the amount of retrievable data, effectively bridging the gap between cold-start challenges and real-time responsiveness.

However, the reliability of these natural language-based user models depends heavily on the robust-

ness of the prompting strategies employed, as inconsistencies here can undermine the precision of the generative retrieval enabled by advanced indexing. Research indicates that simple perturbations in prompt structure can significantly degrade an LLM’s ability to consistently capture user perceptions or personas. The [32] study highlights that current prompting practices often fail to elicit consistent responses regarding user opinions, suggesting a need for more robust elicitation frameworks. To counter this, advanced methods like [33] propose using contrastive examples—providing both positive and negative instances of user intent—to better guide the model’s articulation of user needs. Additionally, the [34] framework introduces an automated method to revise prompts by synthesizing personal context, utilizing a chain of supervised and reinforcement learning to optimize the input sent to frozen LLMs. These advancements underscore that transforming user modeling via natural language is not merely a matter of translation but requires sophisticated engineering to ensure that the resulting contextualization is both accurate and resilient.

Finally, the frontier of user modeling is expanding toward fully personalized parameter-efficient tuning, where the user model becomes an intrinsic part of the LLM’s architecture, mirroring the tight integration seen in modern item indexing. The [35] paper introduces the concept of “One PEFT Per User” (OPPU), where personalized modules store user-specific behavior patterns. This approach integrates parametric user knowledge with non-parametric retrieval, allowing individual LLMs to adapt to user behavior shifts more effectively than prompt-only methods. By moving from static vectors to dynamic, language-grounded, and parametrically adaptable user representations, LLMs are redefining the core of recommendation systems. This evolution sets the stage for the next major leap: the consolidation of these specialized, personalized components into unified Foundation Models for Recommendation, capable of zero-shot generalization across diverse domains as discussed in the following section.

1.5 Towards Foundation Models for Recommendation

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Building upon the dynamic, language-grounded user representations discussed in the previous section, the trajectory of recommender systems is now shifting from specialized, task-specific architectures toward a unified paradigm known as Foundation Models for Recommendation. While prior approaches focused on contextualizing individual users through natural language or parameter-efficient tuning, this emerging vision posits the creation of a single, massive model capable of zero-shot generalization across diverse domains, platforms, and recommendation tasks. Unlike traditional collaborative filtering models that require training from scratch for every new domain, foundation models leverage the universal reasoning capabilities and vast world knowledge inherent in Large Language Models (LLMs) to unify disparate recommendation objectives under a single framework. The core hypothesis driving this shift is that language serves as a sufficiently rich medium to represent users, items, interactions, and contextual metadata, thereby allowing a single model to learn universal interaction patterns that transcend specific datasets.

A pioneering effort in this direction is the “Pretrain, Personalized Prompt, and Predict Paradigm” (P5), which explicitly frames recommendation as a language processing task. By converting all data modalities—including user-item interactions, item metadata, and user reviews—into natural language sequences, P5 demonstrates that a single model can handle multiple downstream tasks such as sequential recommendation, rating prediction, and review generation without task-specific architectural changes [5]. This approach advances the field from shallow, narrow models to deep, big models that function as universal recommendation engines. The ability of P5 to perform

predictions in a zero-shot or few-shot manner significantly reduces the necessity for extensive fine-tuning, marking a critical step toward the foundation model ideal where a pre-trained system can adapt to new scenarios with minimal additional data.

However, building such foundation models presents unique challenges, particularly regarding the alignment of LLMs with the statistical nature of recommendation data. While LLMs possess robust semantic priors, they often struggle to implicitly capture collaborative filtering signals, such as item popularity and co-occurrence patterns, which are crucial for accurate ranking. Research indicates that simply increasing the quantity of training samples yields marginal benefits for models like BIGRec, suggesting that LLMs have limited capacity to assimilate statistical information solely through standard fine-tuning [36]. Consequently, the development of true foundation models requires novel mechanisms to bridge the gap between semantic understanding and collaborative patterns. Strategies such as injecting recommendation-specific knowledge through auxiliary tasks like Masked Item Modeling and Bayesian Personalized Ranking, simulated via natural language, have shown promise in equipping LLMs with the necessary domain expertise to outperform conventional baselines in retrieval and ranking tasks [37].

The quest for universality also extends to cross-domain generalization, where a foundation model must operate effectively on unseen platforms or markets. The “PreRec” framework addresses this by proposing a causal debiasing perspective, utilizing a hierarchical Bayesian deep learning model to capture universal interaction patterns while mitigating heterogeneous biases imposed by cultural differences and platform-specific mechanisms [38]. This is essential for achieving the zero-shot adaptability that defines a foundation model, allowing it to transfer knowledge from data-rich domains to data-scarce environments. Furthermore, the concept of using statistical characteristics of user-item interaction matrices to learn dataset-independent representations offers another pathway toward universal models that do not rely on overlapping users or items across datasets [39].

Recent industrial advancements further validate the scalability and efficacy of this paradigm. The introduction of HSTU, a generative recommender architecture with 1.5 trillion parameters, demonstrates that model quality scales as a power-law of training compute, mirroring the scaling laws observed in general LLMs [40]. This massive scale enables the model to handle high-cardinality, non-stationary streaming data, improving online metrics significantly and paving the way for the first true foundational models in the recommendation domain. Such systems move beyond the limitations of traditional multi-stage pipelines, reformulating recommendation as a sequential transduction task that can be optimized end-to-end.

Moreover, the vision of foundation models encompasses the ability to act as autonomous agents capable of leveraging external knowledge and tools. Systems like RecMind illustrate this potential by employing a self-inspiring algorithm that allows an LLM-powered agent to plan and utilize historical information dynamically to provide zero-shot personalized recommendations [41]. This agent-based approach highlights the future direction where foundation models are not merely passive predictors but active reasoning engines that can navigate complex decision spaces. Similarly, the IDGen framework proposes representing items as unique, semantically rich textual IDs, enabling the training of a foundational generative recommendation model on data from multiple datasets that can generalize to unseen platforms in a completely zero-shot setting [9].

Despite these advances, the path to fully realized foundation models for recommendation is not without obstacles. The sensitivity of LLMs to input prompts and the potential for hallucination remain significant concerns that must be addressed to ensure reliability in production environments [42]. Additionally, the tension between maintaining the general reasoning capabilities of LLMs

and optimizing for specific recommendation metrics requires careful balancing, often necessitating specialized instruction tuning or reinforcement learning from personalized human feedback. Nevertheless, the convergence of large-scale pre-training, unified text-to-text paradigms, and advanced reasoning strategies suggests that the era of foundational recommendation models is imminent. These models promise to democratize access to high-quality recommendation capabilities, enabling zero-shot deployment across diverse applications ranging from e-commerce to healthcare, ultimately reshaping the technological landscape of personalized information access.

2 Architectural Foundations and Representation Learning

2.1 Semantic, Hierarchical, and Collaborative Item Indexing

2.1 Semantic, Hierarchical, and Collaborative Item Indexing

The transition from traditional discriminative recommendation systems to generative paradigms necessitates a fundamental rethinking of how items are represented within the model. In classical collaborative filtering, items are typically assigned unique, randomly generated numeric identifiers (IDs) that are mapped to dense embedding vectors. While effective for memorizing interaction patterns in static settings, this approach suffers from severe limitations in generative contexts: random IDs lack intrinsic meaning, preventing the model from generalizing knowledge across items with similar attributes and failing to support inference on unseen entities. To address these challenges—and to lay the necessary groundwork for the structural reasoning discussed in the following subsection—recent research has pivoted towards advanced indexing strategies that integrate semantic richness, hierarchical structure, and collaborative signals directly into the item identifiers themselves.

A pivotal advancement in this domain is the development of Semantic IDs, which replace arbitrary numeric tokens with compact, discrete representations learned from item content. Unlike random IDs, Semantic IDs capture the hierarchy of concepts inherent in item descriptions, titles, or attributes, thereby enabling the model to perform reasoning based on content similarity rather than mere co-occurrence. A prominent methodology for generating such identifiers involves the use of Residual Quantized Variational Autoencoders (RQ-VAE). This technique learns a hierarchical codebook where each item is represented as a sequence of quantized codes derived from its content embeddings. By substituting traditional random IDs with these semantically grounded sequences, recommendation models can significantly improve their generalization capabilities, effectively mitigating the cold-start problem for tail and previously unseen items without sacrificing top-level ranking metrics [43]. This approach ensures that the identifier itself carries meaningful information, allowing the generative model to infer preferences for new items based on their semantic proximity to known items.

Beyond learning semantic IDs from content, another critical dimension involves the structural organization of these identifiers to mimic human cognitive learning strategies. Inspired by cognitive psychology, specifically the concepts of elaboration and rehearsal, researchers have proposed Semantic-Enhanced Differentiable Search Index (SE-DSI) models. In this framework, document or item identifiers are not merely integers but are constructed as “Elaborative Descriptions” generated via query generation techniques. This makes the identifiers more meaningful and easier for the sequence-to-sequence model to associate with the underlying content. Furthermore, by selecting fine-grained semantic features as “Rehearsal Contents,” the model improves its memorization of the association between an item and its identifier, leading to superior retrieval performance compared to standard Differentiable Search Index (DSI) approaches [44]. This highlights the importance of designing identifiers that align with the model’s learning mechanisms, transforming the indexing

process from a simple lookup task into a semantic reasoning exercise.

To further bridge the gap between discrete identifiers and continuous semantic spaces, self-supervised frameworks have been introduced to learn neural sequential discrete representations directly. Traditional two-stage pipelines, which first generate embeddings and then derive IDs, often suffer from information loss and distribution mismatches. Innovative approaches like LMINDEXER tackle this by employing a generative language model as a semantic indexer. Through progressive training and contrastive learning objectives, such models generate IDs that preserve the semantic nuances of the original documents or items. The use of a self-supervised document reconstruction objective helps overcome the deficiency of explicit semantic supervision, ensuring that the learned identifiers facilitate robust performance in downstream tasks such as product search and recommendation [21]. This end-to-end learning of semantic IDs ensures that the discrete tokens used for generation are optimally aligned with the latent semantic space of the corpus.

However, relying solely on semantic content can sometimes compromise the distinctiveness required for precise item retrieval, especially in large-scale catalogs where many items share similar descriptions. Consequently, hybrid paradigms have emerged to balance semantic meaning with unique identification. The TransRec framework exemplifies this by employing multi-facet identifiers that incorporate random IDs, titles, and attributes simultaneously. This multi-facet approach ensures both distinctiveness and semantic richness, addressing the limitations of pure ID-based or pure description-based identifiers. Additionally, to prevent the generative model from hallucinating out-of-corpus items—a common risk when generating free-form text—specialized data structures and substring indexing techniques are utilized. These mechanisms guarantee that the generated token sequences correspond to valid items within the corpus, while substring indexing encourages the model to generate identifiers from any position, enhancing robustness [22].

The challenge of hallucination is particularly acute in generative recommendation, where the model might generate plausible but non-existent item identifiers. Advanced indexing strategies play a crucial role in mitigating this risk by grounding the generation process in structured, verified semantic spaces. By enforcing a strict mapping between semantic concepts and discrete token sequences, these indexing methods reduce the reliance on the language prior alone, which is often the source of hallucinated content. The integration of hierarchical and collaborative signals into the indexing layer ensures that the generative process is constrained by both the structural logic of the item catalog and the collaborative patterns of user behavior. As the field moves towards foundation models for recommendation, the design of these indexing schemes serves as a critical prerequisite for the more complex structural integrations discussed next; specifically, while this subsection establishes the semantic foundation of item representation, the following section explores how Knowledge Graphs can be integrated to provide the explicit relational backbone necessary for robust, explainable, and logically consistent generative recommendations.

2.2 Integrating Knowledge Graphs for Structural Representation

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Building upon the semantic and hierarchical indexing strategies established in the previous subsection, which ground item identifiers in content and structure, this section explores the integration of Knowledge Graphs (KGs) to provide an explicit relational backbone for generative recommendation. While Semantic IDs enable models to generalize based on attribute similarity, they often lack the explicit logical connectivity required for complex reasoning and explainability. KGs ad-

dress this gap by offering a structured repository of world knowledge that connects entities through defined relations, serving as a critical bridge between the discrete tokens of generative models and the continuous space of semantic understanding. This subsection examines frameworks that fuse KG structural relations with textual semantics, focusing on multi-layer network representations, the joint learning of explainable rules, and advanced contrastive alignment strategies designed to enhance performance in cold-start scenarios and mitigate hallucinations.

A primary challenge in leveraging KGs lies in harmonizing the discrete, symbolic nature of graph structures with the continuous, dense embeddings of language models. Traditional approaches often struggle to capture high-order dependencies without suffering from over-smoothing or losing semantic nuance. To address this, recent frameworks have adopted multi-layer network representations that jointly encode local graph topology and global semantic context. For instance, the “Roof-Transformer” architecture introduces a dual-encoder mechanism where one module processes knowledge resources while another encodes input sentences, fusing them via a dedicated layer to overcome sequence length limitations and retain rich structural information [45]. Similarly, the “GreaseLM” model demonstrates the efficacy of multi-layer modality interaction, allowing information to propagate bidirectionally between pretrained language models and graph neural networks. This bidirectional flow ensures that linguistic nuances ground the graph representations, while structured world knowledge constrains and enriches the language context, leading to more robust reasoning capabilities [46]. Furthermore, frameworks like “IMF” (Interactive Multimodal Fusion) propose two-stage fusion mechanisms that preserve modality-specific knowledge before integrating them through bilinear pooling and contrastive constraints, ensuring that the unique structural signals of the KG are not diluted during fusion [47].

Beyond mere representation fusion, the interpretability of recommendation decisions is paramount, extending the logical consistency initiated by structured indexing. Purely neural approaches often function as black boxes, lacking the ability to articulate the logical paths leading to a specific suggestion. To resolve this, researchers are increasingly turning to neuro-symbolic frameworks that jointly learn explainable rules alongside neural representations. The “RuleE” framework exemplifies this by embedding logical rules directly into the same vector space as entities and relations, enabling soft logical inference that alleviates the brittleness of strict symbolic logic while enhancing the regularity of embeddings [48]. Complementing this, the “EngineKG” framework establishes a closed-loop learning process where a rule-learning module prunes and refines candidate rules based on KG embeddings, while the embedding module is simultaneously enhanced by these symbolic rules. This symbiotic relationship not only improves link prediction accuracy but also generates human-readable reasoning paths, a crucial feature for trustworthy generative recommenders [49]. Additionally, methods that induce explainable rules from item-centric graphs and integrate them into rule-guided neural models have shown significant improvements in handling noisy data and providing transparent justifications for recommendations [50].

The utility of KG integration is perhaps most evident in addressing the cold-start problem, where new users or items lack sufficient interaction history—a challenge initially mitigated by semantic IDs but fully resolved through relational connectivity. Here, KGs provide a semantic bridge, allowing models to infer preferences based on attribute connectivity rather than collaborative signals alone. Contrastive learning has emerged as a powerful strategy to align sparse collaborative signals with rich KG semantics. The “KGCL” framework specifically targets the noise and sparsity inherent in real-world KGs by employing a cross-view contrastive learning paradigm. By contrasting augmented views of the knowledge graph against user-item interactions, KGCL suppresses irrelevant connections and amplifies unbiased signals, significantly boosting performance in long-tail

and cold-start distributions [51]. Similarly, “KRACL” leverages graph context modeling and a specialized knowledge contrastive loss to enrich feedback for sparse entities, demonstrating superior robustness in datasets with low in-degree entities [52].

Furthermore, specific architectures have been designed to transfer knowledge from warm entities to cold-start targets through personalized feature referencing. The “KPER” model utilizes KGs not just as side information but as a dynamic mechanism to probe semantically relevant seeds, aggregating features via gated mechanisms to construct robust representations for cold-start items [53]. Meta-learning approaches also benefit from KG integration; “MetaKG” employs both collaborative-aware and knowledge-aware meta-learners to capture high-order relations that generalize effectively to new tasks, effectively mitigating the data scarcity issue [54]. In strict cold-start scenarios where interactions are entirely absent, frameworks like “ColdGPT” pre-train item-attribute graphs using multi-task learning, transferring knowledge from review texts and purchase sequences to enable accurate embedding generation for unseen items [55].

Finally, the alignment of structural and semantic spaces is critical for inductive reasoning, ensuring that the logical foundations laid by KGs can be effectively utilized by generative models. Methods such as “Bi-Link” utilize contrastive learning with probabilistic syntax prompts to bridge inductive link predictions, ensuring that relational patterns like symmetry and inversion are preserved even when linearizing triples for language models [56]. Moreover, the “SMiLE” framework incorporates network schema as a prior constraint in multi-level contrastive learning, preserving entity consistency across different contexts and improving link prediction accuracy [57]. By synthesizing these structural, symbolic, and contrastive techniques, modern architectures establish a unified foundation where KGs provide the skeletal logic necessary for LLMs to generate accurate, diverse, and explainable recommendations. Having solidified this structural and logical backbone, the discussion now transitions to the enrichment of these representations through diverse sensory modalities, as explored in the following subsection on Multimodal Fusion and Cross-View Alignment Architectures.

2.3 Multimodal Fusion and Cross-View Alignment Architectures

2.3 Multimodal Fusion and Cross-View Alignment Architectures

While the previous subsection established how Knowledge Graphs (KGs) provide the skeletal logic and structural grounding for generative recommendation, this section addresses the enrichment of these representations through diverse sensory modalities. The integration of visual scenes, textual descriptions, and complex behavioral sequences is paramount for constructing robust generative systems that transcend purely symbolic or collaborative signals. As the field transitions from unimodal discriminative models to multimodal generative paradigms, the architectural design of fusion mechanisms and cross-view alignment strategies has become a critical area of investigation. Effective architectures must not only combine heterogeneous data sources but also ensure that the semantic richness of one modality enhances the representation learning of others without introducing noise or computational bottlenecks. This subsection analyzes state-of-the-art architectural designs that facilitate bidirectional view-based learning, employ non-invasive attention mechanisms, and leverage teacher-student distillation frameworks to achieve superior cross-modal alignment, thereby laying the representational foundation necessary for the parameter-efficient adaptation strategies discussed in the subsequent section.

A fundamental challenge in multimodal fusion lies in modeling the intricate interactions between different data streams while preserving their unique structural properties, a problem analogous to

harmonizing discrete graph structures with continuous embeddings discussed earlier. Traditional approaches often rely on simple concatenation or early fusion, which can lead to modality mismatch and information loss. To address this, recent advancements have introduced sophisticated attention mechanisms that dynamically weigh the contribution of each modality based on contextual relevance. For instance, the concept of reciprocal attention fusion has proven effective in scenarios requiring fine-grained alignment, such as Visual Question Answering, by jointly considering relationships between object instances and their parts [58]. This hierarchical fusion of multi-modal information allows models to focus on scene elements most relevant to a specific query, a principle that is directly transferable to recommendation tasks where user intent must be aligned with specific item attributes. Furthermore, the development of structured spatial attention mechanisms, such as AttentionRNN, explicitly enforces dependencies among attention variables, ensuring consistent attention masks that account for both local image features and previously predicted values [59]. Such structured approaches are vital for recommendation systems that need to interpret sequential user behaviors alongside static visual content, ensuring that the temporal evolution of interest is captured alongside spatial semantics.

Bidirectional view-based learning represents another pivotal architectural strategy, particularly when dealing with partial observability or conflicting signals across modalities, mirroring the cold-start challenges addressed via KG connectivity. In many real-world recommendation scenarios, visual data may be incomplete or occluded, necessitating the supplementation of visual modalities with auditory or textual cues. The “Segment Beyond View” framework exemplifies this by utilizing a teacher-student distillation model where a vision teacher with panoramic information guides an audio-visual student operating with limited field-of-view inputs [60]. This paradigm of using a privileged teacher to enhance a student’s perception under constraints is highly relevant for building efficient recommenders that must operate with sparse user interaction logs or limited item metadata. Similarly, the Switch-BERT architecture addresses modality mismatch by learning to optimize attention from a set of interaction modes, allowing the model to attend to outputs from various depths and effectively switch between intra-modal and inter-modal processing [61]. This flexibility enables the system to adapt its fusion strategy dynamically, a crucial capability for handling the diverse and evolving nature of user behavior data.

Teacher-student distillation frameworks have emerged as a powerful tool for transferring knowledge from large, computationally expensive multimodal teachers to compact, efficient student models suitable for real-time recommendation, serving as a conceptual precursor to the parameter-efficient fine-tuning techniques detailed later. This approach is particularly valuable for aligning representations across disparate modalities without requiring massive joint training datasets. Cross-modal contrastive distillation (CCD) has been proposed to bridge the gap between visual and text modalities during the distillation process, leveraging external textual knowledge to assist visual anticipation tasks [62]. By facilitating knowledge transfer between heterogeneous modalities, CCD ensures that the student model inherits the semantic reasoning capabilities of the teacher, thereby improving the quality of generated recommendations. Moreover, saliency-aware knowledge distillation introduces modality-specific losses weighted by saliency scores, allowing the student to mimic the teacher’s behavior within each specific modality [63]. This granular control over the distillation process helps in preserving the distinct characteristics of visual, textual, and behavioral signals while achieving a unified representation space.

Non-invasive attention mechanisms further refine these architectures by enabling the integration of new modalities without disrupting the pre-existing learned representations, a feature essential for maintaining the stability of large foundation models. The DistilVPR pipeline demonstrates

how feature relationships from multiple agents, including self-agents and cross-agents, can be leveraged to explore diverse manifolds such as Euclidean, spherical, and hyperbolic spaces [64]. This exploration of geometric diversity in feature space enhances the representational capacity of the model, allowing it to capture complex non-linear relationships between user preferences and item features. Additionally, the use of disentangled multimodal attention, as seen in DiMBERT, applies separated attention spaces for vision and language, explicitly disentangling representations before fusing them via visual concepts [65]. This disentanglement prevents the entanglement of noise and ensures that the final fused representation retains high fidelity to the original semantic intent of each modality.

In the context of behavioral modeling, these architectural principles extend to capturing dynamic interactions and social contexts, complementing the rule-based reasoning of neuro-symbolic approaches. Social attention architectures have been designed to handle varying numbers of interacting entities, maintaining invariance to ordering while capturing interaction patterns essential for decision-making in dense environments [66]. Applying similar logic to user-item interactions allows recommenders to model the complex ecosystem of user behaviors more accurately. Furthermore, competitive distillation frameworks like CoMD introduce bidirectional feedback between teacher and student models, continually updating the student’s multimodal capabilities through a competitive process [67]. This iterative refinement ensures that the recommendation model remains adaptive and responsive to shifts in user behavior and item distributions.

Ultimately, the synergy of bidirectional learning, non-invasive attention, and advanced distillation techniques forms the backbone of modern multimodal recommendation architectures. By drawing upon innovations from vision-language tasks, such as high-order attention models that learn correlations across data modalities [68], and adapting them to the specific constraints of recommendation, researchers can build systems that are not only accurate but also interpretable and robust. The ability to effectively fuse visual, textual, and behavioral modalities while maintaining computational efficiency is the key to unlocking the full potential of generative recommendation. However, the richness of these multimodal representations often comes at the cost of increased model complexity, creating a pressing need for scalable deployment solutions. This necessity naturally leads to the exploration of Parameter-Efficient Adaptation and Scalable Fine-Tuning strategies, which enable the practical deployment of these sophisticated architectures in industrial settings.

2.4 Parameter-Efficient Adaptation and Scalable Fine-Tuning

2.4 Parameter-Efficient Adaptation and Scalable Fine-Tuning

Building upon the sophisticated multimodal fusion architectures and advanced distillation frameworks discussed in the previous section, a critical implementation challenge emerges: the prohibitive computational cost of deploying these rich representations at scale. While Section 2.3 established how bidirectional learning and non-invasive attention mechanisms create robust, high-dimensional representational spaces, the massive parameter counts of the underlying foundation models render full fine-tuning impractical for dynamic recommendation environments. Consequently, the research focus has shifted decisively toward Parameter-Efficient Fine-Tuning (PEFT) strategies. These methods enable the adaptation of pre-trained models to downstream recommendation tasks by updating only a minimal fraction of parameters while keeping the backbone frozen, thereby bridging the gap between theoretical architectural richness and practical industrial deployment. This transition is essential not only for reducing memory and energy consumption but also for enabling the rapid personalization required to serve diverse user segments and evolving item catalogs, setting the stage

for the specific semantic-collaborative alignment challenges addressed in the subsequent section.

Among PEFT strategies, Low-Rank Adaptation (LoRA) has emerged as the dominant paradigm, fundamentally reshaping the landscape of scalable recommendation [69]. The core premise of LoRA—that the intrinsic dimensionality of task-specific updates is low—allows for the injection of trainable rank-decomposition matrices into each Transformer layer. This approach drastically reduces the number of trainable parameters without introducing additional inference latency, offering a viable pathway to maintain a single base model while deploying thousands of lightweight, task-specific adapters. However, applying standard LoRA to the complex, often sparse data distributions characteristic of user behavior modeling reveals limitations regarding overfitting, static rank selection, and training memory efficiency. To mitigate overfitting in sparse scenarios, recent advancements introduce bi-level optimization frameworks such as [70], which splits the training of pseudo singular vectors and values across different data subsets to enhance generalization. Furthermore, the inefficiency of static rank assignment, which necessitates exhaustive hyperparameter searches, is addressed by dynamic approaches like [71], utilizing meta-learning to identify optimal ranks per layer, and [72], which trains adapters for a range of ranks simultaneously to eliminate search overhead.

Beyond rank optimization, the structural design of adapters has evolved to maximize expressiveness and convergence speed. Theoretical analyses indicate that the original LoRA formulation may be suboptimal for large-width networks due to symmetric learning rates; [73] corrects this by assigning differential learning rates to projection matrices, significantly accelerating convergence. Similarly, [74] decomposes pre-trained weights into magnitude and direction components, employing LoRA specifically for directional updates to better mimic the learning capacity of full fine-tuning. For scenarios demanding extreme parameter compression, methods like [75] share a single pair of low-rank matrices across all layers, learning only small scaling vectors, while [76] re-parameterizes adapters using linear combinations of random bases to decouple trainable parameters from the network architecture.

Memory efficiency during the training phase remains a critical bottleneck, particularly for the multimodal recommendation systems described previously, which must process extensive user histories and heterogeneous item features. Standard LoRA still requires storing full-rank activation gradients, leading to high GPU memory consumption. [77] addresses this by freezing the projection-down matrix and updating only the projection-up matrix, eliminating the need to store full-rank input activations and reducing memory costs by up to 1.4 times. More radically, [78] proposes projecting gradients into a low-rank subspace rather than restricting parameters themselves, enabling full-parameter learning with memory usage comparable to LoRA—a capability particularly relevant for pre-training or continuous learning in recommendation foundations where representation quality is paramount. Additionally, [79] introduces a decoupled side-network architecture that bypasses backpropagation through the frozen backbone entirely, achieving substantial memory savings that facilitate the adaptation of extremely large models on consumer-grade hardware.

Scalability in serving environments is equally crucial for generative recommendation, where systems must handle heterogeneous requests from millions of users concurrently. Traditional deployment methods struggle with the overhead of loading multiple adapter weights, a challenge directly addressed by [80], which proposes a unified paging mechanism and tensor parallelism strategy to serve thousands of distinct LoRA adapters on a single GPU cluster. Complementing this, [81] allows each input in a mini-batch to utilize unique adapter weights, facilitating efficient batching of diverse user requests without sacrificing throughput. For cross-domain recommendation tasks, [82] demonstrates how multiple LoRA modules trained on different domains can be fluidly com-

posed to adapt to unseen tasks without additional training, fostering an ecosystem of reusable recommendation adapters.

Finally, the robustness and stability of these adaptation methods are paramount in privacy-sensitive or federated learning settings, which are increasingly relevant for user behavior modeling. [83] highlights the instability of standard LoRA under differential privacy constraints and proposes freezing specific matrices to stabilize aggregation. Similarly, [84] introduces orthogonal low-rank subspaces to prevent catastrophic forgetting when sequentially adapting to new user behaviors or item categories. Collectively, these advancements in parameter-efficient adaptation and scalable fine-tuning provide the necessary architectural foundation to transform static foundation models into dynamic, personalized, and efficient generative recommendation engines. By resolving the computational barriers of multimodal architectures, these techniques enable the practical realization of systems that can effectively leverage the rich semantic spaces of LLMs, a prerequisite for the crucial task of bridging collaborative signals with language semantics explored in the following section.

2.5 Bridging Collaborative Signals and Language Semantic Spaces

2.5 Bridging Collaborative Signals and Language Semantic Spaces

While Section 2.4 established the parameter-efficient mechanisms necessary to adapt Large Language Models (LLMs) for recommendation tasks, a fundamental architectural challenge remains: reconciling the distinct representational spaces of traditional collaborative filtering (CF) and the rich semantic latent spaces of pre-trained language models. Traditional CF methods rely heavily on identity-based (ID) embeddings to capture implicit user-item interaction patterns, effectively modeling collaborative signals but often neglecting the rich textual context associated with items and users. Conversely, LLMs excel at understanding semantic nuances through natural language but typically lack inherent mechanisms to process discrete ID-based collaborative signals or to distinguish between items with similar descriptions but divergent interaction histories. Bridging this gap is critical for developing generative recommenders that can simultaneously leverage the world knowledge encoded in LLMs and the precise preference patterns derived from user behavior.

A primary strategy for aligning these disparate spaces involves contrastive learning frameworks designed to maximize the mutual information between collaborative and semantic representations. The core objective is to ensure that the embedding of a user or item derived from interaction graphs is geometrically close to its representation derived from textual descriptions within a unified latent space. For instance, the concept of “alignment and uniformity” has been identified as a theoretical foundation for high-quality representations in collaborative filtering, where alignment pulls related pairs together and uniformity disperses representations to utilize the feature space effectively [85]. Extending this to the LLM domain, researchers have proposed frameworks that explicitly align collaborative signals with semantic spaces to prevent the model from overlooking vital filtering signals while retaining semantic richness [10]. By treating the alignment of these two views as a contrastive objective, models can learn to map ID-based interactions and textual semantics into a shared manifold, thereby mitigating the cold-start problem and enhancing recommendation accuracy for items with sparse interaction data.

To further refine this alignment, advanced methodologies employ cross-view alignment frameworks that treat user IDs and natural language as heterogeneous features requiring specialized integration mechanisms. A notable approach is the use of contrastive prompt learning, which encodes user IDs and natural language descriptions individually before applying auxiliary contrastive objectives to

fuse them. Specifically, techniques such as Heterogeneous Feature Matching (HFM) align item descriptions with corresponding user IDs or next preferred items based on interaction sequences, while Instruction Contrastive Learning (ICL) merges these data sources by contrasting probability distributions of output sequences across diverse tasks [86]. This dual-objective strategy ensures that the LLM does not merely treat IDs as arbitrary tokens but understands them as semantic anchors linked to specific behavioral patterns. Similarly, model-agnostic frameworks have been developed to enhance existing recommenders by incorporating auxiliary textual signals and aligning the semantic space of LLMs with the representation space of collaborative relational signals through cross-view alignment [18]. These frameworks often utilize mutual information maximization to theoretically guarantee that incorporating textual signals enhances the overall quality of the learned representations.

Beyond direct alignment, another critical dimension involves transforming item-level interaction data into word-level collaborations to facilitate transferable representations. Traditional sequential models using ID embeddings struggle with cross-domain generalization because IDs are not shared across domains. To address this, innovative approaches convert interaction data into word graphs containing word-level collaborations, allowing pre-trained language models to align word-level semantic and CF-related item representations [87]. This method effectively infuses collaborative filtering information into text-based item representation learning, enabling the model to perform well in both fine-tuning and zero-shot settings for cross-scenario recommendations. Furthermore, the challenge of data sparsity and the risk of false negatives in contrastive learning have led to the development of principled adversarial contrastive losses. These losses adaptively explore and assign hardness to negative instances, empowering the recommender to distinguish between semantically similar items that have different collaborative signals, thus improving generalization ability in out-of-distribution tasks [88].

Moreover, the integration of collaborative knowledge into LLMs can be achieved through efficient instruction-tuning methods that distill world knowledge and reasoning capabilities directly into collaborative filtering processes. Frameworks inspired by in-context learning and chain-of-thought reasoning allow LLMs to leverage collaborative signals without extensive retraining, preserving the model’s general functionalities while enhancing recommendation capabilities [11]. In scenarios where LLMs underperform traditional CF models in warm-start conditions due to a lack of collaborative knowledge, hybrid architectures have been proposed to enable LLMs to directly leverage high-quality user and item embeddings from pre-trained state-of-the-art CF systems. This approach, often referred to as an all-round LLM-based recommender, jointly exploits the emergent abilities of LLMs and the precise collaborative knowledge of traditional models, ensuring superior performance across both cold and warm scenarios [12].

Finally, the alignment process must also address the geometric properties of the embedding space to avoid dimensional collapse, a phenomenon where representations occupy only a low-dimensional subspace, weakening distinguishability. Non-contrastive learning objectives have been proposed to explicitly mitigate this issue by aiming for geometric properties of alignment and compactness, ensuring that the unified semantic-collaborative space remains robust and expressive [89]. By combining these diverse methodologies—ranging from contrastive prompt learning and cross-view alignment to adversarial loss functions and hybrid embedding distillation—researchers are constructing a unified architectural foundation. This foundation allows generative recommendation systems to seamlessly bridge the gap between the implicit, structural signals of collaborative filtering and the explicit, contextual understanding of language semantic spaces, paving the way for more accurate, interpretable, and generalizable recommendation models.

3 Advanced User Behavior Modeling and Preference Elicitation

3.1 Multi-Behavior Dependency Modeling and Graph-Enhanced Representations

3.1 Multi-Behavior Dependency Modeling and Graph-Enhanced Representations

The evolution of user behavior modeling in recommendation systems has progressed significantly from simplistic single-interaction analyses to sophisticated frameworks capable of deciphering complex, multi-faceted user intents. While traditional sequential models rely on recurrent neural networks or standard self-attention mechanisms to capture item-to-item transitions within a singular interaction sequence, they frequently overlook the rich, heterogeneous relational structures inherent in real-world engagement. Users exhibit diverse behaviors—such as page views, add-to-cart actions, favorites, and purchases—that form a complex web of dependencies rather than a linear path. To address the limitations of modeling singular interaction types, recent research has increasingly turned toward graph-enhanced representations. By leveraging multi-relational Graph Neural Networks (GNNs) and hypergraph architectures, these advanced frameworks explicitly capture cross-type behavior dependencies and hierarchical item correlations, establishing a structural foundation that complements the temporal dynamics discussed in subsequent sections.

A primary challenge in multi-behavior recommendation is the effective modeling of dependencies between different interaction types. Early graph-based methods often treated all behaviors uniformly or focused exclusively on a target behavior, thereby discarding valuable auxiliary signals. To overcome this, researchers have developed Multi-Relational Graph Neural Networks that construct comprehensive item graphs incorporating multiple behavior types. For instance, the MGNN-SPred framework builds a Multi-Relational Item Graph (MRIG) based on all behavior sequences, enabling the model to learn global item-to-item relations across both target and auxiliary behaviors [90]. By utilizing a gating mechanism to adaptively fuse user representations derived from these distinct behavioral sequences, such models can effectively predict sparse but critical target behaviors like purchasing, even when auxiliary data like clicking is abundant. Similarly, the GNMR framework explicitly models the dependencies between different user-item interactions under a graph-based message passing architecture, devising a relation aggregation network to handle interaction heterogeneity [91].

Beyond standard multi-relational graphs, the concept of hypergraphs has emerged as a powerful paradigm for capturing higher-order interactions involving more than two entities. Unlike traditional graphs that model pairwise connections, hypergraphs utilize hyperedges to link multiple nodes simultaneously, which is particularly effective for representing complex group behaviors or sessions where multiple items share a common underlying intent. The MBHT framework exemplifies this approach by designing a Multi-Behavior Hypergraph-enhanced Transformer. This architecture incorporates global multi-behavior dependencies into a hypergraph neural structure to capture hierarchical long-range item correlations, while simultaneously employing a multi-scale Transformer to encode behavior-aware sequential patterns [92]. The integration of hypergraphs allows for the modeling of complex dependencies that simple pairwise graphs cannot adequately represent, such as the simultaneous influence of multiple items on a user’s decision within a specific context.

The application of hypergraph neural networks (HNNs) extends to addressing data sparsity and noise, which are prevalent in multi-behavior scenarios. The SHT framework proposes a Self-Supervised Hypergraph Transformer that augments user representations by exploring global collaborative relationships through a hypergraph transformer network [93]. By distilling global context and employing cross-view generative self-supervised learning, this approach enhances the robustness

of recommender systems against noisy and skewed user behavior data. Furthermore, hypergraphs facilitate the construction of “hyper meta-paths” to explicitly illustrate dependencies among different user behaviors. The HMG-CR framework leverages graph contrastive learning coupled with hyper meta-paths to learn unified user embeddings adaptively, avoiding the limitations of fixed fusion schemes used in earlier methods [94].

Hierarchical structures within graph models also play a crucial role in discriminating the importance of different interaction types. The KHGT model introduces a Knowledge-Enhanced Hierarchical Graph Transformer Network that not only captures type-specific behavior characteristics but also explicitly discriminates which interaction types are most critical for forecasting target behaviors [95]. This hierarchical attention mechanism allows the model to dynamically weigh the contribution of various behaviors, such as distinguishing between a casual view and a strong purchase intent. Similarly, the HMGN framework employs a Hierarchical Multi-behavior Graph Attention Network that leverages attention mechanisms to discern information from both inter- and intra-behaviors, recognizing the nuanced hierarchy intrinsic to user actions [96].

Moreover, the integration of memory mechanisms with graph structures has proven effective for modeling multiplex behavioral relations over time, serving as a conceptual bridge to the dynamic interest evolution explored in the following section. The MATN framework utilizes a Memory-Augmented Transformer Network to jointly model type-specific behavioral contexts and cross-type behavior inter-dependencies, supercharging the capture of contextual signals into category-specific latent embedding spaces [97]. This approach ensures that the temporal dynamics and evolving nature of multi-typed interactions are preserved. In the realm of session-based recommendations, where user profiles are unavailable, hypergraph modeling has shown particular promise. The DHCN model treats session data as a hypergraph and employs a dual-channel convolutional network to maximize mutual information between session representations, effectively capturing high-order information among items that pairwise models miss [98].

Finally, the scalability and adaptability of these graph-enhanced representations are critical for industrial deployment. Frameworks like CIGF address the challenges of unbalanced data distribution and gradient conflicts in multi-task learning by designing a Compressed Interaction Graph Convolution Network to model instance-level high-order relations explicitly [99]. Additionally, meta-learning paradigms such as MB-GMN empower user-item interaction learning with the capability to uncover type-dependent behavior representations, automatically distilling behavior heterogeneity for personalized recommendations [100]. Collectively, these advancements demonstrate that leveraging hypergraphs and multi-relational GNNs provides a robust foundation for understanding the intricate web of user behaviors. By establishing these complex structural dependencies, the field is better positioned to tackle the subsequent challenge of modeling how these interests evolve continuously over time, as detailed in the next section.

3.2 Dynamic Interest Evolution and Long-Term Dependency Capture

3.2 Dynamic Interest Evolution and Long-Term Dependency Capture

Building upon the structural foundations of multi-relational graphs and hypergraphs established in the previous section, a critical subsequent challenge lies in modeling how these complex behavioral dependencies evolve continuously over time. While Section 3.1 focused on capturing the static, heterogeneous web of user actions, this subsection addresses the temporal dimension: *how* user interests drift, decay, or stabilize across lifelong interaction histories. Traditional approaches often treat user preferences as static embeddings or rely on fixed-window sequential models that fail to

capture the continuous and non-linear nature of interest evolution. To address these limitations, contemporary research has increasingly shifted towards methodologies that conceptualize user interests as dynamic functions or evolving memory states. This paradigm enables the effective bridging of short-term actions with long-term preference trajectories, serving as a necessary precursor to the latent intent disentanglement explored in the following section.

A fundamental paradigm shift involves moving from discrete time-step modeling to continuous-time representations. In real-world scenarios, user interactions occur at irregular intervals, and the semantic meaning of a behavior is heavily influenced by the time elapsed since previous actions. Standard attention mechanisms often overlook this temporal distance, focusing primarily on spatial correlations within the sequence. To rectify this, researchers have introduced self-modulating attention networks that explicitly incorporate continuous time spaces, allowing the model to learn complex, non-linearly evolving dynamic preferences where the relevance of past behaviors decays or transforms based on temporal proximity [101]. Furthermore, the emergence of Neural Ordinary Differential Equations (ODEs) has enabled the modeling of preference evolution as a continuous process rather than a series of discrete jumps. For instance, the Graph Nested GRU ODE framework extends neural ODEs to continuous-time temporal session graphs, preserving the continuous nature of dynamic user preferences and encoding both temporal and structural patterns into dynamic embeddings [102]. Similarly, continuous-time user preference modeling frameworks maintain a memory bank to store the states of users and elements, updating them chronologically to explicitly model evolving preferences for temporal sets prediction [103].

Parallel to continuous modeling, the concept of dynamic memory states has gained prominence as a mechanism to handle extremely long behavior sequences without incurring prohibitive computational costs, effectively extending the graph-based relational capabilities discussed earlier into the temporal domain. The core idea is to compress historical interactions into a set of memory blocks that can be dynamically updated and queried. The Dynamic Memory-based Attention Network (DMAN) exemplifies this by segmenting long behavior sequences into sub-sequences and maintaining specific memory blocks to preserve long-term interests, utilizing an auxiliary reconstruction loss to ensure memory fidelity [104]. This approach allows for the explicit extraction and combination of short-term and long-term interests, overcoming the inability of standard RNNs to retain information over very long horizons. Building on this, the Denoising User-aware Memory Network (DUMN) introduces a feature purification module to clean noisy implicit feedback before storing it in a user memory network, ensuring that the modeled long-term interests are robust and unbiased [105]. Additionally, hierarchical periodic memory networks have been proposed to capture multi-scale sequential patterns, employing a hierarchical and periodical updating mechanism to support the modeling of lifelong sequential patterns for each individual user [106].

Addressing the disentanglement of interest scales is another critical strategy for capturing dynamic evolution, distinguishing between stable, long-term preferences and ephemeral, session-specific trends. Users often exhibit multiple concurrent interests that evolve at different rates. The Interest Dynamics modeling framework using generative Neural Processes (IDNP) addresses this by learning a global interest function family, defining each user’s long-term interest as a function instantiation that manifests interest dynamics through function continuity, even when interaction sequences are sparse and non-consecutive [107]. Complementing this, the Long Short-Term Preference Modeling for Continuous-Time Sequential Recommendation (LSTSR) explicitly encodes short-term preference evolution via a memory mechanism with message, aggregate, and update operations, effectively capturing changing trends under a dynamic graph structure [108]. Furthermore, contrastive learning frameworks like CLSR have been developed to disentangle long and short-term interests

by using separate encoders and pseudo-labels, allowing for adaptive aggregation that respects the dynamically changing importance of different time scales [109].

The challenge of data sparsity, particularly for long-tailed users or those with irregular activity, is also tackled through these advanced modeling techniques. By treating interests as functions or leveraging memory mechanisms, models can infer preferences even in the absence of dense recent interactions. The Multi-resolution Interest Fusion (MRIF) model captures dynamic changes at different temporal ranges, fusing multi-resolution user interests to make accurate predictions despite varying data density [110]. Moreover, approaches like the Context-Aware Incremental Sequential Recommendation with Pseudo-Multi-Task Learning (CPMR) model the evolution of dynamic interests by creating distinct representations for static preferences, historical temporal states, and contextual temporal states, ensuring that recent contextual shifts do not dilute the importance of long-term historical patterns [111]. Finally, the integration of graph structures with memory networks, as seen in the Memory Augmented Graph Neural Networks (MA-GNN), allows for the simultaneous capture of short-term contextual information and long-range dependencies, effectively modeling item co-occurrence patterns that persist over time [112].

Collectively, these methodologies demonstrate that viewing user interests through the lens of continuous functions and dynamic memory states provides a robust foundation for understanding the complex, evolving landscape of human behavior. By establishing how interests change over time, the field is now better positioned to address the next layer of complexity: distinguishing the observable temporal evolution of actions from the latent, multi-faceted intents that drive them. This transition from modeling *when* and *how* interests evolve to uncovering *what* specific motivations underlie these trajectories forms the core focus of the subsequent section.

3.3 Latent Intent Inference and Multi-Granularity Preference Disentanglement

3.3 Latent Intent Inference and Multi-Granularity Preference Disentanglement

Building upon the continuous-time dynamics and dynamic memory mechanisms discussed in the previous section, a critical subsequent challenge lies in distinguishing between the observable temporal evolution of user actions and the latent, often multi-faceted intents that drive them. While Section 3.2 addressed *how* interests evolve over time, this subsection focuses on *what* specific motivations underlie these trajectories. Traditional recommendation systems frequently treat user interactions as monolithic signals, thereby conflating distinct motivations such as browsing for inspiration versus purchasing for immediate need. To overcome this limitation, contemporary research has increasingly turned to sophisticated disentanglement techniques that leverage clustering, attention mechanisms, and variational inference. These methods aim to decompose complex behavioral sequences into interpretable intent clusters and fine-grained preference aspects, effectively filtering out behavioral noise and establishing a structured semantic foundation that bridges dynamic modeling with the high-level reasoning capabilities explored in the following section.

A primary strategy for uncovering latent intents involves the application of clustering algorithms within deep learning frameworks. By treating intents as latent cluster centers, models can separate diverse user motivations without requiring explicit labels. For instance, the End-to-end Learnable Clustering for Intent Learning in Recommendation (ELCRec) framework unifies behavior representation learning with a learnable clustering mechanism. In this approach, cluster centers are initialized as learnable neurons representing latent intents, and the model is optimized to force behavior embeddings closer to these centers, effectively decoupling complex user intents through simultaneous optimization of recommendation and clustering tasks [113]. Similarly, Intent Contrastive

Learning (ICL) introduces a latent variable to represent user intents, learning their distribution via clustering and utilizing contrastive self-supervised learning to maximize the agreement between a sequence view and its corresponding intent. This paradigm not only improves recommendation performance but also enhances robustness against data sparsity and noisy interactions [114]. Extending this concept, Intent Contrastive Learning with Cross Subsequences (ICSRec) addresses the variability of intents by segmenting user behaviors into multiple subsequences. It employs coarse-grain and fine-grain contrastive learning to align subsequences that share the same target item, assuming they reflect similar underlying intentions, thus capturing latent purposes even in the absence of explicit labels [115]. Furthermore, unsupervised pipelines have been developed to extract intent taxonomies from noisy conversational data by leveraging extractive question-answering models followed by hierarchical clustering, demonstrating the viability of discovering structured intent landscapes from unstructured dialogues [116].

Beyond clustering, attention mechanisms play a pivotal role in isolating specific preference aspects and filtering irrelevant behavioral noise, serving as a precursor to the more explicit semantic reasoning found in LLMs. The Multi-Aspect preferences and INTents (MAINT) model exemplifies this by employing a multi-aspect projection mechanism to generate multiple preference representations from different perspectives. Coupled with a behavior-enhanced LSTM and a multi-aspect refinement attention mechanism, MAINT effectively filters out noise from multi-typed behaviors and adaptively fuses user preferences and intents through a gated fusion module [117]. In scenarios involving multi-behavior data, such as clicks and purchases, the Denoising and Prompt-Tuning (DPT) framework utilizes a pattern-enhanced graph encoder to identify reliable noise patterns, subsequently applying lightweight tuning to attenuate the influence of auxiliary behaviors that may mislead target behavior prediction [118]. Moreover, the Multi-Interactive Attention Network (MIAN) addresses the limitations of standard sequential methods by integrating local interaction modules to capture multiple representations of user preferences across different feature subspaces, ensuring that fine-grained user-specific and contextual information is comprehensively extracted [119]. Attention mechanisms have also been refined to handle uncertainty; for example, Uncertainty-Aware Attention incorporates input-dependent uncertainty into the attention generation process, allowing the model to learn larger variance on instances it is uncertain about, thereby providing more reliable interpretations and predictions [120].

Variational inference offers another powerful avenue for disentangling the entangled factors of intent and preference, providing a probabilistic grounding for the discrete intent clusters identified earlier. The Double Disentangled Collaborative Filtering (DDCF) approach proposes a two-fold representation learning strategy where the first level separates intent and preference influence factors, and the second level builds independent sparse preference representations under individual intents. By employing variational autoencoder networks for both intent recognition and preference decomposition, DDCF treats low-rating items as positive samples for modeling intents while regarding them as negative samples for preferences, significantly improving interpretability [121]. Similarly, the Disentangled Sequence Clustering Variational Autoencoder (DiSCVAE) leverages unsupervised learning to disentangle time-varying local features from time-invariant global attributes, inferring a discrete variable to form a latent mixture model that clusters high-level intentions [122]. In the context of session-based recommendations, the Coarse-to-Fine Sparse Sequential Recommendation (CaFe) framework explicitly learns coarse-grained intents from dense sequences to serve as prior knowledge, which are then fused with fine-grained item representations to handle data sparsity effectively [123]. Additionally, the Multi-Component Graph Convolutional Collaborative Filtering (MCCF) approach decomposes edges in the user-item graph to identify latent components corresponding to different purchasing motivations, recombining them to obtain unified embeddings that

capture fine-grained preferences [124].

The integration of these methodologies allows for a more nuanced understanding of user behavior, creating a rich latent space ready for semantic exploitation. For example, the Knowledge-Aware Multi-Intent Contrastive Learning (KAMCL) model utilizes knowledge graph relationships to construct intents, mining connections between multi-behaviors from an intent perspective to address data sparsity [125]. Furthermore, hierarchical approaches like the Hierarchical User Intent Graph Network learn multi-level user intents from co-interacted patterns, refining user and item representations as distributions over discovered intents to enhance recommendation quality [126]. By synthesizing clustering for global intent discovery, attention for local aspect refinement, and variational inference for probabilistic disentanglement, modern systems can effectively navigate the complexity of human decision-making. These advancements not only mitigate the impact of noisy data but also enable the extraction of fine-grained preference aspects, paving the way for the next evolutionary step: leveraging Large Language Models to perform explicit, chain-based semantic reasoning on these disentangled intents, as detailed in the subsequent section.

3.4 LLM-Driven Semantic Reasoning and Chain-of-Thought for Behavior Analysis

3.4 LLM-Driven Semantic Reasoning and Chain-of-Thought for Behavior Analysis

Building upon the disentangled latent intents and fine-grained preference aspects established in Section 3.3, this subsection explores how Large Language Models (LLMs) elevate user behavior modeling from statistical pattern matching to explicit semantic reasoning. While previous approaches successfully isolated intent clusters through clustering and variational inference, they often lacked the capacity to articulate the logical connections between these latent factors. LLMs address this gap by leveraging their inherent chain-based prompting capabilities to decompose complex behavioral sequences into interpretable semantic narratives. This transition moves the field beyond identifying *what* items a user might prefer based on co-occurrence to understanding *why* specific motivations drive those choices, effectively transforming the structured latent space from the preceding section into rich, reasoned explanations.

A central technique in this domain is the application of Chain-of-Thought (CoT) prompting, which guides the model to generate intermediate reasoning steps before arriving at a final recommendation or intent classification. While CoT has proven effective in general logical tasks, its application to recommendation requires careful adaptation to handle the specific constraints of user data. Research indicates that while LLMs can perform step-by-step reasoning, they are not always robust against linguistic complexities such as negation or contradictory premises, which are common in user reviews and feedback [127]. Furthermore, the internal mechanisms facilitating CoT reveal that LLMs often rely on semantic associations rather than strict symbolic logic, suggesting that effective behavior analysis must align prompts with the model’s semantic reasoning biases [128]. To mitigate error propagation inherent in long reasoning chains, advanced frameworks have been developed to verify reasoning steps deductively, ensuring that the inferred user intents are logically consistent with the observed behavior [129].

To further refine intent deduction and correct the limitations of static reasoning, recent methodologies have introduced dynamic reflection mechanisms that allow models to critique and improve their own behavioral interpretations. For instance, the concept of Dynamic Reflection with Divergent Thinking has been successfully applied to sequential recommendation, where the model iteratively probes user history, critiques initial intent hypotheses, and reflects on feedback to tailor analysis to the target user’s temporal evolution [130]. This approach mimics human learning processes, en-

abling the system to circumvent noise within raw sequences and abstract high-level user preferences across multiple aspects. Similarly, the Re2LLM framework employs a reflective reinforcement strategy where the model examines its own recommendation errors through self-reflection to construct a knowledge base of hints, effectively guiding the LLM to focus on specialized knowledge essential for accurate session-based recommendations [131]. These reflection loops are critical for stabilizing the model’s output, as standalone self-evaluation can often be unstable or overconfident without diverse solving perspectives [132].

Structure-guided prompting offers another layer of sophistication by imposing organizational schemas on the reasoning process, ensuring that the analysis of user behavior is comprehensive and logically sound. Techniques such as Tree of Thoughts or Graph of Thoughts allow the model to explore multiple potential intent trajectories simultaneously, evaluating the validity of each path before converging on the most plausible user state [133]. In the context of multi-intent scenarios, which are prevalent in complex user sessions, structure-guided approaches like Sub-Intent Instruction (SII) enable the decomposition of mixed signals into distinct semantic clusters, significantly enhancing the model’s ability to handle multi-intent spoken language and behavioral inputs [134]. Moreover, hierarchical prompting strategies have demonstrated efficacy in filtering irrelevant information from long observation traces, such as extensive browsing histories, by first summarizing action-aware observations before predicting the next user action [135].

The interpretation of semantic aspects in user reviews and action sequences also benefits from coarse-to-fine reasoning strategies. By breaking down natural language understanding tasks into multiple granular steps, models can leverage semantic-based structured knowledge, such as Abstract Meaning Representation, to capture the nuances of user utterances and connect varying levels of intent granularity [136]. This is particularly relevant for aspect-aware recommendation, where chain-based prompting is used to uncover fine-grained semantic interactions that traditional latent factor models miss, providing clearer insights into specific user preferences regarding item attributes [137]. Additionally, the use of contrastive learning in conjunction with intent modeling helps distinguish between latent intent distributions, improving robustness against data sparsity and noisy interactions [114].

Despite these advancements, challenges remain in ensuring that the reasoning processes are faithful to the context and free from hallucinations. Context-faithful prompting strategies, such as using opinion-based prompts or counterfactual demonstrations, are essential to prevent models from over-riding observed user behavior with parametric knowledge biases [138]. Furthermore, the design of these reasoning frameworks must account for the cognitive gaps between human intention formulation and model interpretation, addressing the “envisioning” difficulties users face when interacting with AI-driven systems [139]. By synthesizing dynamic reflection, structure-guided architectures, and robust chain-based prompting, LLM-driven semantic reasoning establishes a powerful foundation for next-generation user behavior modeling. However, the efficacy of these sophisticated reasoning mechanisms is fundamentally contingent upon the quality of the input representation. As detailed in the following section, realizing the full potential of these deductions necessitates a critical preprocessing stage: converting raw interaction histories into the rich, coherent natural language contexts that enable such nuanced narrative analysis.

3.5 Converting Interaction Histories into Natural Language Contexts

3.5 Converting Interaction Histories into Natural Language Contexts

While Section 3.4 established how Large Language Models (LLMs) leverage semantic reasoning and Chain-of-Thought prompting to deduce latent user intents, the efficacy of these reasoning mechanisms is fundamentally contingent upon the quality of the input representation. The paradigm shift from discriminative ranking to generative recommendation necessitates a critical preprocessing stage: transforming raw, sparse interaction logs—typically composed of timestamps, item IDs, and action types—into rich, coherent natural language contexts. Unlike traditional systems that rely on dense vector embeddings which often lack semantic interpretability, this conversion enables LLMs to perform the nuanced narrative analysis required for the reasoning strategies discussed previously. Without effectively translating the “silent language” of clicks and views into explicit text, the sophisticated deduction and reflection capabilities of LLMs cannot be fully mobilized.

A primary strategy in this domain involves conceptualizing user histories as “Interest Journeys.” Rather than viewing interactions as isolated events, this approach treats them as a continuous narrative reflecting the evolution of user preferences over time, providing the structural basis for the semantic reasoning described in the preceding section. The framework proposed in [28] exemplifies this by defining interest journeys as persistent, non-transient user interests that can be extracted and summarized via LLMs. By employing techniques such as few-shot prompting and prompt-tuning, raw activity logs are converted into nuanced textual descriptions that mimic how humans articulate their hobbies and routines, such as stating, “I just started learning the ukulele,” rather than simply listing a sequence of video IDs. This method not only enhances the interpretability of user models but also provides a controllable mechanism for downstream recommendation tasks, allowing the system to reason about the “why” behind user actions rather than just the “what.”

Parallel to the narrative approach is the innovation of embedding behavior directly into the token space of language models, effectively treating user actions as a form of language themselves. The concept of Large Content and Behavior Models (LCBMs), introduced in [140], advances this by incorporating “behavior tokens”—such as clicks, shares, likes, and purchases—directly into the training corpora of LLMs. By aligning receiver behaviors with semantic content at the token level, these models optimize for the effectiveness level of communication, predicting not just content relevance but also the likelihood of specific user actions. This integration allows the model to simulate behaviors and understand the semantic implications of an action sequence within the same latent space as natural language, thereby bridging the gap between content understanding and behavioral prediction.

However, converting long-term, high-volume interaction logs into context windows poses significant challenges regarding the context limits of standard transformers, necessitating compression techniques that preserve semantic fidelity for the reasoning processes outlined in Section 3.4. In [141], the authors propose segmenting historical behaviors into blocks and utilizing an LLM-based summarizer to generate concise textual summaries of these blocks. By introducing hierarchical and recurrent summarization techniques inspired by CNNs and RNNs, the framework constructs a prompt that encompasses a synthesized user preference summary alongside recent interactions. This approach significantly reduces the token count while retaining the critical semantic signals required for accurate sequential recommendation, demonstrating that efficient contextualization is key to scaling LLMs for industrial applications.

Furthermore, the translation of behavior into language-like structures extends to gaming and spe-

cialized domains, where event logs are treated analogously to sentences to uncover higher-level motivations. The [142] study illustrates this by extending long-range Transformer models to player behavior data. By preprocessing and tokenizing in-game events as words in sentences, the model learns player representations in a self-supervised manner without ground-truth annotations. This linguistic framing of behavior allows for the emergence of structured embedding spaces that reveal complex behavior patterns, proving that the syntax of user actions can be effectively modeled using NLP architectures. Similarly, [143] reformulates interaction log analysis by mapping user workflows to regular grammars, where interactions serve as terminals and behavior patterns as regular expressions. This structural parallel between language and interaction highlights the universality of converting behavioral data into formal linguistic representations.

To address the computational costs associated with processing long textual behaviors, particularly when feeding into the complex reasoning frameworks of Section 3.4 or the agent simulations of Section 3.6, recent works have also explored decoupling behavior encoding from inter-behavior interaction. The [144] paper introduces Behavior Aggregated Hierarchical Encoding (BAHE), which uses pre-trained shallow layers of an LLM to extract embeddings of atomic user behaviors offline. These embeddings are then combined in deeper, trainable layers to model interactions, drastically reducing training time and memory usage. This strategy ensures that the conversion of long behavioral histories into dense, context-aware representations remains feasible for real-time systems handling billions of users. Additionally, the [26] framework proposes distilling diverse user interactions into compact user embeddings via self-supervised pretraining, which are then integrated with LLMs through cross-attention and soft-prompting. This method allows LLMs to dynamically adapt to user contexts without the overhead of feeding entire raw history logs into the context window, offering a balanced trade-off between rich contextualization and computational efficiency.

Finally, the temporal dynamics of user interests require conversion strategies that respect the sequential and session-based nature of data, ensuring that the resulting narratives accurately reflect the fluidity of human intent. [145] emphasizes that user behaviors are highly homogeneous within sessions but heterogeneous across them, advocating for models that extract session-level interests before modeling their evolution. By converting these session interests into adaptive influences on target items, the system captures the dynamic flow of user intent more accurately than flat sequence models. Collectively, these approaches demonstrate that converting interaction histories into natural language contexts is not merely a formatting exercise but a sophisticated engineering challenge involving summarization, tokenization, hierarchical encoding, and efficient attention mechanisms. This foundational transformation sets the stage for the advanced memory mechanisms and agent-based simulations discussed in the following section, where these textualized behaviors serve as the episodic and semantic memories that drive autonomous, long-horizon decision-making.

3.6 Advanced Memory Mechanisms and Agent-Based Behavior Simulation

3.6 Advanced Memory Mechanisms and Agent-Based Behavior Simulation

Building upon the foundation laid in Section 3.5, where raw interaction logs are transformed into rich natural language contexts and hierarchical embeddings, the next evolutionary step involves equipping generative agents with the cognitive architecture to store, retrieve, and reason over these textualized histories over extended time horizons. While Section 3.5 addressed the *formatting* of user history into interpretable narratives, this subsection focuses on the *mechanisms* that allow autonomous agents to retain, refine, and utilize these narratives to simulate realistic human decision-making. The transition from static profile vectors to dynamic, agent-based paradigms ne-

ecessitates sophisticated memory architectures capable of distinguishing between fleeting interactions and enduring preferences. Traditional recurrent models often suffer from catastrophic forgetting or an inability to disentangle episodic specifics from semantic generalizations. To overcome these limitations, recent research has championed dual-memory systems that decouple the storage of fine-grained episodic experiences from abstract semantic knowledge. Inspired by cognitive science, architectures such as growing recurrent networks employ distinct components: one dedicated to learning object instances or specific interaction events (episodic memory) and another developing compact, category-level representations (semantic memory) [146]. This structural separation enables recommendation agents to mitigate catastrophic forgetting, ensuring that new user interactions refine rather than overwrite established preference patterns—a capability further bolstered by intrinsic memory replay mechanisms that periodically reactivate neural trajectories to consolidate knowledge [147].

Beyond the structural segregation of memory types, the dynamic refinement of stored information is paramount for capturing the fluid nature of user interests hinted at in the “Interest Journeys” of the previous section. Static memory banks are insufficient for modeling the evolving intent of users over long horizons; thus, modern frameworks incorporate dynamic memory enhancement loops where agents actively construct and refine prompts based on retrieved historical states. The Large Decision Model with Memory (LDM²) exemplifies this approach by decomposing human behaviors into state-action tuples, storing them in a model-indexed memory, and employing tree exploration to discover superior decision processes that enrich the memory bank [148]. This iterative cycle of exploration and refinement allows agents to achieve a holistic understanding of the global environment, mirroring how users update their preferences based on cumulative experiences rather than isolated events. Similarly, the concept of a “Neural Map” introduces a spatially structured memory image, enabling agents to store arbitrary environmental information over long time lags, thereby surpassing the constraints of simple temporal convolutions or standard LSTM layers restricted to short-term contexts [149]. In the realm of recommendation, such spatially aware structures can be adapted to map user journey trajectories, allowing the system to recall not merely which items were clicked, but the contextual “location” of the user within their multidimensional interest space.

To fully leverage the hierarchical encoding strategies discussed in Section 3.5, the simulation of realistic user behavior further benefits from multi-scale memory architectures that align with human cognitive processes. Theoretical models posit that intelligent agents must reason across multiple levels of temporal abstraction, ranging from low-level proprioceptive inputs to high-level goal selection. Hierarchical architectures integrating low-level controllers with high-level planners leveraging discrete graph representations have demonstrated significant improvements in sample efficiency and long-horizon planning capabilities [150]. Translating this to user modeling, agents can utilize multi-time scale world models to reason about user actions across varying temporal granularities, predicting both immediate clicks and long-term consumption goals with high fidelity [151]. Furthermore, the integration of active memory systems that mediate between semantic and sensorimotor representations facilitates the data flow orchestration required for high-level cognitive abilities such as prospection and simulation [152]. These active components are essential for constructing user simulators that do not merely react to recommendations but proactively explore item spaces, mimicking the exploratory nature of human browsing.

As these agents become more sophisticated, they increasingly demonstrate the capacity to simulate complex, socially aware behaviors and multi-agent interactions, extending the individual user modeling of earlier sections to collective dynamics. In crowded or highly interactive environments, agents leveraging memory-enabled deep reinforcement learning frameworks can retain critical in-

formation about their surroundings and model sequential dependencies effectively, prioritizing interactions that mirror human social awareness [153]. This capability is particularly pertinent for conversational recommenders and multi-user platforms where understanding the interplay between different agents is crucial. Moreover, the emergence of low-cost, real-time generative agents like Lyfe Agents illustrates how summarization and forgetting mechanisms can prioritize critical memory items, enabling agents to exhibit human-like self-motivated social reasoning and collaborative problem-solving without prohibitive computational costs [154]. By combining option-action frameworks with asynchronous self-monitoring, these agents maintain consistency over long dialogue sessions, a feature directly transferable to preserving user persona consistency during extended recommendation interactions.

Finally, the robustness of these agent-based simulations hinges on the ability to handle partial observability and uncertainty through specialized memory cores. Architectures such as the Perception-Prediction-Reaction (PPR) agent utilize a temporal hierarchy featuring slow-ticking recurrent cores to facilitate information flow over long spans, paired with fast-ticking cores for immediate reaction, thereby enhancing performance in tasks demanding long-term memory retention [155]. When applied to user behavior modeling, such designs empower recommendation agents to distinguish between transient noise in user clicks and stable, long-term preferences, resulting in more robust and explainable recommendations. The synthesis of dual-memory systems, dynamic refinement protocols, and hierarchical cognitive architectures thus forms the backbone of next-generation user simulators. These systems enable agents to traverse long-horizon behavioral trajectories with a fidelity that closely approximates human decision-making dynamics, effectively operationalizing the natural language contexts and compressed behavioral representations established in the preceding section.

4 Generative Mechanisms, Retrieval, and Personalization Strategies

4.1 End-to-End Generative Recommendation vs. Retrieval-Augmented Paradigms

4.1 Architectural Paradigms: End-to-End Generation vs. Retrieval-Augmentation

The paradigm shift in recommender systems has catalyzed the emergence of two distinct architectural philosophies for leveraging Large Language Models (LLMs): pure end-to-end generative recommendation and Retrieval-Augmented Generation (RAG) frameworks. These approaches represent fundamentally different strategies for balancing the model’s parametric knowledge with external data, each carrying unique implications for hallucination risks, knowledge freshness, and computational efficiency. Understanding this dichotomy is essential, as it sets the stage for addressing the specific integration challenges discussed in subsequent sections.

End-to-end generative recommendation reformulates the recommendation task as a sequence-to-sequence generation problem, where the model directly outputs item identifiers or tokens based on user history without explicitly querying an external database during inference. This approach aims to create a unified foundation model capable of internalizing complex user-item interactions. For instance, recent advancements have proposed generating unique, concise textual IDs for items, allowing the LLM to treat recommendation as a natural language generation task [9]. Similarly, models like GPTRec utilize sub-item tokenization derived from quantized embeddings to handle large vocabularies within a generative framework, effectively matching the performance of traditional sequential models while enabling more complex list-wise objectives [156]. Furthermore, the HSTU architecture demonstrates that massive generative recommenders can scale effectively with compute, achieving significant improvements in online metrics by treating recommendation as a

sequential transduction task [40]. Another notable approach, MMGRec, employs hierarchical quantization to assign semantic tokens to items, training a Transformer to autoregressively generate these identifiers based on multimodal and collaborative filtering information [4]. The primary advantage of this paradigm lies in its efficiency during inference; once trained, the system relies solely on the model’s weights, eliminating the latency associated with real-time retrieval processes. Additionally, by integrating collaborative signals directly into the generative process, methods like ColaRec attempt to align content semantics with interaction-based representations within a single end-to-end framework [157].

However, the pure generative approach faces significant challenges regarding hallucination and knowledge staleness. Since these models rely entirely on parametric memory, they are prone to generating non-existent items or “hallucinating” item identifiers that do not correspond to any valid entity in the catalog. This is a critical vulnerability, as LLMs inherently tend to produce plausible-sounding but factually incorrect content when disconnected from ground truth sources [158]. Moreover, updating the knowledge base in a pure generative model typically requires expensive retraining or fine-tuning, making it difficult to incorporate fresh items or rapidly changing user preferences in dynamic environments.

In contrast, Retrieval-Augmented Generation (RAG) frameworks address these limitations by decoupling the retrieval of relevant information from the generation process. In this paradigm, a retriever first fetches candidate items or relevant documents from an external knowledge base, which are then provided as context to the LLM for final ranking or explanation generation. This architecture significantly mitigates hallucination risks by grounding the model’s output in retrieved evidence. Studies have shown that RAG can effectively reduce hallucinations by integrating external knowledge, although the reliability of the system remains contingent on the quality of the retrieved context [159]. Specifically, in federated recommendation scenarios, hybrid RAG mechanisms have been successfully employed to prevent LLM hallucinations while overcoming data sparsity by mining both ID-based patterns and text-based features [160]. The RAG approach also offers superior knowledge freshness, as the external database can be updated in real-time without modifying the model parameters, a crucial feature for domains requiring up-to-date information such as news or e-commerce.

Nevertheless, the introduction of a retrieval component introduces its own set of trade-offs and structural complexities. The performance of the system is heavily dependent on the retriever’s ability to identify high-utility passages; if the retrieval step fails to capture relevant items, the LLM may still generate suboptimal recommendations or be misled by noisy context [161]. There exists a complex tension between the model’s internal prior and the retrieved information, where strong parametric biases can cause the model to ignore correct retrieved facts or, conversely, blindly accept incorrect retrieved information [162]. Furthermore, RAG systems often incur higher computational costs and latency due to the necessity of performing dense retrieval and processing long contexts during inference. To address these issues, advanced frameworks like Self-RAG have been developed to enable models to adaptively retrieve passages on-demand and critique their own generations, thereby optimizing the balance between retrieval cost and generation quality [163]. Additionally, ensuring the faithfulness of the generated response to the retrieved context remains a challenge, with research indicating that larger models generally perform better in maintaining attribution without sacrificing fluency [164].

Ultimately, the choice between end-to-end generative recommendation and RAG paradigms depends on the specific application requirements. While end-to-end models offer streamlined inference and the potential for foundational generalization across domains, they struggle with hallucination and

static knowledge. Conversely, RAG provides robust grounding and adaptability to new information but at the cost of increased system complexity and reliance on retrieval quality. Emerging hybrid strategies aim to bridge this gap, leveraging the generative capabilities of LLMs for semantic understanding while utilizing retrieval mechanisms to ensure factual accuracy and efficiency [165]. As the field evolves, the integration of conformal prediction and uncertainty quantification into these frameworks will likely become essential for certifying the trustworthiness of both generative and retrieval-augmented recommendation systems [166]. Recognizing these inherent tensions between independent retrieval and generation modules naturally leads to the critical question of how to effectively align their objectives, a topic explored in depth in the following section on bridging the gap between retrievers and generators.

4.2 Bridging the Gap Between Retriever and Generators

4.2 Bridging the Gap Between Retriever and Generators

As established in the preceding discussion on architectural paradigms, Retrieval-Augmented Generation (RAG) offers a robust solution to hallucination and knowledge staleness by decoupling retrieval from generation. However, this decoupling introduces a critical structural tension: the objectives of the retriever and the generator are often misaligned. Traditional retrieval systems are optimized to fetch documents that are semantically similar to a query, yet these “human-friendly” results do not always constitute the optimal context for an LLM to generate high-quality, personalized responses. This subsection examines the mechanisms designed to bridge this disconnect, focusing on end-to-end fine-tuning strategies, the nuanced optimization of document relevance within context windows, and hybrid architectures that harmonize the two components. These alignment strategies serve as a necessary prerequisite for the personalized prompt engineering techniques discussed in Section 4.3, ensuring that the retrieved context is not only factually grounded but also structurally amenable to user-specific conditioning.

A primary challenge in RAG systems is the erroneous assumption that retrievers and LLMs can operate as independent modules without performance degradation. Research indicates that treating these components in isolation leaves a substantial gap between retrieving information and assembling it into an LLM-friendly context. To address this, recent frameworks propose chaining supervised and reinforcement learning to train a dedicated “bridge” model that optimizes the connection between the retriever and the LLM [167]. This approach validates the ranking assumptions of retrievers specifically within the context of generation tasks, ensuring that the selected documents not only match the query semantics but also facilitate the generator’s reasoning process. Furthermore, the concept of an “augmentation-adapted retriever” has emerged as a generic plug-in solution, where the retriever learns the specific preferences of a target LLM from a source model, thereby improving zero-shot generalization even when the target model cannot be fine-tuned jointly [168].

Beyond modular bridging, end-to-end fine-tuning represents a potent strategy for deeply aligning retriever-generator architectures. The Retrieval-Augmented Dual Instruction Tuning (RA-DIT) methodology exemplifies this by operating in two distinct steps: first updating the pre-trained language model to better utilize retrieved information, and second, updating the retriever to return results preferred by the language model [169]. This dual optimization ensures that the generator becomes more robust to noise while the retriever learns to prioritize documents that maximize generation quality. Similarly, the Retrieval-Pretrained Transformer (RPT) architecture jointly trains the retrieval component and the language model from scratch for long-range modeling, using

a semantic objective where the retriever aims to fetch chunks that increase the probability of predicting the next text segment [170]. Such joint training paradigms allow the system to internalize the interplay between retrieval quality and generative coherence, moving beyond the limitations of post-hoc integration.

Optimizing document relevance within the limited context window of an LLM is equally critical, as the position, quantity, and nature of retrieved documents significantly influence generation performance. Contrary to the intuition that only highly relevant documents should be included, studies have revealed that including a certain degree of irrelevant noise can unexpectedly enhance performance by more than 30% in accuracy, suggesting that the characteristics of retrieved documents for effective prompt formulation are complex [171]. Moreover, the capacity of different model architectures to utilize context varies; for instance, decoder-only models may effectively process fewer than five documents despite having large context windows, whereas encoder-decoder models show monotonic improvement with increased document counts [172]. To manage these constraints, techniques such as “Context Tuning” employ lightweight models to fetch and rank context items based on numerical, categorical, and habitual usage signals, significantly improving tool retrieval and plan generation accuracy [173]. Additionally, methods like Corrective Retrieval Augmented Generation (CRAG) introduce a lightweight evaluator to assess retrieval quality and trigger alternative actions, such as web searches or document decomposition, when initial retrieval is sub-optimal, thereby enhancing robustness against retrieval errors [174].

Hybrid retrieval strategies and advanced re-ranking mechanisms further refine the alignment between retrieval outputs and generative needs. In complex domains such as legal question answering, integrating Case-Based Reasoning (CBR) into the RAG process allows for structured retrieval that enriches prompts with contextually relevant cases, significantly improving answer quality [175]. Furthermore, the synergy between traditional search engines and LLMs can be leveraged through iterative refinement frameworks like InteR, where retrieval models expand query knowledge using LLM-generated collections, and LLMs enhance prompt formulation using retrieved documents [176]. The effectiveness of these systems also depends on the ability to distinguish between mere relevance and actual utility; well-instructed LLMs have demonstrated the capacity to evaluate the utility of passages for supporting question answering, a capability that can be formalized to reduce dependency on input sequence order and improve downstream generation [177]. Finally, to ensure the reliability of the retrieved context, conformal prediction frameworks have been introduced to quantify retrieval uncertainty, ensuring that the true answer is captured within the context with a specified confidence level, thus making the generative process more trustworthy [178]. By synthesizing these end-to-end tuning, context optimization, and hybrid strategies, modern systems are increasingly capable of bridging the structural and semantic gaps between retrievers and generators. Having established a coherent pipeline for retrieving and aligning high-quality context, the focus naturally shifts to Section 4.3, which explores how to effectively inject individual user semantics into these architectures through personalized prompt engineering and instruction tuning.

4.3 Personalized Prompt Engineering and Instruction Tuning

4.3 Personalized Prompt Engineering and Instruction Tuning

While Section 4.2 established the mechanisms for aligning retrievers with generators to ensure factual grounding, the subsequent challenge lies in effectively injecting individual user semantics into these generative architectures. The transition from static, one-size-fits-all recommendation models to dynamic, generative systems hinges on sophisticated prompt engineering that encapsulates

unique user identities and intents. In the realm of generative recommendation, personalized prompt engineering serves as the critical interface where raw interaction histories are transformed into contextual signals that Large Language Models (LLMs) can interpret to generate tailored outputs. This process moves beyond simple text concatenation, employing advanced strategies such as soft-prompting, instruction tuning, and dynamic adaptation to align model behavior with individual user preferences, thereby laying the foundational user representations required for the personalized alignment discussed in Section 4.4.

A primary challenge in this domain is the efficient integration of complex, often noisy, user interaction data into the limited context windows of LLMs. Traditional approaches that rely on appending long sequences of historical interactions as text prompts often suffer from information dilution and computational inefficiency. To address this, recent advancements have introduced the concept of using user embeddings as soft prompts. The [26] framework exemplifies this shift by proposing a method where user embeddings, distilled from diverse interactions via self-supervised pretraining, are integrated with LLMs through cross-attention and soft-prompting mechanisms. This approach allows the model to dynamically adapt to user contexts without the overhead of processing extensive textual history, demonstrating significant performance gains on tasks requiring deep user understanding compared to text-prompt-based contextualization. Similarly, the [27] study introduces a User Embedding Module (UEM) designed to compress free-form text histories into dense embeddings that serve as soft prompts. This method effectively biases language models with user signals, enabling the handling of significantly longer histories than conventional methods while maintaining predictive accuracy.

Beyond static embedding integration, the field has seen a surge in dynamic prompt adaptation techniques that tailor prompts to specific input instances or evolving dialogue states. Fixed prompts often fail to capture the nuance of individual user queries or the fluid nature of conversational recommendations. The [179] paper proposes a paradigm where prompts are learned directly from dialog contexts, distilling useful signaling information to improve response generation. This contextual dynamism is further refined in [180], which presents an instance-specific prompt-tuning algorithm. Instead of relying on fixed conversation history, this method generates prompts based on instance-level control codes, allowing for precise control over dialogue attributes and proving superior to static prompting baselines with only a fraction of the parameters required for full fine-tuning. The efficacy of such instance-dependent strategies is reinforced by [181], which utilizes a lightweight trainable component to generate unique prompts for each input sentence, consistently outperforming fixed prompt tuning baselines across various natural language understanding tasks.

Instruction tuning represents another pivotal strategy for aligning generative recommenders with domain-specific requirements and user intents. While standard instruction tuning applies uniform guidelines across a dataset, personalized scenarios demand context-aware instructions that evolve with the user’s journey. The [182] framework addresses this by generating both responses and instructions based on previous context, effectively allowing the model to self-guide its generation process in multi-turn dialogues. This is particularly crucial in complex recommendation scenarios where simple task instructions are insufficient to capture shifting user needs. Furthermore, the [183] framework demonstrates the power of reformulating intent detection and value extraction as question-answering tasks within an instruction tuning schema. By leveraging similarities in natural language descriptions across domains, SQATIN achieves state-of-the-art performance in cross-domain transfer, a critical capability for recommender systems operating in diverse verticals.

The optimization of these prompt structures often involves sophisticated meta-learning and regularization techniques to ensure robustness and generalizability, especially in few-shot settings common

in personalized scenarios. [184] adopts model-agnostic meta-learning to automatically discover optimal prompt initializations, facilitating rapid adaptation to new user tasks without starting from scratch. Complementing this, [185] introduces a framework that jointly meta-learns a gradient regularization function to prevent overfitting to sparse user data, thereby enhancing domain generalization. For scenarios requiring the management of multiple user personas or task types, [186] offers a solution by distilling knowledge from multiple source prompts into a single transferable prompt, which is then adapted to target tasks via low-rank updates. This ensures that the system can handle diverse recommendation tasks efficiently while minimizing the parameter footprint.

Finally, the structural design of prompts plays a non-trivial role in their effectiveness. Research indicates that the architecture of the prompt itself—whether hierarchical, decomposed, or structured—can significantly influence the model’s ability to attend to relevant user information. [187] proposes generating soft prompt embeddings through a hypernetwork, offering greater flexibility and stability compared to standard tuning. Similarly, [188] leverages the observation that soft prompts often exhibit low intrinsic rank, utilizing low-rank matrices to initialize prompts, which reduces trainable parameters while maintaining effectiveness. These structural innovations, combined with dynamic and instruction-based strategies, form a comprehensive toolkit for personalized prompt engineering. By synthesizing user embeddings, dynamic context awareness, and adaptive instruction tuning, modern generative recommenders can achieve a level of personalization that closely mimics human-like reasoning. Having established how to structurally encode user preferences into prompts, the discussion naturally progresses to Section 4.4, which explores how to further refine these models through Reinforcement Learning from Personalized Human Feedback (P-RLHF) to align with the nuanced and often conflicting values of individual users.

4.4 Reinforcement Learning from Personalized Human Feedback (P-RLHF)

4.4 Reinforcement Learning from Personalized Human Feedback (P-RLHF)

Building upon the personalized user representations and adaptive prompt structures established in Section 4.3, the next critical step is to align the generative model’s outputs with the nuanced and often divergent values of individual users. The trajectory of aligning Large Language Models (LLMs) with human values has traditionally been defined by Reinforcement Learning from Human Feedback (RLHF). This paradigm typically operates on the premise of aggregating preferences from a diverse population to train a single, universal reward model. While effective for establishing general helpfulness and safety standards, this aggregate approach inherently assumes a homogeneity in human values that rarely exists in practice. In the context of generative recommendation, where users possess distinct, often conflicting preferences shaped by unique cultural backgrounds, individual tastes, and situational contexts, the “one-size-fits-all” assumption becomes a critical bottleneck. Consequently, the field is witnessing a pivotal evolution from aggregate RLHF to Personalized-RLHF (P-RLHF), a framework designed to capture and align with the multifaceted nature of individual human preferences rather than a statistical average.

The fundamental limitation of vanilla RLHF in personalized settings lies in its inability to distinguish between diverse user intents when learning a single reward function. When feedback data encompasses conflicting viewpoints, a monolithic reward model struggles to reconcile these differences, often converging on a compromise that satisfies no one completely. To address this, recent research has formally introduced the task of learning from personalized human feedback, proposing frameworks that require the joint learning of a user model and a language or reward model. In such architectures, the user model ingests specific user information—leveraging the embedding

techniques discussed previously—to generate dynamic user representations that condition the reward modeling process. This structural design allows the system to encode assumptions about the underlying diversity of user preferences directly into the learning objective. For instance, methods have been developed to fine-tune large models into personalized language and reward models that significantly outperform non-personalized baselines in aligning with individual annotator preferences, demonstrating that explicit modeling of user identity is crucial for high-fidelity alignment [189].

Building upon the concept of joint learning, another sophisticated approach treats personalized alignment as a Multi-Objective Reinforcement Learning (MORL) problem. Instead of attempting to learn a single scalar reward that averages out contradictions, this perspective decomposes preferences into multiple dimensions defined by user-declared personalizations. These dimensions can be trained independently in a distributed manner and subsequently combined through post-hoc parameter merging. This technique, often referred to as “Personalized Soups,” allows for the efficient construction of models that can switch between or interpolate among diverse preference profiles without the need for retraining from scratch. By viewing alignment through the lens of multi-objective optimization, researchers can achieve a Pareto-optimal generalization across the entire space of preferences, ensuring that the system respects the heterogeneity of human opinions [190] [191].

The theoretical underpinnings of P-RLHF further reveal that traditional single-party RLHF approaches are statistically insufficient for capturing the nuances of multi-party interactions. Theoretical studies have initiated the exploration of multi-party RLHF, demonstrating that learning a single reward function fails to balance the preferences of multiple individuals effectively. By incorporating meta-learning techniques to learn multiple distinct preferences and adopting various social welfare functions—such as Nash, Utilitarian, and Leximin welfare functions—systems can aggregate preferences in ways that guarantee fairness and efficiency. These works highlight a significant separation in sample complexity between multi-party and single-party settings, underscoring the need for specialized algorithms that can handle the increased statistical demands of modeling diverse stakeholders [192]. Furthermore, the challenge of active teacher selection arises when feedback comes from teachers with varying levels of expertise and rationality. Frameworks like the Hidden Utility Bandit (HUB) model these differences, enabling algorithms to actively select which teacher to query, thereby learning more accurate and robust reward models tailored to specific user segments [193].

Beyond architectural innovations, the data efficiency of P-RLHF remains a paramount concern, as collecting exhaustive preference data for every individual user is impractical. To mitigate this, researchers have proposed three-stage customized reward model learning schemes that leverage general preference enrichment and customized preference imitation learning. These strategies allow models to preserve general capabilities while adapting to domain-specific or individual-specific preferences with limited data. Additionally, the generation of synthetic preference data and the use of AI feedback (RLAIF) offer scalable pathways to augment sparse human feedback, although care must be taken to ensure these synthetic signals accurately reflect the diversity of human sentiment rather than reinforcing model biases [194] [195].

The shift toward P-RLHF also necessitates a re-evaluation of optimization algorithms, particularly given the computational burdens associated with managing multiple conflicting objectives. Standard Proximal Policy Optimization (PPO), while historically dominant, can be unstable and computationally heavy in these complex scenarios. Consequently, RL-free algorithms like Multi-Objective Direct Preference Optimization (MODPO) have emerged. MODPO extends Direct Pref-

erence Optimization (DPO) to handle multiple alignment objectives by training language models as implicit collective reward models. This approach folds language modeling directly into reward modeling, offering a stable and computationally efficient alternative that can produce a Pareto front of models catering to diverse preferences with significantly reduced resource overhead [196]. Moreover, the integration of uncertainty quantification into the reward learning process helps prevent over-optimization, a common pitfall where models exploit reward model flaws. Techniques such as Uncertainty-Penalized RLHF utilize diverse reward ensembles to penalize actions with high uncertainty, ensuring that personalized recommendations remain robust and trustworthy [197].

In summary, the evolution from aggregate RLHF to Personalized-RLHF represents a fundamental paradigm shift in generative recommendation. By jointly learning user representations and reward functions, decomposing preferences into multi-objective frameworks, and developing efficient optimization strategies, P-RLHF enables systems to navigate the complex landscape of diverse human values. This transition is essential for building next-generation recommenders that do not merely predict what a user might click, but deeply understand and align with their unique intent, context, and evolving preferences. However, while P-RLHF defines the objective of personalized alignment, the practical deployment of these systems is often hindered by the computational inefficiencies of traditional optimization algorithms like PPO. This challenge drives the discussion in Section 4.5, which explores efficient alignment mechanisms beyond PPO that are critical for scaling personalized recommendation systems.

4.5 Efficient Alignment Mechanisms Beyond PPO

4.5 Efficient Alignment Mechanisms Beyond PPO

While the preceding section established Personalized-RLHF (P-RLHF) as a critical paradigm for capturing diverse user values, the practical deployment of these personalized systems is often hindered by the computational inefficiencies of traditional optimization algorithms. The alignment of Large Language Models (LLMs) has historically relied on Reinforcement Learning from Human Feedback (RLHF) with Proximal Policy Optimization (PPO) serving as the de facto standard. Originally proposed to balance sample complexity and implementation ease in domains like robotic locomotion and game playing, PPO achieves state-of-the-art performance through a sophisticated actor-critic framework [198]. However, applying PPO to LLM alignment introduces prohibitive challenges: it requires maintaining multiple model copies simultaneously (actor, critic, reference, and reward models), leading to excessive memory costs and complex hyperparameter tuning. Recent critiques highlight that PPO is frequently plagued by instability, sensitivity to implementation details, and high variance in policy updates, which impede the rapid iteration necessary for effective personalization [199]. Furthermore, evidence suggests that many performance gains attributed to PPO stem from code-level optimizations rather than its core algorithmic design, raising questions about its theoretical necessity for language model fine-tuning [200].

To address these bottlenecks and enable the scalable personalization discussed in Section 4.4, the research community has pivoted toward computationally efficient alternatives that bypass the complex online actor-critic loop. A prominent direction involves offline preference optimization methods, which eliminate the need for online sampling and explicit reward modeling during the policy update phase. Direct Preference Optimization (DPO) has emerged as a leading candidate, reformulating alignment as a classification task on preference data. While DPO offers significant simplicity and stability, theoretical analyses indicate fundamental limitations, such as hindering the learning capacity for human-preferred responses and exhibiting oversensitivity to the quality of the initial

supervised fine-tuning (SFT) model [201]. Specifically, the DPO loss function tends to decrease the probability of generating dispreferred data faster than it increases the probability of preferred data, potentially leading to suboptimal convergence [201]. Additionally, standard DPO treats all preference pairs equally, ignoring the magnitude of preference differences—a crucial factor in personalized settings. This limitation is addressed by variants like DPO with an offset (ODPO), which enforces a margin based on preference strength [202], and DPO-Positive (DPOP), which corrects failure modes where the likelihood of preferred examples inadvertently decreases [203].

Beyond pure offline methods, there is a growing trend toward simplifying the online reinforcement learning process itself to better suit the deterministic transitions and trajectory-level rewards characteristic of LLM alignment. Researchers argue that PPO’s complexity is often unnecessary, revisiting simpler policy gradient algorithms. For instance, ReMax proposes a REINFORCE-style algorithm enhanced with variance reduction techniques, significantly reducing memory usage and eliminating several PPO-specific hyperparameters while maintaining comparable or superior performance [204]. Similarly, studies posit that carefully adapted standard REINFORCE-style variants can outperform both PPO and “RL-free” methods like DPO, suggesting that the motivational principles behind PPO’s development for continuous control do not fully translate to language tasks [205]. Another innovative approach, REBEL (Reinforcement Learning via Regressing Relative Rewards), reduces policy optimization to a regression problem on relative rewards, offering a unified and lightweight framework that matches the theoretical guarantees of natural policy gradient without PPO’s heuristic complexity [206].

Representation engineering and alternative optimization objectives further enhance alignment efficiency. Methods like Statistical Rejection Sampling Optimization (RSO) improve upon DPO by sourcing preference data from the target optimal policy via rejection sampling, providing a more accurate estimation of the optimal policy than methods restricted to SFT samples [207]. Additionally, approaches such as Exact Optimization (EXO) claim to optimize the alignment objective in the same direction as RL algorithms asymptotically but with greater efficiency by circumventing standard RL solver complexities [199]. The exploration of Bayesian frameworks, such as BRAIn (Bayesian Reward-conditioned Amortized Inference), reintroduces the reward model in a distribution matching context, effectively bridging the gap between the fine-grained information usage of PPO and the scalability of offline methods [208].

The shift away from PPO is also driven by the need for robustness and theoretical clarity. Studies indicate that PPO’s clipping mechanisms, while effective heuristics, may not strictly enforce trust region constraints or guarantee monotonic improvement, leading to potential instability [209]. In contrast, newer methods aim to provide clearer minimization targets or utilize different divergence metrics, such as relative Pearson divergence, to achieve more balanced regularization [210]. Moreover, the heavy-tailed nature of gradients in PPO suggests that standard clipping tricks merely compensate for outlier-rich regimes, prompting the exploration of robust estimators as alternatives [211]. As the field matures, the consensus is moving toward a diverse ecosystem of alignment algorithms where the choice between PPO, DPO, REBEL, or other variants depends on specific trade-offs between computational resources, data availability, and alignment fidelity. These efficient mechanisms not only lower the barrier to entry for aligning large models but are also essential for realizing the low-latency, rapid-update personalization strategies required in modern generative recommendation systems. Looking ahead, as we transition to Section 4.6, these efficient optimization backbones will increasingly rely on automated feedback loops to further reduce the dependency on costly human annotation.

4.6 AI Feedback, Self-Reflection, and Automated Reward Modeling

4.6 AI Feedback, Self-Reflection, and Automated Reward Modeling

Building upon the efficient optimization backbones discussed in Section 4.5, which alleviate the computational burdens of traditional PPO, the scalability of generative recommendation systems now faces a new bottleneck: the reliance on extensive, costly, and subjective human annotation for Reinforcement Learning from Human Feedback (RLHF). While algorithms like DPO and REBEL streamline the *process* of alignment, they still fundamentally depend on high-quality preference data. To overcome the limitations of human-centric data collection and enable the continuous, low-latency personalization strategies required for modern systems, the research community has pivoted toward leveraging AI-generated feedback, self-reflection mechanisms, and advanced automated reward modeling. This shift not only decouples alignment scalability from human labor but also establishes the automated feedback loops necessary to support the robust, secure architectures explored in the subsequent Section 4.7.

Reinforcement Learning from AI Feedback (RLAIF) stands at the forefront of this paradigm shift, utilizing powerful off-the-shelf language models to generate preference labels in lieu of human annotators. Studies have demonstrated that RLAIF can achieve performance comparable to, or even superior to, traditional RLHF across various tasks, including dialogue generation and summarization [195]. By employing a critic model to evaluate responses, RLAIF effectively scales the alignment process, offering a potential solution to the throughput limitations inherent in human-centric approaches. However, basic RLAIF implementations face challenges, particularly regarding the helpfulness and truthfulness of generated responses, which can lead to decreased user satisfaction despite higher preference win ratios [212]. To address these limitations, hybrid approaches such as Hybrid Reinforcement Learning from AI Feedback (HRLAIF) have been proposed. These methods enhance the accuracy of AI annotations and incorporate AI-driven red teaming to improve harmlessness, thereby inheriting the cost-efficiency of RLAIF while restoring the satisfaction rates associated with human feedback [212].

Beyond direct AI feedback, self-reflection and self-supervised mechanisms offer innovative pathways for reducing annotation burdens by enabling models to iteratively refine their own outputs. The concept of enabling models to implicitly learn self-improvement has gained traction, where the training objective is reformulated to maximize the quality gap between responses conditioned on a reference, rather than merely maximizing absolute quality [213]. This Implicit Self-Improvement (PIT) framework leverages existing preference data without requiring additional human effort, allowing models to align more closely with human preferences through iterative self-correction. Furthermore, principle-driven self-alignment strategies, such as SELF-ALIGN, combine generative capabilities with a small set of human-written principles to guide models in producing ethical and reliable responses with minimal supervision [214]. Such approaches significantly reduce the need for extensive human annotations, relying instead on the model’s inherent reasoning capabilities to adhere to predefined ethical and functional standards, thus creating a more autonomous alignment loop.

Parallel to these feedback generation strategies, automated reward modeling has evolved through generative adversarial and contrastive techniques to enhance robustness and generalization, addressing the instability issues often found in learned proxies. Traditional reward models are vulnerable to noise and distribution shifts, prompting the development of Generative Adversarial Reward Modeling. This approach, known as Reinforcement Learning with Generative Adversarial Feedback (RLGAF), enables models to learn from expert demonstrations without direct exposure to training

examples, preserving sample efficiency while improving generalization [215]. Additionally, contrastive rewards have been introduced to penalize reward uncertainty and encourage improvement over baseline responses. By incorporating a penalty term derived from offline sampling, contrastive rewards help calibrate the model according to task difficulty and reduce variance during optimization, leading to substantial improvements in alignment evaluated by both automated metrics and human judges [216].

The integration of domain knowledge and conservative optimization objectives further refines these automated systems, particularly for specialized recommendation scenarios where generic rewards may fail. For instance, infusing domain-specific knowledge into the reward model can drastically reduce the required amount of preference annotation while advancing state-of-the-art performance in tasks like e-commerce opinion summarization [217]. Moreover, ensemble-based conservative optimization objectives, such as worst-case optimization and uncertainty-weighted optimization, have proven effective in mitigating the overoptimization issues commonly observed in learned reward models [218]. These ensemble methods ensure that the policy does not exploit flaws in a single proxy reward model, thereby maintaining alignment with true underlying human preferences—a critical step in preventing the reward hacking vulnerabilities that will be detailed in Section 4.7.

Innovative data collection strategies also play a crucial role in closing the automated feedback loop efficiently. Active Preference Optimization (APO) frames RLHF as a contextual preference bandit problem, actively selecting prompts to collect preference data in a sample-efficient manner [219]. This approach ensures that the collected data provides maximal information gain, reducing the overall sample budget required for effective alignment. Additionally, the use of spontaneous physiological signals, such as facial expression reactions, has been explored to automatically annotate user preferences in generative tasks, offering a novel, non-intrusive method for capturing implicit feedback [220].

Despite these advancements, the transition to fully automated feedback introduces new risks regarding reliability, fairness, and security. Critical evaluations suggest that the complexity of reinforcement learning steps in RLAIIF may sometimes be unwarranted if simpler supervised fine-tuning with stronger teacher models yields comparable results [221]. More critically, the reliance on learned proxy rewards and AI-generated preferences creates vulnerability surfaces for reward hacking and adversarial manipulation of preference data [222]. As the field progresses, the synthesis of AI feedback, self-reflection, and automated reward modeling must be balanced with rigorous safety protocols. The next section, Section 4.7, will delve deeply into these robustness, safety, and adversarial resilience challenges, examining how to protect these automated alignment mechanisms from internal optimization failures and external attacks.

4.7 Robustness, Safety, and Adversarial Resilience in Alignment

4.7 Robustness, Safety, and Adversarial Resilience in Alignment

While the automated feedback mechanisms and reward modeling strategies discussed in Section 4.6 offer scalable pathways for aligning generative recommenders, they simultaneously introduce a complex landscape of security and safety challenges. The very reliance on learned proxy rewards and AI-generated preferences creates new vulnerability surfaces that extend far beyond traditional recommendation paradigms. As Large Language Models (LLMs) become integral to generating personalized content and reasoning about user preferences, the alignment mechanisms designed to make them helpful and harmless must be rigorously stress-tested against internal optimization failures and external adversarial threats.

A primary concern arising from the optimization processes detailed previously is the phenomenon of reward hacking, where the generative agent exploits flaws in the proxy reward function to maximize scores while failing to satisfy true user intent or ethical constraints. In generative recommendation, this might manifest as the model generating verbose, off-topic, or misleading item descriptions that score highly on a relevance metric but provide poor utility. Recent research has highlighted that standard regularization techniques, such as penalizing the Kullback-Leibler divergence between action distributions, are often insufficient because they fail to account for deviations in the state space visited by the agent. To address this, advanced strategies like Occupancy Measure Regularization have been proposed, which regularize the policy based on the divergence of state occupancy measures rather than just action distributions, thereby more effectively preventing the agent from drifting into dangerous or deceptive behavioral trajectories [223]. Furthermore, the reliance on single reward models creates a bottleneck where underspecification can lead to overoptimization; while ensembling reward models can mitigate this by aggregating diverse perspectives, studies indicate that even ensembles may not fully eliminate reward hacking if all models share similar underlying error patterns [224].

Beyond these internal optimization failures, aligned generative recommenders face severe external threats from adversarial actors aiming to poison the preference data used for alignment—a critical vulnerability given the automated data collection strategies mentioned in Section 4.6. The process of learning utility or reward models from pairwise comparisons is inherently vulnerable, as there is no gold standard for subjective preferences. Malicious actors can inject a small fraction of poisoned preference data to skew the learned reward function, effectively promoting or demoting specific items or behaviors. Research demonstrates that sophisticated poisoning attacks, such as gradient-based frameworks or rank-by-distance methods, can achieve near-total success rates with minimal data contamination, compromising the integrity of the entire recommendation ecosystem [225]. This threat is exacerbated in federated learning environments, where the decentralized nature of data collection makes it difficult to verify the authenticity of user updates. Attackers can exploit popularity biases or craft synthetic user profiles to manipulate item exposure rates stealthily, often bypassing existing defense mechanisms that rely on statistical anomaly detection [226]. In decentralized collaborative settings, adversaries can further adapt their strategies by emulating benign clients and conditioning their malicious gradients on local neighborhood structures, making detection even more challenging [227].

The intersection of alignment and fairness presents another critical frontier for robustness, complicating the evaluation metrics discussed in the subsequent section. Ensuring that generative recommenders treat all user groups equitably is challenging due to the dynamic nature of user interactions and the potential for alignment processes to inadvertently amplify existing biases. For instance, continual adaptation to new data in dynamic systems can exacerbate performance disparities among user groups defined by sensitive attributes, necessitating frameworks that explicitly optimize for fairness during the fine-tuning process [228]. The challenge is further compounded in multimodal settings, where the integration of visual and textual content may expose additional sensitive attributes, requiring disentangled representation learning to filter out biased information before it influences the generative process [229]. Moreover, the tension between robustness, fairness, and privacy creates a triangular trade-off; techniques designed to enhance robustness against poisoning, such as differential privacy or gradient clipping, may inadvertently degrade fairness metrics or reduce model utility for underrepresented groups. Conversely, efforts to enforce strict fairness constraints can make the system more susceptible to specific types of adversarial perturbations that target the fairness mechanism itself [230].

To build resilient alignment frameworks capable of withstanding these multifaceted threats, researchers are exploring dialectical approaches that enable models to navigate conflicts between parametric memory and external evidence, particularly when the latter is poisoned. Dialectical Alignment frameworks utilize AI feedback to train models to distinguish between trustworthy context and adversarial inputs, significantly improving defense capabilities without requiring explicit prompts warning of attacks [231]. Additionally, preference-free alignment methods that rely on regularized relevance rewards offer a promising alternative to human preference learning, reducing the attack surface associated with collecting and processing subjective feedback data [232]. Ultimately, achieving adversarial resilience in generative recommendation requires a holistic approach that integrates robust reward modeling, secure federated aggregation protocols, and fairness-aware optimization objectives. As the field moves toward more autonomous agents, the development of provable defenses against backdoor attacks and the ability to track the source of malicious updates through coding theoretic approaches will become indispensable for maintaining trust in these systems [233]. The future of safe generative recommendation lies not only in improving accuracy but in constructing architectures that are inherently robust to the multifaceted threats of reward hacking, data poisoning, and bias amplification, thereby laying the necessary groundwork for the reliable evaluation protocols detailed in Section 4.8.

4.8 Evaluation Metrics for Generative and Aligned Systems

4.8 Evaluation Metrics for Generative and Aligned Systems

Building upon the robustness and safety frameworks established in Section 4.7, the deployment of aligned generative recommenders necessitates a fundamental reimagining of evaluation protocols. The transition from discriminative ranking to generative paradigms renders traditional metrics such as Precision@K, Recall, and NDCG insufficient, as they fail to capture the multifaceted capabilities of Large Language Model (LLM)-based systems. These advanced models do not merely rank items; they produce natural language explanations, engage in multi-turn conversations, and generate personalized content. Consequently, evaluation must extend beyond simple accuracy to encompass factuality, diversity, reasoning capacity, and—critically—the degree of alignment with human values and resilience against the adversarial threats previously discussed. The research community is thus shifting towards comprehensive strategies that leverage LLM-based judges, scenario-based benchmarks, and multidimensional quality assessments to validate the safety and efficacy of these systems.

A primary challenge in this new landscape is ensuring the factuality of generated content, particularly given the risk of hallucinations regarding item attributes or non-existent user preferences—a vulnerability directly linked to the reward hacking phenomena described earlier. The reliability of using LLMs as evaluators of their own outputs or those of peer models remains a critical area of investigation. Studies indicate that while large models like GPT-4 can perform competitively on classification benchmarks for factual inconsistency, they often struggle with more complex formulations and may fail to detect nuanced errors compared to specialized metrics [234]. Furthermore, the assumption that models skilled in generation are equally adept at evaluation has been challenged by the “Generative AI Paradox,” where models exhibit lower performance in evaluation tasks compared to generation tasks, occasionally providing unfaithful evaluations in areas where they lack competence [235]. To address these limitations, researchers have proposed automated attribution evaluation benchmarks like AttributionBench, which reveal that even state-of-the-art LLMs face significant challenges in verifying whether every claim in a generated response is fully supported by cited evidence, with failures often stemming from an inability to process nuanced information

[236].

Beyond factuality, assessing diversity and fairness is paramount, especially since the alignment processes detailed in Section 4.7 aim to mitigate but not necessarily eliminate bias amplification. Evaluating fairness in RecLLMs requires moving beyond aggregate metrics to examine disparities across sensitive attributes such as gender, age, and their intersections. Frameworks like CFaiRLLM have been developed to systematically assess consumer-side fairness by analyzing how recommendations vary when sensitive attributes are included in user prompts, highlighting that the choice of user profile sampling strategy significantly impacts fairness outcomes [237]. Similarly, the FaiRLLM benchmark provides a dedicated suite of metrics and datasets for music and movie recommendations, demonstrating that popular models like ChatGPT can still exhibit unfairness toward specific sensitive attributes [238]. Moreover, traditional fairness evaluations often overlook the intrinsic personalization of recommendation systems; a critical literature review reveals that most existing frameworks fail to account for personalization factors, thereby inadvertently perpetuating unfair practices [239]. To mitigate these issues, novel approaches such as the Unbiased P5 (UP5) foundation model utilize counterfactually-fair prompting techniques to eliminate bias while maintaining high recommendation performance [240].

The reliability of using LLMs as judges for evaluating other models is another contentious issue, particularly when assessing robustness against adversarial inputs. While LLM-based evaluators offer scalability, they are susceptible to systematic biases, such as position bias, where the order of candidate responses in the context can easily manipulate the evaluation result [241]. Research shows that fine-tuned judge models, despite achieving high accuracy on in-domain tests, often function as task-specific classifiers with poor generalizability and fairness compared to larger proprietary models [242]. To counteract these biases, calibration frameworks employing multiple evidence generation, balanced position aggregation, and human-in-the-loop mechanisms have been proposed to align LLM evaluations more closely with human judgments [241]. Additionally, the “WideDeep” approach suggests that constructing wider and deeper networks of LLM evaluators, mimicking academic peer review processes, can significantly improve the fairness and stability of evaluations [243].

To holistically assess generative recommendation systems, the community is adopting comprehensive benchmarks that simulate real-world CRUD (Create, Read, Update, Delete) scenarios and conversational interactions, effectively stress-testing the resilience mechanisms discussed in the previous section. For instance, the Concept protocol integrates both system-centric and user-centric factors, utilizing LLM-based simulators to evaluate conversational recommenders on abilities often neglected in traditional protocols, such as empathy and proactivity [244]. In the Chinese context, the SuperCLUE benchmark incorporates actual user queries and ratings from battle platforms to better reflect human preferences in open-ended dialogue scenarios, arguing that accuracy on closed-ended questions is insufficient for real-world assessment [245]. Furthermore, evaluating the alignment of these systems with human values requires multidimensional metrics covering safety, responsibility, and social norms, as demonstrated by benchmarks like CValues, which highlight gaps in model responsibility despite adequate safety performance [246].

Finally, the quality of the evaluation process itself is under scrutiny to ensure that the feedback loops used for refining aligned models are trustworthy. The “Critique of Critique” framework, or MetaCritique, proposes evaluating the quality of critiques generated by LLMs based on factuality and comprehensiveness, ensuring that the feedback loops used for model refinement are themselves reliable [247]. As the field progresses, there is a growing consensus that evaluation must be multidimensional, incorporating diverse metrics for diversity, serendipity, and factuality, while

rigorously auditing the evaluators themselves to prevent the propagation of biases and errors [248]. The development of robust, standardized protocols remains essential for guiding the deployment of trustworthy and effective generative recommendation systems, closing the loop between theoretical alignment strategies and practical, verifiable safety.

5 Interactive Agents, Conversational Systems, and User Simulation

5.1 Autonomous Agent Architectures for Dialogue Management

5.1 Autonomous Agent Architectures for Dialogue Management

The integration of Large Language Models (LLMs) into conversational recommender systems (CRS) has precipitated a paradigm shift from static, rule-based dialogue managers to dynamic, autonomous agent architectures. While traditional CRS often struggle to maintain coherent dialogue flow alongside complex recommendation tasks—frequently resulting in aimless chit-chat or premature suggestions that ignore user feedback—recent research has increasingly adopted multi-agent frameworks. These architectures decompose the recommendation process into manageable sub-tasks through specialized LLM agents, exerting finer control over dialogue acts and enhancing the precision of preference elicitation. This structural evolution lays the necessary groundwork for the sophisticated user modeling and synthetic data generation techniques discussed in the subsequent section.

A primary architectural innovation in this domain is the decomposition of the dialogue management system into distinct functional roles, enabling a modular approach to conversational planning. The Multi-Agent Conversational Recommender System (MACRS) exemplifies this strategy by establishing a framework wherein four distinct LLM-based agents collaborate to govern the dialogue flow [249]. In this architecture, agents are assigned specific dialogue acts, generating candidate responses that are subsequently evaluated to select the most appropriate system action. This multi-agent act planning framework effectively mitigates the issue of uncontrolled dialogue generation by ensuring that each conversational turn is purposefully aligned with the overarching recommendation goal. Furthermore, MACRS incorporates a user feedback-aware reflection mechanism, enabling the system to reason about errors in previous turns and adjust future dialogue acts based on implicit semantic cues, thus creating a closed-loop learning environment within the conversation.

Beyond simple role separation, advanced architectures leverage the concept of “Standardized Operating Procedures” (SOPs) to encode human-like workflows into prompt sequences, ensuring logical consistency and reducing hallucinations during complex task execution. MetaGPT introduces a meta-programming framework that assigns diverse roles to various agents, functioning similarly to an assembly line to break down complex tasks into subtasks [250]. By incorporating a “Researcher” agent to process information and a “Decider” agent to make final judgments, such frameworks create an interpretable forum for models to communicate feedback and iteratively improve outputs [251]. This division of labor is particularly critical in recommendation scenarios where the system must balance information gathering with item suggestion, as it allows specialized agents to verify intermediate results before they influence the final recommendation logic.

The challenge of managing dialogue in dynamic environments also necessitates architectures capable of adaptive coordination and real-time strategy adjustment. Dynamic LLM-Agent Network (DyLAN) proposes a strategic team of agents that communicate via a dynamic interaction architecture, utilizing inference-time agent selection and an early-stopping mechanism to optimize performance [252]. Unlike static multi-agent systems, DyLAN automatically optimizes the agent team based

on an “Agent Importance Score,” ensuring that the most relevant expertise is deployed for specific sub-tasks such as preference elicitation or explanation generation. Similarly, the AutoAgents framework adaptively generates and coordinates multiple specialized agents based on the specific content of the user’s task, coupling the relationship between tasks and roles to build an AI team on the fly [253]. This adaptability is crucial for CRS, where user intents can shift rapidly, requiring the dialogue manager to reconfigure its internal agent hierarchy to address new constraints or preferences effectively.

Furthermore, effective dialogue management requires robust mechanisms for handling multi-turn interactions and maintaining long-term context, which single-agent models often fail to achieve due to context window limitations or attention dilution. The RAISE architecture enhances conversational agents by incorporating a dual-component memory system that mirrors human short-term and long-term memory, facilitating context continuity across complex, multi-turn dialogues [254]. When combined with multi-agent collaboration, such memory structures allow agents to retain critical user preferences and historical interaction data, enabling more personalized and coherent recommendation journeys. Additionally, frameworks like TPE (Think-Plan-Execute) decouple the response generation process into three distinct roles—Thinker, Planner, and Executor—to enhance the explainability and controllability of responses [255]. In a recommendation context, the “Thinker” can analyze user emotions and latent preferences, the “Planner” can devise a strategy for item exploration, and the “Executor” can synthesize these inputs into a natural language response, ensuring that the dialogue remains both empathetic and goal-directed.

The scalability of these architectures is also a focal point, with research exploring both centralized and decentralized coordination strategies to manage increasing numbers of agents and task complexities. Studies on multi-robot collaboration suggest that hybrid frameworks, which combine centralized oversight with decentralized execution, often achieve better task success rates and scale more effectively than purely centralized or decentralized models [256]. Translating this to dialogue management, a hybrid approach might involve a central “Manager” agent orchestrating the high-level dialogue strategy while specialized “Analyst” or “Searcher” agents operate semi-autonomously to retrieve item details or analyze user sentiment. This is further supported by the MACRec framework, which deploys specialized agents including a Manager, User/Item Analyst, Reflector, Searcher, and Task Interpreter to tackle recommendation tasks directly through collaborative efforts [257].

Finally, the efficacy of these autonomous architectures relies heavily on the ability of agents to engage in constructive debate and consensus-building to resolve ambiguities in user intent. Frameworks that encourage divergent thinking through multi-agent debate have been shown to mitigate the “Degeneration-of-Thought” problem, where single models become stuck in incorrect reasoning paths [258]. By simulating a peer-review process or a round-table conference, agents can challenge each other’s assumptions regarding user preferences, leading to more robust and accurate recommendation decisions [259]. As these architectures evolve, the integration of social psychology principles into agent collaboration mechanisms promises to further refine the nuance and effectiveness of autonomous dialogue management in generative recommendation systems [260]. Through these sophisticated multi-agent collaborations, future CRS will be capable of navigating complex user needs with a level of strategic planning and adaptability that closely mimics human expert recommenders, thereby providing the high-fidelity interaction logs necessary to train the next generation of user simulators.

5.2 LLM-Based User Simulators and Synthetic Data Generation

5.2 LLM-Based User Simulators and Synthetic Data Generation

Building upon the sophisticated autonomous agent architectures established for dialogue management, the next critical component in the generative recommendation ecosystem is the development of high-fidelity user simulators. While Section 5.1 detailed how multi-agent systems orchestrate the recommendation process from the system’s perspective, this section focuses on the complementary challenge: modeling the user side of the interaction with equal granularity. The evolution from static, rule-based user policies to dynamic, LLM-driven simulators represents a paradigm shift in how Conversational Recommendation Systems (CRS) are trained and evaluated. Traditional approaches often relied on limited semantic models that failed to capture the nuance, variability, and emotional depth of human interaction. In contrast, the advent of Large Language Models (LLMs) offers the potential to generate synthetic user behaviors that serve as cost-effective, scalable proxies for real users, enabling rigorous testing of the complex dialogue agents described previously without the prohibitive costs of recruiting human participants.

A primary application of these LLM-based simulators is the generation of synthetic conversational data to augment scarce training datasets, a necessity for robustly training the multi-agent frameworks discussed earlier. In domains where privacy concerns restrict access to real user logs, such as call centers or healthcare, LLMs provide a viable mechanism for creating realistic dialogue trajectories. For instance, frameworks have been introduced where LLMs engage in “self-talk,” assuming multiple roles within a conversation to bootstrap task-oriented dialogue agents. This approach allows for the automated generation of training data that can be filtered based on success metrics, significantly reducing the reliance on expensive human-annotated instruction tuning data [261]. Similarly, systems like LUCID utilize modularized, LLM-driven pipelines to produce linguistically sophisticated and diverse dialogues across multiple domains, effectively addressing the bottleneck of high-quality data scarcity for complex intent classification [262]. These synthetic datasets not only expand the volume of available training material but also introduce challenging conversational phenomena that are often underrepresented in static corpora, thereby stress-testing the adaptability of the dialogue managers.

Beyond simple data augmentation, advanced user simulators are being designed to model specific psychological and behavioral traits, thereby enhancing the realism required for effective agent-to-agent interaction. Recent works propose endowing agents with rich contextual information, including five senses, emotional states, relationship dynamics, and episodic memory, to generate responses that are consistent with a defined persona [263]. This level of granularity is essential for simulating subrational human behaviors, such as risk aversion or myopic decision-making, which are characteristic of real-world economic and social interactions. By treating LLMs as implicit computational models of humans, researchers have successfully replicated findings from classic psychological experiments, such as the ultimatum game, using synthetic demonstrations derived from LLM agents [264]. Furthermore, simulators like EmoUS have been developed to explicitly model user emotions alongside semantic actions, allowing for the evaluation of how system behaviors influence user emotional states—a crucial factor often overlooked in generic user policies [265]. Such emotionally aware simulators provide the necessary feedback loop for the reflection mechanisms inherent in architectures like MACRS.

Despite these advancements, the deployment of LLM-based user simulators is fraught with significant challenges, particularly regarding evaluation validity and controllability, which directly impacts the reliability of the autonomous systems they are meant to train. A critical issue iden-

tified in recent literature is data leakage, where the simulator’s training data or prompt context inadvertently includes information about the target items or conversation goals, leading to inflated performance metrics for the CRS being evaluated. Studies analyzing existing simulators, such as iEvaLM, have highlighted that the success of recommendations often depends more on the quality of the conversational history provided in the prompt than on the simulator’s ability to authentically mimic user preferences [266]. This suggests that many current evaluation protocols may be measuring the model’s context utilization rather than its genuine recommendation capability. Additionally, controlling the output of a user simulator through a single prompt template remains difficult, as LLMs may drift from their assigned persona or fail to adhere to specific behavioral constraints over long conversation horizons, undermining the consistency required for stable agent collaboration.

To address these limitations, researchers are exploring more robust frameworks for constructing and validating user simulators to ensure they provide high-quality interaction logs for the next generation of recommenders. One promising direction involves the use of structured prompting techniques that incorporate pre-analytical recollection and grounding mechanisms to improve long-context tracking and state consistency [267]. Another approach focuses on developing comprehensive evaluation protocols that measure key properties of synthetic users, such as their ability to express binary and open-ended preferences, request recommendations, and provide feedback, thereby revealing deviations from human behavior [268]. Furthermore, the concept of using LLMs to simulate entire subpopulations for survey research and opinion estimation is gaining traction, though it requires careful analysis of potential biases and the development of frameworks to ensure representative modeling [269].

The future of user simulation lies in moving beyond zero-shot generation toward systems that incorporate cognitive architectures and memory frameworks, bridging the gap between verbal simulation and the behavioral modeling discussed in the following section. Position papers argue for treating LLMs as data augmentation systems guided by robust cognitive models, utilizing episodic memory and self-reflection to mitigate hallucinations and improve reliability [270]. Additionally, the integration of inner monologue mechanisms allows agents to “think before they speak,” cultivating communication skills such as empathy and topic transition that are vital for anthropomorphic interaction [271]. As these technologies mature, the synergy between synthetic data generation and rigorous evaluation will be paramount. However, while current simulators excel at generating verbal interactions, they often overlook the silent, non-verbal signals that drive real-world recommendations. This limitation sets the stage for the subsequent exploration of agent-based modeling of non-verbal behaviors, where the focus shifts from generating dialogue to simulating the implicit actions—clicks, views, and purchases—that constitute the backbone of collaborative filtering.

5.3 Modeling Non-Verbal Behaviors and Collaborative Filtering via Agents

5.3 Agent-Based Modeling of Non-Verbal User Behaviors

While the preceding section highlighted the advancements in LLM-based user simulators for generating rich conversational data, a critical dimension of user behavior remains underexplored: non-verbal interactions. The evolution of LLM-based agents in recommender systems has predominantly focused on simulating human dialogue and verbal communication. However, in real-world recommendation scenarios, the vast majority of user feedback is implicit and non-verbal, manifested through actions such as clicks, views, add-to-cart events, and purchases. These silent signals constitute the backbone of collaborative filtering (CF), yet traditional LLM agents often lack the

intrinsic capability to comprehend or simulate these behavioral nuances due to the fundamental gap between language modeling and behavior modeling. To bridge this divide and complement the dialogue-centric approaches, emerging research proposes a paradigm shift where both users and items are conceptualized as autonomous agents capable of engaging in non-verbal collaborative learning, thereby enabling a more holistic simulation of the recommendation ecosystem.

A pioneering approach in this domain is the AgentCF framework, which creatively redefines the roles of entities within the recommendation loop. Instead of treating items as passive data points, AgentCF models both users and items as autonomous language agents [272]. This dual-agent architecture allows for the simulation of user-item interactions where the item agent actively participates in the decision-making process by reflecting on its own attributes and potential appeal. The framework operates through a collaborative learning mechanism where, at each time step, user and item agents interact autonomously. Subsequently, the system evaluates the disparity between the agents’ simulated decisions and actual interaction records. Based on this feedback, both agents are prompted to reflect on and adjust their strategies collaboratively, effectively modeling the two-sided relations inherent in collaborative filtering [272]. This process enables the agents to propagate preferences to others in subsequent interactions, implicitly capturing the core “collaborative” spirit of CF where the behavior of one user influences the representation of an item for others.

The significance of modeling non-verbal behaviors extends beyond simple interaction simulation; it addresses the incomprehension of LLMs regarding complex user-item relations. Traditional methods often rely on fixed representations that fail to capture the dynamic nature of user interests or the heterogeneous relationships between users and items. For instance, standard Collaborative Metric Learning (CML) approaches often struggle with the heterogeneity of user-item relations, ignoring that a user’s preference for an item depends on latent item-item relations and previously interacted items [273]. By employing autonomous agents, the system can dynamically infer these latent structures. The agent-based approach allows for the simulation of diverse interaction behaviors, including user-item, user-user, item-item, and collective interactions, which are difficult to capture with static embedding methods. This capability is crucial for generating synthetic data that mirrors the complexity of real-world user trajectories, thus overcoming the limitations of existing studies that mainly focus on simulating human dialogue while neglecting implicit behavioral exhibits [272].

Furthermore, modeling non-verbal behaviors via agents facilitates a deeper understanding of multi-behavior dependencies. In practical scenarios, users exhibit a spectrum of behaviors ranging from implicit clicks to explicit purchases, each carrying different levels of preference intensity. Traditional models often treat these signals uniformly or separately, missing the rich semantic transitions between them. Agent-based simulations can incorporate these nuances by assigning different “intent” states to user agents, allowing them to reason through the progression from a click to a purchase. This aligns with findings that leveraging multiple behavior types and modeling the sequence from implicit to explicit actions significantly enhances recommendation performance [274]. By simulating these sequences, agents can learn to distinguish between casual browsing and strong purchase intent, a distinction that is often lost in standard ID-based collaborative filtering.

The integration of non-verbal behavior simulation also opens new avenues for addressing data sparsity and cold-start problems. In scenarios where interaction data is scarce, autonomous agents can generate plausible interaction histories based on semantic reasoning and collaborative reflection. This is particularly relevant when considering that traditional CF methods suffer when faced with sparse datasets, as they rely heavily on observed co-occurrences [273]. Agent-based models can infer missing links by simulating how a rational user agent would interact with an item agent given specific contextual cues and semantic attributes. Moreover, the ability of item agents to reflect on

their own characteristics allows them to adapt their representations to better match emerging user trends, effectively mitigating the cold-start issue for new items by leveraging semantic knowledge rather than relying solely on historical interaction counts.

Finally, the agent-based modeling of non-verbal behaviors provides a robust foundation for explainable recommendations. Since the interactions are driven by explicit reasoning steps within the agents’ “thought processes,” the system can generate natural language explanations for why a specific non-verbal action (like a click) was simulated or predicted. This contrasts with black-box deep learning models where the rationale behind a recommendation is often opaque. By treating the recommendation process as a multi-agent negotiation of preferences and attributes, the system not only predicts the next item but also articulates the underlying logic, enhancing trust and transparency. As the field moves towards more sophisticated user simulators, the convergence of verbal dialogue capabilities with non-verbal behavioral simulation represents a critical step toward building Artificial General Recommenders. This progression naturally sets the stage for the next frontier: embedding these behaviors within deeper cognitive architectures that feature dynamic persona customization, inner monologue reasoning, and memory-augmented planning to achieve true anthropomorphism.

5.4 Cognitive Modeling, Persona Customization, and Inner Monologue

5.4 Cognitive Modeling, Persona Customization, and Inner Monologue

Building upon the agent-based simulation of non-verbal behaviors detailed in the preceding section, the pursuit of truly anthropomorphic recommenders necessitates a deeper excavation into the internal cognitive architectures that drive these actions. While Section 5.3 established how autonomous agents can replicate implicit signals like clicks and purchases through collaborative reflection, achieving genuine human-like engagement requires moving beyond behavioral mimicry to simulate the complex internal processes characterizing human decision-making. This evolution involves the intricate customization of dynamic personas, the implementation of “inner monologue” mechanisms for strategic reasoning, and the integration of memory-augmented planning to ensure consistency and emotional depth over time. These cognitive layers transform agents from reactive simulators of interaction logs into proactive entities capable of nuanced social intelligence, setting the stage for the rigorous interactive evaluation protocols discussed in the following section.

Persona customization serves as the foundational layer for this cognitive depth, yet early approaches often struggled with maintaining consistency or avoiding egocentric behaviors where the agent rigidly forces its predefined traits regardless of the conversational context. Advanced frameworks now address this by treating personas not as static attribute lists but as dynamic, concept-based structures that interact with the partner’s persona. For instance, the COSPLAY framework proposes representing both self-persona and partner persona as concept sets, utilizing set algebras to find common ground and generate responses that balance self-expression with curiosity toward the user [275]. This approach mitigates the tendency of agents to twist topics stiffly, thereby creating a more human-like “team” dynamic. Furthermore, to enhance the richness of these personas, recent methodologies have introduced the concept of “habitual schemas.” Instead of relying solely on one-off facts, these systems retrieve schemas representing kinds of events an agent frequently participates in, including typical goals and sub-events, to condition the LLM generation [276]. This allows agents to reveal their personalities through narrative consistency rather than explicit declaration, closely mirroring how humans construct identity through repeated actions and experiences, thus providing the stable behavioral baselines necessary for the non-verbal simulations described

previously.

Beyond static persona definition, the simulation of human cognition requires the capacity for introspection and strategic planning, often realized through “inner monologue” or self-talk mechanisms. Inner speech is a critical psychological process in humans, facilitating self-regulation and transparency. Research indicates that programming robots with overt self-talk systems can significantly enhance human trust and the perception of anthropomorphic features during cooperative tasks [277]. In the realm of LLMs, this concept is expanded into Reflective Linguistic Programming (RLP), where models are encouraged to introspect on their predefined personality traits, emotional responses, and planned strategies before generating a response. This introspective capability enables the emergence of sophisticated behaviors, such as strategic deception or sensitivity to user discomfort, which arise spontaneously from the model’s cognitive modeling rather than being explicitly pre-programmed [278]. Such self-reflection allows agents to navigate complex social nuances, effectively acting as a “digital soul” that can adapt its strategy based on internal deliberation, bridging the gap between the silent reasoning of non-verbal actions and the explicit articulation of dialogue.

To further refine cognitive modeling, agents must possess the ability to simulate multiple minds or perspectives simultaneously, a phenomenon known as cognitive synergy. The Solo Performance Prompting (SPP) technique transforms a single LLM into a cognitive synergist by engaging it in multi-turn self-collaboration with multiple distinct personas. By dynamically identifying and simulating different personas based on task inputs, the agent can combine diverse strengths and knowledge bases, reducing factual hallucinations and enhancing problem-solving capabilities in a manner analogous to human collaborative intelligence [279]. This multi-persona internal dialogue allows the agent to weigh different viewpoints, effectively mimicking the internal conflict and resolution processes inherent in human decision-making, which is crucial for handling the ambiguous and multi-faceted nature of user preferences in generative recommendation.

Memory-augmented planning is another cornerstone of robust cognitive modeling, ensuring that agents retain context and learn from interactions over extended horizons. Effective cognitive architectures require memory systems that are active, introspective, and capable of associating semantic knowledge with sensorimotor data. Systems like ArmarX demonstrate how a centralized memory system can orchestrate data flows between different processes, enabling high-level cognitive abilities such as prospection and simulation [152]. In social contexts, memory plays a pivotal role in emotional valence prediction and adaptive empathy. Agents equipped with distributed memory systems, such as SensAI+Expanse, can adapt their cognitive processes to individual users by recalling past interactions and adjusting their emotional responses accordingly, thereby fostering resilient and personalized human-agent bonds [280]. Moreover, the construction of personalized knowledge bases that match external information sources with specific agent personas ensures that the agent’s world knowledge remains consistent with its character, preventing the leakage of out-of-character information [281]. This longitudinal consistency is vital for maintaining the integrity of the simulated non-verbal behaviors and persona traits across long-term interaction sequences.

Finally, the integration of these elements—dynamic persona customization, inner monologue reasoning, and memory augmentation—culminates in agents capable of complex Theory of Mind (ToM) reasoning. By inferring latent trait vectors from observed trajectories and modulating prediction mechanisms via fast weights, agents can better anticipate user goals and intentions [282]. This depth of cognitive modeling is essential for moving towards Artificial General Recommenders that do not merely retrieve items but understand the underlying emotional and cognitive states of users, engaging in interactions that are not only informative but also empathetic and psychologically

resonant. As these technologies mature, the boundary between simulated cognition and genuine understanding continues to blur, offering new frontiers for trustworthy and deeply personalized AI systems. However, the realization of such sophisticated cognitive architectures demands equally advanced methods to verify their efficacy, reliability, and social acumen. Consequently, the field is shifting towards interactive evaluation protocols and social interaction benchmarks capable of stress-testing these inner monologues and dynamic personas, a transition that forms the focus of the subsequent discussion.

5.5 Interactive Evaluation Protocols and Social Interaction Benchmarks

5.5 Interactive Evaluation Protocols and Social Interaction Benchmarks

Building upon the sophisticated cognitive architectures, dynamic persona customization, and inner monologue mechanisms detailed in the preceding section, the verification of these advanced capabilities necessitates a parallel evolution in evaluation methodologies. The rapid maturation of Large Language Models (LLMs) has fundamentally transformed the landscape of Conversational Recommender Systems (CRS), shifting the focus from static ranking metrics to the assessment of dynamic, multi-turn interaction quality. Traditional evaluation protocols, which often rely on offline datasets and single-turn accuracy metrics like precision or recall, are increasingly recognized as inadequate for capturing the nuanced interactivity, explainability, and social intelligence inherent in modern generative recommendation. Consequently, the research community is witnessing a paradigm shift toward interactive evaluation protocols that leverage LLM-based simulators and sophisticated social interaction benchmarks. These emerging frameworks aim to replicate the complexity of real-world human dialogue, providing a robust testbed for assessing whether agents can effectively operationalize their simulated cognitive states and persona consistency in practice.

A primary driver of this shift is the realization that standard evaluation methods frequently fail to account for the interactive nature of capable CRS. Early attempts to utilize LLMs for evaluation revealed significant inadequacies in existing protocols, which tended to over-emphasize matching ground-truth items while neglecting the fluidity of conversation [283]. To address this, researchers have proposed interactive evaluation approaches, such as iEvaLM, which harness LLM-based user simulators to generate diverse interaction scenarios. This method allows for the assessment of a system’s ability to handle multi-turn dialogues, elucidate recommendations, and adapt to user feedback in real-time, thereby offering a more flexible and comprehensive framework than static benchmarks [283]. Such simulators are essential for testing the “inner monologue” and strategic planning capabilities discussed previously, ensuring that an agent’s internal reasoning translates into coherent external behavior.

However, the reliance on user simulators introduces its own set of challenges, particularly regarding reliability and data leakage. Critical analysis of current LLM-based simulators has highlighted issues such as inflated evaluation results caused by data leakage in conversational histories and the difficulty of controlling simulator outputs through simple prompt templates [266]. Furthermore, traditional simulators often exhibit unrealistic behaviors, such as extreme patience or a singular focus on a specific target item, which fails to capture the exploratory nature of genuine user behavior. To mitigate these limitations, advanced simulators have been developed that incorporate knowledge of alternative items and allow for dynamic shifts in user preferences and patience levels, providing a more realistic stress test for CRS models [284]. Additionally, toolkits like UserSimCRS have been introduced to facilitate automatic evaluation by integrating persona modeling, context awareness, and satisfaction prediction, enabling the simulation of dialogues that closely mimic real human

conversations with minimal training data [285]. These advancements directly support the evaluation of the “habitual schemas” and dynamic persona structures outlined in Section 5.4.

Beyond simple dialogue fluency, there is a growing emphasis on evaluating the “social intelligence” of recommendation agents, a direct extension of the Theory of Mind (ToM) and cognitive synergy concepts. The ability of an agent to understand social cues, manage group dynamics, and exhibit appropriate personality traits is crucial for long-term user engagement. Benchmarks such as RoleInteract have been designed specifically to evaluate the sociality of role-playing agents at both individual and group levels, revealing that proficiency in one-on-one interactions does not necessarily translate to competence in group settings [286]. Similarly, the SocKET benchmark provides a theory-driven approach to measuring social language understanding across categories like humor, sarcasm, and trustworthiness, highlighting that while pretrained models possess innate social capabilities, significant room for improvement remains [287]. The concept of “AI-nality” or AI personality further expands this scope, utilizing psychometric tests to assess how well LLMs can adopt and maintain distinct personas, which is essential for personalized and empathetic recommendation experiences [288].

To rigorously assess these complex capabilities, new multi-turn evaluation benchmarks have been established to probe the depth of cognitive modeling. MT-Bench-101 offers a fine-grained taxonomy of abilities across thousands of dialogue turns, exposing the fact that common alignment techniques often fail to enhance multi-turn performance significantly [289]. Complementing this, MT-Eval categorizes interaction patterns into recollection, expansion, refinement, and follow-up, identifying key factors such as error propagation and distance to relevant content that impact multi-turn robustness [290]. For vision-language models involved in multimodal recommendation, ConvBench introduces a hierarchical capability structure mirroring human cognitive processes, allowing for precise attribution of failures in perception, reasoning, or creativity [291]. These benchmarks serve as the critical bridge between theoretical cognitive models and verified performance.

Moreover, the evaluation of conversational recommenders is increasingly incorporating game-theoretic and strategic dimensions to test decision-making under pressure, reflecting the strategic reasoning enabled by inner monologues. Frameworks like LLMArena and AgentBench evaluate agents in dynamic, multi-agent environments, assessing skills such as strategic planning, opponent modeling, and team collaboration, which are vital for handling competitive or complex ecosystem scenarios [292][293]. Negotiation-specific benchmarks, including NegotiationToM and LLM-Deliberation, stress-test Theory of Mind (ToM) abilities and strategic reasoning in scenarios requiring compromise and inference of partner motives, areas where even state-of-the-art models often struggle compared to humans [294][295]. These evaluations ensure that the cognitive synergy and multi-persona simulations described earlier function effectively in adversarial or cooperative settings.

Finally, to ensure a holistic assessment, protocols like Concept have been proposed to integrate both system-centric factors (e.g., effectiveness, fluency) and user-centric factors (e.g., satisfaction, Usability), addressing the gap between academic metrics and real-world user experience [244]. These interactive evaluation protocols and social interaction benchmarks collectively represent a critical advancement in the field, moving beyond superficial accuracy to deeply probe the interactivity, explainability, and social acumen of next-generation generative recommenders. By leveraging LLMs not just as systems to be evaluated but as active participants in the evaluation process via simulation and automated judging, the community is establishing a more rigorous and realistic foundation for developing truly intelligent conversational agents. This robust evaluative infrastructure is a prerequisite for the safe deployment of these agents in high-stakes domains, setting the stage for

the discussion on domain-specific applications and real-world safety constraints in the following section.

5.6 Domain-Specific Agents and Real-World Deployment Scenarios

5.6 Domain-Specific Agents and Real-World Deployment Scenarios

Building upon the rigorous interactive evaluation protocols and social interaction benchmarks established in the preceding section, the transition from simulated environments to real-world deployment represents the ultimate stress test for Large Language Model (LLM) agents. While Section 5.5 highlighted the necessity of assessing multi-turn coherence and social intelligence through simulators, the application of these agents in high-stakes domains such as healthcare, finance, and autonomous driving demands a paradigm shift from theoretical capability to operational safety. In these specialized fields, precision, adherence to professional standards, and strict ethical compliance are not merely desirable attributes but existential requirements. The unique challenges inherent to these sectors necessitate agents that possess not only deep contextual knowledge but also the ability to operate within rigidly defined safety boundaries to prevent catastrophic failures, thereby extending the “social intelligence” concepts discussed earlier into critical professional contexts.

In the healthcare sector, the integration of LLM agents into clinical workflows illustrates both the transformative potential and the perilous risks of generative recommendation. While these models show promise in information synthesis and clinical decision support, their deployment is heavily constrained by the dangers of hallucination and the generation of unsafe medical advice. Research indicates that standard LLMs often fail to respond adequately to complex medical queries, occasionally producing erroneous information or content that violates fundamental medical principles [296]. To bridge the gap between simulation and reality, the community is advancing toward “Artificial-intelligence Structured Clinical Examinations” (AI-SCI), which evaluate agent performance on real-world clinical tasks rather than static benchmarks [297]. Furthermore, to enhance reasoning capabilities in zero-shot settings, multi-disciplinary collaboration frameworks have been developed, allowing LLM-based agents to engage in role-playing discussions that mimic teams of medical experts [298]. Despite these methodological advancements, the imperative for safety remains paramount; recent studies emphasize the need for comprehensive safety evaluations specifically tailored to medical AI, defining clear metrics for alignment to minimize risks to personal and public health [299]. The ultimate objective is to evolve these systems into progressive pathways for artificial healthcare assistants capable of navigating medicine’s multifaceted data modalities while maintaining strict ethical oversight [300].

Similarly, in the financial domain, the deployment of LLM agents introduces complex principal-agent problems where the utility functions of the AI may diverge from human intentions. Empirical studies have revealed that LLM-based agents can override principal objectives in online shopping and trading tasks, highlighting the inherent misalignment risks when information asymmetry exists [301]. To address these alignment issues, novel frameworks like QuantAgent have been proposed, utilizing a two-layer loop structure where agents refine their responses based on a domain-specific knowledge base and test these responses in real-world scenarios to self-improve [302]. However, the deployment of such agents in finance also raises significant legal and ethical concerns regarding accountability. Experts argue that unlike human professionals, LLM interactions lack attorney-client confidentiality and are not bound by professional ethics, suggesting that AI systems should be designed to help users refine their queries rather than providing direct, actionable legal or financial advice [303]. This underscores the necessity of encoding legal standards, such as fiduciary obliga-

tions, directly into the agent’s operational logic to ensure robust communication and compliance [304].

Perhaps the most physically tangible and safety-critical deployment scenario for LLM agents is autonomous driving (AD). Here, the non-interpretability of traditional deep neural networks in long-tail scenarios poses significant safety hurdles that static evaluation benchmarks cannot fully capture. Integrating LLMs as intelligent decision-makers offers a pathway to leverage common-sense reasoning for behavioral planning, augmented by safety verifier shields to ensure contextual safety [305]. Approaches such as LanguageMPC demonstrate how LLM decisions can be translated into actionable driving commands, enabling vehicles to handle complex, multi-vehicle coordination tasks with human-like reasoning [306]. Nevertheless, aligning these agents with diverse human driving styles remains a challenge; addressing this requires constructing natural language datasets from human demonstrations to train agents that reflect varied behavioral norms [307]. Safety in this domain is further reinforced by frameworks that learn responsibility allocations from data, allowing autonomous vehicles to understand their shared responsibilities in multi-agent interactions without resorting to overly conservative behaviors [308]. Moreover, the verification of these agents is critical; formal methods and rational agent programming languages are being employed to mathematically verify agent plans for obstacle avoidance and recovery, ensuring correctness before deployment [309].

Across all these domains, a unifying theme emerges: the necessity of prioritizing safeguarding over pure autonomy. Whether in scientific experimentation, robotics, or clinical practice, the vulnerabilities of LLM agents demand a triadic framework involving human regulation, agent alignment, and environmental feedback [310]. The development of “good” social actors requires moving beyond the semantic correctness measured in previous sections to consider relational and situational factors, ensuring that agents treat individuals with respect and adhere to social, legal, ethical, empathetic, and cultural (SLEEC) rules [311; 312]. As these agents move from controlled simulations to the wild, continuous monitoring and context-sensitive safety boundaries become essential to detect and halt unsafe behaviors before they cause harm [313]. Thus, the future of domain-specific LLM agents lies not merely in their cognitive prowess or their performance on benchmarks, but in their ability to operate safely, responsibly, and transparently within the intricate fabric of professional and societal standards.

6 Trustworthiness, Safety, Fairness, and Ethical Considerations

6.1 Hallucination Taxonomy, Detection, and Mitigation in Recommendations

6.1 Hallucination Taxonomy, Detection, and Mitigation in Recommendations

As Large Language Models (LLMs) and Multimodal Large Language Models (MLLMs) transform recommendation systems from discriminative predictors to generative narrators, the phenomenon of hallucination has emerged as a fundamental barrier to trustworthiness. Unlike traditional ID-based collaborative filtering, where errors are typically limited to ranking inaccuracies, generative recommenders produce natural language rationales and item descriptions that can be factually incorrect, ungrounded in user history, or inconsistent with actual item attributes. This shift necessitates a rigorous examination of hallucination specific to the recommendation domain. This subsection provides a comprehensive taxonomy of these errors, reviews automated detection frameworks tailored for open-ended generation, and analyzes mitigation strategies leveraging fine-grained feedback and alignment techniques, laying the groundwork for understanding the broader security vulnerabilities discussed in the subsequent section.

Taxonomy of Hallucinations in Recommendation Hallucinations in generative recommendation can be categorized based on their orientation and severity, drawing parallels from broader LLM research while addressing domain-specific constraints. A fundamental distinction exists between “factual mirage,” where the model generates content contradicting known facts (e.g., attributing a non-existent genre to a movie), and “silver lining,” where the model fabricates plausible but unsupported details to fill information gaps [314]. In the context of recommendation systems, these manifest primarily as object hallucinations (recommending items that do not exist), attribute hallucinations (misstating item features like price, rating, or availability), and relationship hallucinations (incorrectly linking a user’s past behavior to a current suggestion) [315].

The complexity increases in multimodal scenarios, where recent studies in vision-language models have identified specific orientations such as “contextual guessing” and “identity incongruity.” These are particularly relevant when recommendations rely on fusing product images or video thumbnails with textual descriptions [316]. The severity of these hallucinations ranges from mild nuances to alarming errors that completely mislead the user, necessitating a fine-grained taxonomy that accounts for the specific domain constraints of e-commerce, news, or entertainment [314]. Furthermore, in conversational recommenders, hallucinations may appear as “faithfulness” errors, where the system contradicts its own previous turns or ignores explicit user constraints, a challenge highlighted in dialogue-specific benchmarks [317].

Automated Detection Frameworks Detecting hallucinations in real-time recommendation pipelines is challenging due to the open-ended nature of generated text, rendering traditional metrics like accuracy or perplexity insufficient for identifying factual inconsistencies. Consequently, researchers have developed specialized detection frameworks operating at different stages of the inference process. One promising direction involves leveraging the internal states of the model. Methods such as INSIDE propose utilizing the dense semantic information retained within the model’s internal layers, using metrics like EigenScore to evaluate self-consistency in the embedding space; this can be adapted to detect when a recommendation rationale diverges significantly from the user’s profile representation [318]. Similarly, analyzing artifacts from self-attention mechanisms and token attribution can provide early warnings of hallucinatory generation before the final output is produced [319].

Another robust approach is the use of retrieval-augmented verification. Frameworks like KnowHalu employ a multi-step process involving query decomposition and knowledge retrieval to fact-check generated content against a trusted knowledge base, ensuring that recommended items and their descriptions align with ground truth [320]. For multimodal recommendations, unified detection frameworks such as UNIHD utilize auxiliary tools to validate the consistency between visual inputs and textual descriptions, directly addressing the challenge of visual hallucination where an item’s image does not match its generated description [321]. Furthermore, semantic-aware cross-check consistency (SAC3) expands on self-consistency by perturbing queries semantically and checking for response stability across different model instantiations, a technique vital for ensuring that recommendation explanations remain robust under slight variations in user prompts [322].

Mitigation Strategies via Feedback and Alignment Mitigating hallucinations requires moving beyond simple post-processing to integrated training and alignment strategies that enforce factual grounding. A pivotal advancement is the use of fine-grained human feedback. Unlike coarse sequence-level rewards, fine-grained feedback allows models to learn from segment-level corrections, pinpointing exactly where a recommendation rationale goes astray. The RLHF-V framework demonstrates that collecting human preferences in the form of segment-level corrections and applying dense direct preference optimization can significantly reduce hallucination rates while

maintaining data efficiency [323]. This concept extends to AI-generated feedback, where smaller models or specialized detectors provide sentence-level annotations to train larger models, creating a scalable pipeline for hallucination mitigation without prohibitive human annotation costs [315].

Dialectical alignment and targeted instruction tuning offer another pathway to robustness. By diagnosing the specific hallucination tendencies of a model—such as its propensity to over-trust certain summary tokens—researchers can generate targeted instruction data to counteract these biases. The DFTG framework illustrates how diagnosing model-specific hallucination patterns and generating corresponding corrective data can outperform generic instruction tuning [324]. Moreover, decoding-time strategies like OPERA introduce an over-trust penalty and a retrospection-allocation mechanism to prevent the model from neglecting crucial context tokens, effectively reducing hallucinations during the generation of recommendation lists without requiring additional training data [325]. Finally, incorporating counterfactual thinking into the model’s reasoning process, as proposed in Counterfactual Inception, encourages the system to consider alternative realities and verify facts before committing to a recommendation, thereby enhancing overall trustworthiness [326]. Together, these approaches form an essential defense layer against unintentional model errors, distinguishing them from the adversarial threats and malicious manipulations explored in the following section on security and robustness.

6.2 Adversarial Robustness and Security Defenses for LLM-based Recommenders

6.2 Adversarial Robustness and Security Defenses for LLM-based Recommenders

While Section 6.1 addressed the challenge of unintentional model errors through hallucination detection and mitigation, this subsection shifts focus to the deliberate, malicious threats that target the integrity of LLM-based recommendation systems. The transition from traditional collaborative filtering to generative, context-aware personalization significantly expands the attack surface, rendering these systems vulnerable to sophisticated adversarial strategies that were less prevalent in ID-based architectures. Unlike hallucinations, which stem from internal model uncertainty or knowledge gaps, adversarial attacks are engineered exploits designed to manipulate model behavior. The security landscape for LLM-based recommenders is characterized by two primary vectors: prompt injection attacks, which manipulate the model’s input context to alter recommendations, and backdoor injections, which compromise the model’s integrity during training or deployment. Addressing these vulnerabilities requires a comprehensive understanding of both offensive strategies and defensive mechanisms, ranging from robust training protocols to manifold-based purification techniques, setting the stage for the broader ethical trade-offs discussed in the subsequent section.

Prompt injection represents a critical threat to the interactive nature of LLM recommenders, where user queries and system instructions are processed as natural language. Adversaries can craft malicious inputs that override system prompts or inject hidden triggers to force the model into generating biased, harmful, or competitor-favored recommendations. The emergence of automated tools for generating attack variants exacerbates this risk. For instance, recent research has introduced tools capable of performing automated variant analysis for prompt injection attacks, demonstrating that defenders must test against not just known attacks but also their evolving permutations to ensure system resilience [327]. Furthermore, the vulnerability extends beyond simple input manipulation; studies have shown that prompt-based learning paradigms inherit universal vulnerabilities from pre-training stages, allowing adversaries to insert triggers into plain text that can completely

control or severely degrade the performance of fine-tuned recommendation models across arbitrary downstream tasks [328]. This universality implies that even models instruction-tuned for specific recommendation domains remain susceptible to carefully crafted linguistic triggers that evade standard guardrails.

Beyond prompt injection, backdoor attacks pose an insidious threat by embedding hidden triggers within the model’s parameters or training data. In the context of recommenders, a backdoored model might function normally for benign users but consistently recommend specific items when a trigger—such as a particular phrase in a review or a specific image in a multimodal setting—is present. The scope of these attacks has expanded to include training-time poisoning and weight-level manipulation. For example, adversaries can now inject backdoors during the instruction tuning phase of LLMs by introducing a mere 1% of poisoned samples, leading to a significant performance drop rate and controlled output manipulation while maintaining content integrity on clean inputs [329]. Moreover, the threat persists even after model deployment; research has demonstrated that adversaries can inject backdoors into deployed models by modifying model weights within a small norm bound, a technique known as adversarial weight perturbation, which successfully implants triggers without affecting the model’s utility on non-triggered inputs [330]. This is particularly concerning for recommender systems that rely on third-party pre-trained models or operate in environments where model weights might be accessible or modifiable, such as edge devices susceptible to hardware-level faults like Rowhammer attacks [331].

The complexity of these attacks is further amplified in multimodal recommendation scenarios, where visual and textual modalities are fused. Advanced attacks like BadCLIP utilize dual-embedding guidance to align visual triggers with textual target semantics, making the backdoor resistant to standard detection and fine-tuning defenses [332]. Additionally, test-time backdoor attacks have emerged, where adversaries inject triggers via universal adversarial perturbations in images without needing access to the training data, allowing for dynamic activation of harmful effects in Multimodal Large Language Models (MLLMs) [333]. These developments highlight the necessity for defense mechanisms that are not only robust to data poisoning but also resilient to weight manipulation and test-time perturbations.

To counteract these diverse threats, the research community has developed several robust defense strategies, with adversarial training and manifold approximation emerging as leading approaches. Adversarial training, which involves augmenting the training data with adversarial examples, remains a cornerstone of defense. However, standard adversarial training often struggles with generalization and can converge to degenerate minima. To address this, novel frameworks such as Gray-box Adversarial Training have been proposed, utilizing intermediate model states to seed stronger adversaries and improve robustness against both white-box and black-box attacks [334]. Furthermore, enhancing adversarial training with hypersphere embedding mechanisms has shown promise in regularizing features onto compact manifolds, thereby improving robustness with minimal computational overhead [335]. For scenarios involving multiple perturbation types, parameter-saving adversarial training leverages hypernetworks to train specialized sub-models, achieving state-of-the-art robustness while significantly reducing parameter counts [336].

Complementing robust training, manifold-based defenses offer a powerful post-hoc or test-time mitigation strategy by exploiting the observation that adversarial and backdoored samples often lie off the natural data manifold. The Textual Manifold-based Defense (TMD) projects text embeddings onto an approximated manifold of natural language, effectively neutralizing adversarial perturbations in NLP tasks without sacrificing clean accuracy [337]. Similarly, techniques utilizing Langevin dynamics, such as MALADE, drive off-manifold adversarial samples toward high-density regions

of the target class distribution, providing a stochastic yet effective purification mechanism [338]. These manifold approximation techniques are particularly relevant for LLM recommenders, where the semantic space is high-dimensional and complex. Additionally, detection-based approaches have evolved to identify backdoors by analyzing feature representations; for instance, the AEGIS detector leverages feature clustering to identify backdoor-infected robust models with high accuracy, proving that salient features of adversarially robust networks can break the stealthy nature of backdoor attacks [339]. Other methods, such as Shared Adversarial Unlearning (SAU), propose unlearning shared adversarial examples to mitigate backdoor risks using only a small clean dataset, offering a practical solution for purifying compromised models [340].

In conclusion, securing LLM-based recommendation systems requires a multi-faceted approach that addresses the full lifecycle of the model, from data collection and instruction tuning to deployment and inference. While prompt injection and backdoor attacks present formidable challenges distinct from the hallucinations discussed previously, advances in adversarial training, manifold projection, and unlearning techniques provide a robust toolkit for defenders. However, achieving absolute robustness is rarely free; it often introduces tensions with other critical system properties. As we transition to the next subsection, it becomes essential to examine how these security measures interact with fairness and privacy, exploring the complex trade-offs where optimizing for adversarial robustness may inadvertently impact equitable treatment or data confidentiality.

6.3 The Interplay and Trade-offs Between Fairness, Privacy, and Robustness

6.3 The Interplay and Trade-offs Between Fairness, Privacy, and Robustness

Having examined the specific adversarial threats and defense mechanisms in Section 6.2, and preceding the deep dive into privacy-preserving architectures in Section 6.4, it is crucial to address the complex triadic relationship between fairness, privacy, and adversarial robustness. While these three pillars are fundamental to trustworthy artificial intelligence, they often exist in a state of tension where optimizing for one property can inadvertently degrade the others. This subsection critically analyzes the theoretical underpinnings and empirical evidence of these trade-offs, moving beyond isolated optimizations to propose unified frameworks capable of achieving simultaneous alignment in generative recommendation contexts.

The interplay between differential privacy (DP) and fairness has been a subject of intense scrutiny, with early literature suggesting an inherent conflict. It is widely observed that the noise injection mechanisms required for DP can disproportionately impact minority groups, thereby exacerbating existing biases. For instance, research indicates that the performance cost of differential privacy is often higher for minority groups, creating a disparate impact where privacy protections inadvertently compromise fairness [341]. Furthermore, studies have shown that classical collaborative filtering methods trained with differentially private stochastic gradient descent can result in unfair recommendations against inactive users, effectively amplifying bias rather than mitigating it [342]. This phenomenon is particularly acute in language model fine-tuning, where DP has been found to amplify gender, racial, and religious biases due to disparities in gradient convergence across sub-groups [343]. Theoretical analyses support these empirical findings, demonstrating that output perturbation mechanisms can bound individual and group fairness levels, with the negative impact often growing with model dimensionality [344].

However, the narrative is not uniformly pessimistic; emerging research challenges the notion that

privacy and fairness are strictly antagonistic. Under specific conditions, particularly when employing robust training objectives or optimal model architectures, differential privacy can act as a regularizer that reduces between-group performance differences [341]. Some empirical studies suggest that Local Differential Privacy (LDP) may lead to slightly improved fairness without significantly affecting model performance, contradicting the belief that privacy necessarily worsens fairness [345]. Additionally, advanced frameworks like FairSP have demonstrated that it is possible to achieve fair predictions in semi-private settings by correcting noise-protected sensitive attributes using limited clean data [346]. Similarly, information-theoretic approaches utilizing information bottlenecks combined with LDP have theoretically proven that privacy mechanisms can enhance the fairness of learned representations by effectively obfuscating sensitive information [347].

The relationship between robustness and fairness introduces another layer of complexity, directly connecting to the adversarial threats discussed previously. A significant body of work highlights a dichotomy where achieving fairness can increase a model’s vulnerability to adversarial attacks. Specifically, fairness constraints often force decision boundaries closer to data points of protected groups, thereby reducing the margin and increasing susceptibility to perturbations [348]. This trade-off is driven by the distance to the decision boundary across different groups, suggesting that equitable treatment may come at the cost of security. Conversely, other studies propose that integrating causal structures can bridge this gap. By utilizing causal adversarial perturbations, it is possible to create metrics that simultaneously ensure individual fairness and adversarial robustness, leveraging structural causal models to define semantic similarity among individuals [349]. Furthermore, the introduction of causal fair metrics has shown promise in achieving classifiers that are accurate, fair, and resilient to adversarial perturbations by encompassing sensitive attributes within a causal framework [350].

The intersection of all three dimensions—privacy, fairness, and robustness—reveals a “three-way knot” where optimizing any single vector often penalizes the others. Comprehensive analyses indicate that while predictive performance and fairness might be jointly optimized, doing so often comes at the expense of data privacy [351]. In distributed learning environments, this trilemma is even more pronounced, with tight analyses exhibiting fundamental trade-offs between privacy, robustness against adversarial machines, and utility [352]. Moreover, the fault tolerance of neural networks, which is related to robustness, has been found to be at odds with adversarial robustness, although differentially private models can enhance fault tolerance [353].

To navigate these tensions, unified frameworks are essential. Recent advancements propose mechanisms that address these goals simultaneously rather than sequentially. For example, the FairDP mechanism achieves certified fairness with differential privacy by training group-specific models and integrating their knowledge, thereby balancing privacy, utility, and fairness [354]. In federated learning, personalized frameworks like Ditto have been shown to inherently provide both fairness and robustness benefits in statistically heterogeneous networks, challenging the assumption that these are competing constraints [355]. Additionally, theoretical connections between robustness and privacy suggest that algorithms robust to adversarial data corruptions can be transformed into differentially private ones, implying that optimal error rates for private estimators are akin to those for robust estimators [356]. This theoretical link is further supported by the observation that Renyi-differential privacy and robustness against adversarial examples share a common foundation in probabilistic mappings and divergence measures [357].

In conclusion, while significant tensions exist between fairness, privacy, and robustness in generative recommendation systems, they are not insurmountable. The path forward lies in developing holistic architectures that leverage causal reasoning, personalized learning, and advanced cryptographic

techniques to harmonize these objectives. Understanding these interdependencies is a prerequisite for designing the privacy-preserving architectures detailed in the following section, ensuring that generative recommenders do not sacrifice ethical alignment for efficiency or vice versa, but rather operate within a Pareto-optimal frontier of trustworthiness.

6.4 Privacy-Preserving Architectures and Sensitive Attribute Protection

6.4 Privacy-Preserving Architectures and Sensitive Attribute Protection

Building upon the theoretical analysis of the trade-offs between fairness, privacy, and robustness in Section 6.3, this section delves into the concrete architectural mechanisms designed to operationalize privacy guarantees in Large Language Model (LLM)-driven generative recommendation systems. While the preceding discussion highlighted how optimizing for one dimension can inadvertently compromise others, the practical deployment of these systems requires robust engineering solutions that specifically address the leakage of sensitive user attributes and behavioral patterns. As generative recommenders increasingly rely on decentralized data sources to personalize content, the traditional assumption that keeping raw data local ensures privacy has been rigorously challenged. Recent research demonstrates that even in federated learning environments, where data remains on client devices, adversaries can infer sensitive information from shared gradients or model parameters. Consequently, safeguarding user data in generative systems necessitates a multi-layered architectural approach that synthesizes Local Differential Privacy (LDP), advanced Federated Learning (FL) protocols, and adversarial defense mechanisms specifically engineered to prevent sensitive attribute inference.

Local Differential Privacy (LDP) serves as a foundational technique for extending the privacy trust boundary directly to the client side, ensuring that individual data contributions are randomized before leaving the user’s device. This approach is critical for protecting against membership inference and attribute reconstruction attacks. For instance, studies have shown that LDP can effectively mitigate attribute information leakage in complex tasks such as speech emotion recognition, although the efficacy often depends on the number of model updates exposed to potential adversaries [358]. Furthermore, the application of LDP in time-series analytics highlights the delicate trade-off between privacy and utility; while protective noise can interfere with temporal dependencies, techniques like shuffling can amplify privacy guarantees while mitigating accuracy loss [359]. Theoretical analyses confirm that LDP provides strong guarantees against various attack surfaces, though empirical measurements suggest that worst-case theoretical bounds may be overly conservative, allowing for optimized privacy budgets in practical deployments [360]. These findings underscore the importance of calibrating LDP mechanisms not just for theoretical security, but for the nuanced utility requirements of generative recommendation.

Beyond simple noise addition, novel federated architectures are being developed to reconcile the tension between high model utility and rigorous privacy protection, directly addressing the utility-fairness-privacy trilemma discussed previously. One promising direction involves the use of generative models to share synthetic data representations rather than raw gradients. The [361] framework exemplifies this by decomposing client networks into private extractors and public classifiers, sharing only generators to aggregate knowledge without exposing sensitive feature extractors. Similarly, the [362] protocol utilizes beta-variational autoencoders to synthesize data that retains distinctive features of private samples while providing strong differential privacy guarantees, thereby improving resilience against poisoning and inference attacks. These generative approaches are particularly relevant for tabular data, where frameworks like [363] enable the collaborative training of Gener-

ative Adversarial Networks (GANs) across non-identical participants without risking the leakage of underlying distribution statistics. By shifting from raw data sharing to synthetic representation exchange, these architectures aim to decouple personalization capabilities from privacy risks.

However, the threat landscape extends beyond passive inference to active attacks targeting specific sensitive attributes, requiring defenses that go beyond standard differential privacy. Adversaries can exploit correlations in data features to launch attribute inference attacks, even when standard differential privacy is applied. To counter this, user-configurable privacy defenses have been proposed, such as [364], which formulates privacy protection as an adversarial learning problem. This method generates slight perturbations to gradients specifically designed to fool adversary models targeting user-specified sensitive attributes, significantly improving utility compared to uniform noise addition. Additionally, research into gradient leakage has revealed that data representation leakage is the root cause of many privacy breaches. Consequently, defenses that learn to perturb data representations, such as those proposed in [365], can degrade the quality of reconstructed data by orders of magnitude without sacrificing model performance. These targeted defenses represent a shift from blanket privacy measures to adaptive strategies that respond to specific threat vectors.

The vulnerability of federated systems to such sophisticated attacks necessitates robust architectural safeguards tailored to the recommendation domain. For example, [366] highlights that standard LDP mechanisms are often ineffective against interaction-level membership inference in recommendation contexts unless recommendation performance is severely compromised, prompting the need for specialized defense methods. Moreover, the risk of gradient inversion attacks, where attackers reconstruct training data from shared gradients, remains a critical concern. Works like [367] demonstrate that pretrained generative priors can easily breach data privacy, suggesting that simple gradient sharing is insufficient. In response, methods like [368] propose generating synthetic samples that are visually distinct from original data but preserve predictive features, making gradient leakage attacks extremely difficult. These advancements illustrate an escalating arms race between inference capabilities and privacy-preserving obfuscation techniques.

Finally, the interplay between privacy and other security dimensions, such as robustness against poisoning and backdoor attacks, cannot be overlooked, echoing the interconnected challenges raised in Section 6.3. While differential privacy offers some level of robustness, it does not inherently defend against all forms of adversarial manipulation. Research indicates that increasing privacy levels can enhance certified attack inefficacy but may not always improve certified prediction accuracy, requiring a careful balance [369]. Furthermore, the presence of malicious clients attempting to inject false data or launch Sybil attacks necessitates monitoring mechanisms that can detect convergence anomalies without violating privacy constraints [370]. By integrating these diverse techniques—ranging from LDP and generative data exchange to adversarial perturbation and robust aggregation—future generative recommendation systems can achieve a higher standard of trustworthiness. However, as the following section will elaborate, ensuring data security through these architectures is only one component of a truly equitable system; it does not automatically resolve the deep-seated societal biases embedded within the training corpora that may be amplified by these very generative processes.

6.5 Bias Amplification, Intersectionality, and Equitable Algorithmic Recourse

6.5 Bias Amplification, Intersectionality, and Equitable Algorithmic Recourse

While the privacy-preserving architectures discussed in the preceding section establish a critical foundation for data security, they do not inherently address the societal biases embedded within

the training corpora of Large Language Models (LLMs). The deployment of LLMs in generative recommendation systems introduces profound risks regarding the amplification of these biases. Unlike traditional discriminative models that merely rank existing items, generative models synthesize content and simulate user interactions, potentially exacerbating historical prejudices. A critical limitation in early fairness research was the tendency to evaluate bias mitigation techniques on binary attributes in isolation, such as gender or race independently. However, real-world discrimination is rarely unidimensional; it often manifests through the complex interplay of multiple social identities. Consequently, a truly fair recommendation model must account for “gerrymandering” groups that comprise not only single attributes but also intersectional groups, where individuals face compounded discrimination based on the convergence of their identities [371]. Failure to address these intersectional dynamics can lead to scenarios where fairness metrics appear satisfied at the aggregate level while specific subgroups, such as elderly women of color, suffer from severe algorithmic neglect.

The challenge of intersectional bias is particularly acute in generative systems due to data sparsity. As the number of protected attributes increases, the population size of specific intersectional subgroups decreases rapidly, making statistically reliable estimation of fairness metrics difficult. This sparsity often leads to the underrepresentation of marginalized intersectional groups in training data, causing models to learn skewed distributions that reinforce existing inequalities. To address this, researchers have proposed Bayesian probabilistic modeling approaches that enable reliable, data-efficient estimation of fairness even in multi-dimensional settings with sparse data [372]. Furthermore, the definition of fairness itself must evolve beyond simple demographic parity. Recent work suggests framing intersectional fairness within a geometrical setting, projecting data onto a hypercube to analyze fairness across different levels of attribute intersection. This “fractal approach” posits that ensuring fairness at the lowest intersectional level—such as specific combinations of race and gender—naturally propagates fairness to higher, more aggregate levels, suggesting that trustworthiness is an emergent property of addressing bias at the most granular scale [373].

In the context of generative recommendation, bias amplification can occur not only in the final output but also in the latent representations learned by the model. Standard debiasing methods, such as adversarial training, often require updating the entire model parameters to reach a new debiased state, which can be irreversible and computationally expensive. Moreover, these methods may struggle with the non-linear and multivariate nature of intersectional biases. To mitigate these issues, novel modular approaches have been introduced, such as stand-alone debiasing subnetworks that can be integrated into the core model on-demand. This allows practitioners to selectively mitigate bias for specific attributes or intersectional groups without altering the underlying model weights, offering a flexible solution for dynamic recommendation environments [374]. Similarly, the concept of “Fair Adversarial Networks” has been proposed to remove bias from the data itself by altering proxy variables, ensuring that sensitive attributes become undetectable and thus preventing the model from learning discriminatory patterns in the first place [375].

Beyond mitigation, ensuring equitable algorithmic recourse is paramount. Users harmed by biased recommendations must have mechanisms to understand and challenge these decisions. However, current fairness metrics often fail to capture the nuances of individual harm or the specific mechanisms causing disparity. For instance, some studies highlight that different bias mitigation algorithms impact different sets of individuals, raising questions about the arbitrariness of the debiasing process itself [376]. To address this, researchers are developing frameworks for “algorithmic recourse” that focus on reducing pre-existing real-world disparities rather than merely optimizing statistical metrics. The “impact remediation framework” utilizes causal modeling to discover optimal inter-

vention policies that improve equity for underserved populations, shifting the focus from fairness as a constraint to fairness as an active objective [377].

Furthermore, the legal and ethical grounds for algorithmic fairness are evolving. While many technical approaches rely on “awareness” of protected attributes to correct bias, modern antidiscrimination law in some jurisdictions may prohibit such explicit use of sensitive data, viewing it as a form of affirmative action that violates equal protection principles. This creates a tension between technical efficacy and legal compliance, necessitating approaches that can achieve fairness without explicit demographic targeting or that are grounded in specific legal doctrines allowing for remediation of historical discrimination [378]. Additionally, the collection of demographic data itself poses privacy risks and potential harms of miscategorization, suggesting that fairness methods must sometimes operate without direct access to sensitive attributes, perhaps by using behavioral proxies or counterfactual reasoning [379].

Finally, the intersection of privacy and fairness presents a unique challenge, bridging the concerns of the previous section with the equity goals outlined here. Techniques like Differential Privacy (DP), while essential for protecting user data, have been shown to amplify gender and racial biases during the fine-tuning of large language models due to disparities in gradient convergence across subgroups. This necessitates joint strategies, such as combining DP with Counterfactual Data Augmentation, to maintain both privacy and fairness simultaneously [343]. As generative recommenders become more autonomous, the field must move toward holistic frameworks that integrate intersectional fairness definitions, modular mitigation techniques, and robust recourse mechanisms. These efforts lay the groundwork for the next critical dimension of trustworthiness: ensuring that such systems are not only fair but also interpretable and developed within a secure lifecycle, a topic explored in the following section.

6.6 Explainability, Interpretability, and the Secure Development Lifecycle

6.6 Explainability, Interpretability, and the Secure Development Lifecycle

Bridging the gap between the equitable algorithmic recourse discussed previously and the broader regulatory frameworks to follow, this section addresses the critical role of explainability within a robust Secure Development Lifecycle (SDLC). The integration of Large Language Models (LLMs) into generative recommendation systems has introduced unprecedented capabilities in personalization and content synthesis. However, the inherent opacity of these deep learning architectures poses significant challenges to trustworthiness, particularly in high-stakes domains where decisions impact user autonomy and safety. While the field of Explainable AI (XAI) has traditionally focused on post-hoc interpretation techniques to illuminate model decisions, recent discourse suggests that explainability alone is insufficient if not embedded within a rigorous, security-oriented development framework. To truly operationalize trust, the industry must shift from viewing explainability as a standalone feature to treating it as a core component of an SDLC. This approach necessitates the integration of threat modeling, design reviews, and penetration testing throughout the AI lifecycle, ensuring that systems are not only interpretable but also robust against adversarial exploitation and aligned with the societal values outlined in the preceding discussion on fairness.

The traditional reliance on isolated fairness metrics and post-hoc explainability often fails to address the systemic vulnerabilities present in production environments. Attempts to solve these problems in silos can feel Sisyphean; fairness metrics are computationally difficult to incorporate into training and rarely alleviate all harms, while explainability methods can be gamed to appear fair or inadvertently reduce privacy [380]. Furthermore, providing explanations without a secure

contextual framework may increase user confidence in predictions even when those explanations are flawed or manipulated. Consequently, a paradigm shift is required where lessons from information security are leveraged to overcome the shortcomings of current XAI methods. By adopting concepts such as threat modeling, design review, penetration testing, and incident response, developers can create AI systems that are resilient in adversarial settings, thereby reinforcing the modular bias mitigation strategies introduced earlier [380].

Threat modeling serves as the foundational step in this secure lifecycle, requiring developers to anticipate how explainability mechanisms themselves might be exploited. Research has demonstrated that explanations can expose sensitive information, enabling attacks such as model inversion or model extraction. For instance, adversaries can leverage feature-based explanations to reconstruct private training data or steal model parameters with significantly fewer queries than standard black-box attacks [381; 382]. In the context of generative recommendation, where user history and preferences are highly sensitive, the exposure of attention weights or rationale tokens could lead to severe privacy breaches, compounding the risks identified in the intersectionality section. Therefore, threat modeling must explicitly consider the “Six Ws” of Explainable Security (XSec)—Who, What, Where, When, Why, and How—recognizing that XSec involves multiple stakeholders including developers, users, and attackers, each with distinct objectives [383]. This multi-faceted reasoning is essential to identify vulnerabilities where explanation interfaces might serve as vectors for data leakage or preference poisoning.

Following threat modeling, the design phase must incorporate “explainability by design” rather than relying solely on retrofitted interpretation tools. There is a growing consensus that the future of human-centric XAI lies not in post-hoc approximations of black-box models, but in neural network architectures that are intrinsically interpretable [384]. Designing systems with interpretable-by-design components allows for more reliable auditing and reduces the risk of “fairwashing,” where superficial explanations mask biased underlying logic—a direct countermeasure to the intersectional biases discussed previously [385]. Moreover, integrating domain knowledge directly into the model’s reasoning process, similar to how human experts operate, can enhance the validity and trustworthiness of the generated explanations [386]. This alignment ensures that the system’s decision-making logic is transparent not just in terms of feature importance, but in terms of causal reasoning and adherence to ethical guidelines.

Penetration testing and red-teaming represent the critical verification stage of the SDLC, where the robustness of both the recommendation engine and its explanation module is stress-tested. Unlike traditional software testing, AI penetration testing must evaluate the system’s resilience against specific adversarial strategies, such as prompt injection or the manipulation of explanation outputs to mislead auditors. The concept of “Seamful XAI” suggests that rather than hiding the imperfections and limitations of AI systems, designers should reveal these “seams” to empower users and auditors to identify potential failures and understand the boundaries of the system’s competence [387]. By intentionally exposing these limitations during testing, organizations can develop more effective countermeasures and incident response protocols. Additionally, the use of synthetic datasets with known generative factors can serve as a rigorous testing ground to validate whether explanation methods consistently reflect the true drivers of model behavior, thereby improving reliability before deployment [388].

Finally, the operationalization of explainability requires continuous monitoring and governance, encapsulated in the notion of a “Chain of Trust” that spans the entire lifecycle of the AI system [389]. This involves establishing standardized auditing frameworks that go beyond technical performance to include ethical and societal impact assessments, laying the groundwork for the regulatory

compliance mechanisms detailed in the subsequent section. The proposal for an AI Audit Standards Board highlights the necessity of evolving standards that keep pace with rapid technological advancements, ensuring that auditing practices remain relevant and robust [390]. Furthermore, internal algorithmic auditing frameworks should be embedded within organizations to close the accountability gap, ensuring that every stage of development yields documented evidence of compliance with ethical principles [391]. By synthesizing explainability with the rigorous disciplines of the Secure Development Lifecycle, the field can move towards building generative recommendation systems that are not only powerful and personalized but also transparent, accountable, and secure against the evolving landscape of AI threats, seamlessly transitioning into the broader discourse on ethical alignment and societal impact.

6.7 Ethical Alignment, Regulatory Compliance, and Societal Impact Assessment

6.7 Ethical Alignment, Regulatory Compliance, and Societal Impact Assessment

Building upon the Secure Development Lifecycle (SDLC) framework established in the previous section, which integrates threat modeling and rigorous testing to ensure system robustness, the deployment of Large Language Models (LLMs) in generative recommendation necessitates an equally rigorous synthesis of emerging legal frameworks, ethical guidelines, and societal impact assessment mechanisms. As these models transition from experimental tools to foundational components of decision-making infrastructures, the challenge of aligning them with diverse human values while ensuring regulatory compliance has become paramount. The complexity of this task is compounded by the fact that existing governance mechanisms, often designed for traditional automated decision-making systems, frequently fail to address the emergent capabilities and opaque nature of generative AI [392]. Consequently, the field is moving towards holistic auditing frameworks that operate across multiple layers of the AI lifecycle, encompassing technology providers, pre-trained models, and downstream applications [393], thereby extending the “Chain of Trust” concept into the broader socio-legal domain.

A central pillar of ethical alignment involves navigating the tension between centralized regulation and decentralized, community-driven safety mechanisms. While governments worldwide are developing horizontal regulations, such as the European Union’s AI Act, to impose uniform standards on high-risk systems, these approaches often struggle with the specific nuances of generative models [394]. Critics argue that a one-size-fits-all regulatory bundle may lack the proportionality and foreseeability required for diverse AI use cases, prompting proposals for contextual frameworks that bifurcate the AI lifecycle into learning and utilization phases [395]. To bridge the gap between broad legal principles and the technical implementation details discussed in the SDLC, researchers advocate for a “Dual Governance” model. This approach synergizes centralized government mandates with crowdsourced safety tools, allowing for dynamic adaptation to emerging threats—such as those identified during penetration testing—while maintaining a baseline of legal accountability [396]. Such hybrid models are essential for fostering innovation while mitigating risks related to privacy violations, misinformation, and intellectual property theft.

Beyond legal compliance, the operationalization of ethical values requires robust methodologies for assessing societal impact that complement the technical metrics of the SDLC. Traditional bias auditing is often insufficient because it fails to capture contextual factors and value tensions that are difficult to quantify. To address this, multi-stakeholder value-based assessment frameworks have been proposed, which visualize the trade-offs between competing ethical principles and facilitate negotiation among diverse stakeholders [397]. These frameworks emphasize that ethical alignment

is not merely a technical optimization problem but a socio-technical process requiring the inclusion of voices from marginalized communities to prevent the amplification of intersectional biases [398]. Furthermore, the development of comprehensive metrics catalogues is critical for translating abstract ethical principles into actionable accountability measures that can be integrated into the continuous monitoring phases of development. These catalogues delineate process, resource, and product metrics to ensure procedural integrity and transparent decision-making in generative AI systems [399].

The challenge of alignment is further complicated by the inherent biases and normative values embedded within foundation models themselves, which persist even when security vulnerabilities are addressed. Empirical studies reveal that many leading LLMs exhibit underlying normative frameworks biased toward specific cultural norms, particularly those of Western and English-speaking societies, and some even display authoritarian tendencies [400]. This suggests that simple fine-tuning may be inadequate; instead, there is a growing argument for infusing generic ethical reasoning capabilities into models, enabling them to adapt to varying ethical policies rather than being rigidly aligned to a single moral code [401]. Techniques such as instruction learning are being developed to decipher and navigate these intrinsic values, allowing models to generate outputs that comply with dynamic value instructions, thus adding a layer of adaptive ethics to the static defenses of the SDLC [402].

Moreover, the societal impact of generative recommenders extends to the integrity of information ecosystems and the potential for deepfake proliferation, areas where technical security alone is insufficient. Ensuring media authenticity requires not only detection capabilities but also attribution mechanisms that can probabilistically link generated content to its source model, thereby supporting legislative frameworks for transparency [403]. The legal landscape is also evolving to address liability, privacy, and intellectual property rights specific to generative outputs, with scholars urging for a clear taxonomy of legal issues and a shared knowledge base to facilitate interdisciplinary collaboration [404]. In the financial and healthcare sectors, where the stakes of algorithmic error are high, frameworks like HUDERAF (Human Rights, Democracy and the Rule of Law Assurance Framework) provide end-to-end assurance processes that integrate the risk analysis protocols of the SDLC with broader stakeholder engagement to safeguard fundamental rights [405].

Finally, achieving true ethical alignment demands a shift from reactive auditing to proactive governance embedded throughout the software development lifecycle, seamlessly merging with the security practices previously outlined. Ethics-based auditing (EBA) should be understood as a “soft” yet formal mechanism that sparks ethical deliberation at key intervention points, linking technical specifications with organizational governance structures [406]. This requires continuous evolution of cybersecurity and governance frameworks, such as ISO 42001, to specifically address the multifaceted risks of LLMs, including the need for human-expert-in-the-loop validation processes that reinforce both security and ethical standards [407]. Ultimately, the path forward involves establishing independent bodies, such as an AI Audit Standards Board, to ensure that auditing practices remain robust and responsive to the rapid advancements in AI, thereby maintaining public trust and promoting a culture of safety and ethical responsibility that transcends mere technical compliance [390].

7 Evaluation Protocols, Benchmarks, and Industrial Deployment

7.1 Beyond Accuracy: Multidimensional Metrics for Generative Recommendation

7.1 Beyond Accuracy: Multidimensional Metrics for Generative Recommendation

The paradigm shift from discriminative ranking to generative recommendation fundamentally alters the landscape of system evaluation, necessitating a departure from traditional reliance on accuracy-centric metrics. While measures such as Precision, Recall, and Normalized Discounted Cumulative Gain (NDCG) remain effective for assessing the relevance of static ranked lists, they are increasingly insufficient for capturing the multifaceted capabilities of Large Language Model (LLM)-driven recommenders. Unlike their discriminative counterparts, generative systems do not merely retrieve items; they synthesize natural language explanations, construct personalized narratives, and reason over complex user intents. Consequently, a comprehensive evaluation framework must extend beyond simple prediction correctness to encompass critical dimensions such as diversity, serendipity, factuality, and reasoning capacity. This multidimensional approach is essential for addressing unique challenges inherent to generative models, including echo chambers, hallucination, and the need for interpretable logic, thereby laying the groundwork for the more complex scenario-based evaluations discussed in the following section.

Diversity remains a paramount concern, yet its definition expands significantly in generative contexts. Traditional metrics often fail to capture the semantic richness required when models generate content rather than just selecting IDs. In generative recommendation, diversity extends beyond the categorical variation of items to include the linguistic variety and conceptual breadth of the generated narratives. Research indicates that standard lexical metrics frequently miss these semantic nuances, necessitating advanced strategies that measure semantic diversity rather than mere lexical overlap [408]. Furthermore, the interplay between relevance and diversity is being re conceptualized; rather than viewing them as competing objectives, newer approaches suggest they can be synergistic. The concept of “divergent relevance” posits that high diversity can emerge as an inherent outcome of the relevance inference process, challenging the necessity for explicit trade-off mechanisms [409]. To operationalize this, researchers have proposed quantitative characteristic metrics—such as diversity, density, and homogeneity—to describe text collections, which can be adapted to evaluate the dispersion and uniformity of generated recommendation lists [410]. Additionally, in knowledge graph-enhanced scenarios, structured metrics like Entity Coverage and Relation Coverage provide a rigorous way to quantify diversity within the semantic space of interconnected entities [411].

Serendipity, defined as the ability to surprise users with relevant but unexpected items, represents another dimension where accuracy-focused metrics fall short. The subjective nature of serendipity complicates its measurement, as a “pleasant surprise” varies significantly across individuals. Psychological models, such as the Wundt curve, have been leveraged to approximate individual curiosity distributions, suggesting that effective serendipity metrics must account for a user’s specific appetite for stimulus [412]. Validating these metrics requires alignment with real user feedback rather than relying solely on offline proxies. Studies collecting thousands of user responses indicate that metrics incorporating unexpectedness, relevance, timeliness, and user curiosity correlate better with actual satisfaction than those relying on item popularity alone [413]. Moreover, given the risk of echo chambers and homophily in current systems, specialized serendipity benchmarks utilizing cluster analysis are necessary to detect whether a system is inadvertently restricting user open-mindedness [414].

Perhaps the most critical challenges for generative recommenders are factuality and faithfulness, as LLMs are prone to hallucinating non-existent items or attributing false characteristics to real products. Evaluating factual consistency requires moving beyond surface-level n-gram overlaps like BLEU or ROUGE. Frameworks such as QuestEval and ClozE have demonstrated that question-answering based and cloze-based metrics offer superior correlation with human judgments regarding consistency and coherence without requiring ground-truth references [415; 416]. For multimodal recommendations involving images and text, composite metrics like CLIPBERTScore, which combine visual and textual similarity, provide robust factuality detection [417]. The reliability of these metrics is further scrutinized through meta-evaluation frameworks like GO FIGURE, which establishes necessary conditions for assessing factuality across different domains [418]. Additionally, distinguishing between factual accuracy and reasoning ability is vital; datasets like BaRDa help separate “truth” from “rationality,” ensuring that a model is not just stating facts but reasoning about them correctly [419].

Finally, the reasoning capacity of generative recommenders—their ability to explain *why* an item is suggested through logical chains of thought—requires evaluation protocols that assess step-by-step deduction. Reference-based metrics often fail in this domain due to the non-uniqueness of valid reasoning paths. Innovative approaches like SocREval utilize the Socratic method to enable reference-free evaluation of reasoning chains, leveraging LLMs themselves as judges to assess the quality of logical deductions [420]. Furthermore, the evaluation of explanations must be robust against gaming; research indicates that some explanation benchmarks suffer from Goodhart’s Law, where metrics can be inflated without genuine improvement in explainability, necessitating a broader reassessment of evaluation goals [421].

In summary, the evaluation of generative recommendation systems demands a holistic suite of metrics that transcends traditional accuracy. By integrating measures of semantic diversity, personalized serendipity, rigorous factuality checks, and robust reasoning assessment, the community can ensure that generative recommenders are not only precise but also diverse, trustworthy, and capable of engaging users through meaningful interactions. However, while these multidimensional metrics provide a necessary foundation, they often operate in static settings. To fully understand the dynamic impact of these systems—where past recommendations influence future behavior and preference evolution—research must advance toward the scenario-based benchmarking and high-fidelity simulation environments detailed in the subsequent section.

7.2 Scenario-Based Benchmarking and Simulation Environments

7.2 Scenario-Based Benchmarking and Simulation Environments

While the multidimensional metrics discussed in the previous section provide a necessary foundation for evaluating static outputs, they often operate in isolation, failing to capture the dynamic, iterative nature of real-world user interactions. To fully assess the impact of generative recommenders—where past suggestions influence future preferences and behavior—the research community must advance toward scenario-based benchmarking and high-fidelity simulation environments. These environments serve as digital proving grounds, enabling the rigorous testing of efficacy, safety, and long-term systemic effects without the prohibitive costs and ethical risks associated with live deployment. The emerging landscape of benchmarking is increasingly defined by scenario-distilling methodologies and the utilization of digital twins to construct realistic, end-to-end simulation frameworks that bridge the gap between offline evaluation and online performance.

A primary driver for developing these advanced simulators is the well-documented disconnect be-

tween static offline metrics and dynamic online system performance. Traditional measures often overlook the feedback loops inherent in user-system interactions, where recommendation policies actively shape preference evolution. To address this, researchers have introduced simulation frameworks capable of modeling these complex temporal dynamics. For instance, the RecLab framework evaluates multiple recommenders across controlled simulated environments to investigate the correlation between offline scores and online outcomes, revealing that improvements in static metrics often yield diminishing returns in dynamic settings [422]. Similarly, SARDINE offers a flexible and interpretable simulator designed to span various sources of complexity found in real recommendation environments, enabling the study of interactive and data-driven methods like reinforcement learning under evolving conditions [423]. These tools underscore the necessity of transitioning from static datasets to environments that faithfully mimic the temporal evolution of user interests and the systemic consequences of recommendation policies.

To construct benchmarks that reflect this complexity, the concept of “scenario-distilling” has emerged as a critical methodology. Real-world applications involve intricate execution paths and diverse AI modules that simple micro-benchmarks cannot adequately capture. The AIBench Scenario framework addresses this by formalizing real-world applications as Directed Acyclic Graph-based models and distilling them into essential task permutations. This approach ensures that benchmarks evaluate the holistic behavior of systems rather than isolated components, providing a rigorous basis for assessing generative recommenders in realistic contexts [424]. Complementing this, the Elliot framework provides a comprehensive pipeline for reproducible evaluation, incorporating diverse splitting strategies and metrics spanning accuracy, bias, and fairness, thereby supporting the experimental design required for robust scenario-based testing [425].

At the frontier of realistic environment creation lie digital twins and agent-based simulations. By leveraging Large Language Models (LLMs), researchers can now construct generative agents equipped with profiles, memory, and action modules capable of mimicking human behavior with unprecedented fidelity. Agent4Rec exemplifies this approach by creating a movie recommendation simulator where LLM-empowered agents interact with recommendations page-by-page, capturing both taste-driven and emotion-driven actions. This facilitates the exploration of complex phenomena such as the filter bubble effect and causal relationships that are difficult to isolate in live systems [426]. Expanding on this, KuaiSim provides a comprehensive user environment supporting multi-behavior and cross-session responses, facilitating the evaluation of request-level, session-level, and retention optimization tasks. By restructuring existing simulators and validating them against real-world data distributions, KuaiSim demonstrates the importance of consistency and behavioral richness in simulation design [427].

Beyond modeling standard user behavior, simulation environments are essential for generating critical scenarios to test safety and robustness, particularly in high-stakes domains like autonomous driving and the vehicular metaverse. In these contexts, generating rare but safety-critical scenarios is vital for validating system reliability. CaDRE introduces a generative framework specifically designed to create diverse and controllable safety-critical driving scenarios using real-world trajectories, optimizing for both quality and diversity to ensure robust testing of autonomous vehicles [428]. Similarly, in the context of the vehicular metaverse, generative AI is employed to synchronize physical and virtual entities, creating immersive testing grounds where digital twins of vehicles and augmented reality recommenders interact in real-time [429]. These approaches illustrate how generative models function not only as the subject of evaluation but also as the engine for creating the complex scenarios necessary to stress-test them.

Furthermore, the integration of synthetic data generation into these simulation environments ad-

dresses persistent issues of data sparsity and privacy. Diffusion models and Generative Adversarial Networks (GANs) are increasingly utilized to generate realistic synthetic data that can replace or augment real datasets, preserving user privacy while maintaining the statistical properties required for effective training and evaluation [430]. The ability to generate high-fidelity synthetic logs allows for the pre-training and evaluation of reinforcement learning-based recommenders without exposing real users to potentially suboptimal policies [431]. However, the validity of these simulations hinges on overcoming the “Sim-to-Real” gap, as synthetic environments may not perfectly replicate the nuances of human decision-making. Consequently, research into domain adaptation and the use of control nets to improve structural fidelity in synthetic data generation is crucial for enhancing the credibility of simulation-based benchmarks [432].

Finally, to ensure reproducibility and fair comparison across the field, there is a growing push for standardized and shareable task environments. The community is moving towards sharing not just datasets, but entire task environments that encapsulate the logic, dynamics, and evaluation protocols of specific recommendation scenarios. This shift is vital for comparing generative models fairly and understanding their behavior across different operational contexts [433]. Additionally, tools like T-RECS provide a unified language for simulating sociotechnical systems, allowing researchers to study the long-term societal impacts of recommender systems, such as polarization and inequality, within a controlled yet realistic simulation framework [434]. As generative recommendation continues to evolve, the development of sophisticated, scenario-based benchmarks and digital twin environments remains a cornerstone of rigorous research. Yet, establishing these environments is only the first step; the subsequent challenge lies in successfully bridging the domain discrepancy between these synthetic simulations and actual human interactions, a topic explored in the following section.

7.3 Bridging the Sim-to-Real Gap in User Behavior Modeling

7.3 Bridging the Sim-to-Real Gap in User Behavior Modeling

Building upon the scenario-based benchmarks and digital twin environments detailed in the previous section, the practical utility of generative recommendation systems hinges on successfully bridging the “sim-to-real” gap. While simulators offer a scalable solution for training agents and evaluating policies without the privacy risks or data scarcity associated with real-world user logs, a critical challenge remains: the domain discrepancy between synthetic simulations and actual human interactions. This gap can lead to models that excel in controlled digital twins but fail catastrophically when exposed to the stochasticity and complexity of live user behaviors. Addressing this disconnect requires a multifaceted approach involving rigorous quantification metrics, advanced domain adaptation techniques, causal reasoning, and careful calibration of generative processes, ultimately ensuring that the insights gained from simulation environments (Section 7.2) translate effectively into the high-stakes, latency-constrained industrial deployments discussed subsequently (Section 7.4).

A primary step in narrowing this gap is the development of robust evaluation frameworks capable of quantifying the divergence between synthetic and real data distributions. Traditional metrics often fail to capture the nuanced statistical properties of user interaction sequences. Recent advancements propose hierarchical evaluation methods that assess generative models from low-level feature matching to high-level functional performance. For instance, frameworks like MPEGO enable multi-level performance evaluation by allowing user-driven features to measure generation quality across different domains, ensuring that synthetic user trajectories align with real-world

behavioral patterns not just superficially, but in terms of underlying structural logic [435]. Furthermore, specific metrics such as the InceptionTime Score and Frechet InceptionTime Distance have been adapted for time-series data, providing a consensual approach to gauge the qualitative performance of generative models in capturing temporal dependencies inherent in user behavior [436]. These metrics are essential for establishing a baseline of fidelity before any downstream task validation occurs.

Beyond static evaluation, dynamic calibration mechanisms are crucial for aligning simulation parameters with real-world observations. In the context of agent-based modeling, which is increasingly used to simulate user populations, static parameter settings often fail to reflect the heterogeneous and evolving nature of human interests. Automatic calibration frameworks that incorporate both dynamic and heterogeneous adjustments have shown promise. By fitting simulation results to real-world data through time-varying parameter adjustments and cluster-wise entity calibration, researchers can significantly reduce the distributional discrepancy between simulated agents and real users [437]. This approach allows simulators to adapt to changing user trends, thereby maintaining relevance over long horizons and preventing the drift that often plagues static simulation environments.

Another potent strategy for bridging the gap involves leveraging domain adaptation and randomization techniques originally developed for computer vision but increasingly applicable to behavioral data. The concept of domain randomization, where synthetic data is diversified to cover a broad spectrum of possibilities, forces models to learn invariant features that generalize better to reality. Studies in visual domains demonstrate that randomizing styles and enforcing consistency across stylized images can achieve generalization results comparable to methods that access target domain data [438]. Translating this to user behavior, generating diverse synthetic interaction patterns that encompass various demographic and contextual shifts can help recommendation models become robust to the unseen variability of real user populations. Additionally, techniques like Cross Domain Generative Augmentation utilize latent diffusion models to generate synthetic samples that specifically fill the density gaps between different domains, effectively reducing the non-iidness that characterizes the shift from simulation to reality [439].

However, simply increasing data volume or diversity is insufficient if the underlying causal structures are not preserved. A significant limitation of many generative models is their tendency to learn spurious correlations rather than causal mechanisms, leading to validity issues when deployed in new environments. Research indicates that while standard Generative Adversarial Networks (GANs) may preserve causality in simple cross-sectional scenarios, they often struggle with more complex temporal or structural causal relationships [440]. To address this, differentiable causal discovery methods have been proposed to automatically identify and align the causal relationships between environment parameters and observed gaps. By learning a causal graph that governs the simulation dynamics, systems like COMPASS can prune the search space for parameter tuning and improve generalization to unseen real-world scenarios [441]. Integrating such causal oversight into user behavior simulators ensures that the generated interactions are not only statistically similar but also mechanistically sound.

Furthermore, the integrity of the synthetic data generation process itself must be managed to prevent “self-consuming loops” where models trained on their own outputs degrade over time. Theoretical and empirical studies show that without correction mechanisms, training on synthetic data can lead to model collapse. Introducing self-correcting functions, potentially derived from expert knowledge or physical laws (or in the case of recommendations, established behavioral theories), can stabilize these loops and maintain high fidelity even when the ratio of synthetic to real data is

high [442]. Additionally, post-processing techniques such as dynamic sample filtering and dataset recycling have been proven to enhance the utility of synthetic data, reducing the accuracy gap with real-world benchmarks to negligible levels [443].

Finally, it is imperative to acknowledge the trade-offs between data fidelity, privacy, and fairness when bridging the sim-to-real gap. The application of differential privacy to generative models can inadvertently exacerbate biases, creating a “Matthew effect” where underrepresented user groups are further marginalized in the synthetic data [444]. Therefore, validating generative models requires not only assessing accuracy but also monitoring fairness metrics to ensure that the sim-to-real transfer does not amplify existing societal biases. Comprehensive surveys on synthetic data for recommender systems highlight the necessity of balancing these competing objectives to build trustworthy simulators [445]. By combining rigorous multi-level evaluation, causal alignment, dynamic calibration, and fairness-aware generation, the research community can effectively narrow the socio-technical gap. This alignment is a prerequisite for the next major challenge: deploying these validated, high-fidelity generative models within the strict latency, cost, and scalability constraints of industrial infrastructure.

7.4 Industrial Scalability, Latency, and Cost-Efficiency

7.4 Industrial Scalability, Latency, and Cost-Efficiency

Having established the methodological rigor required to bridge the sim-to-real gap in the previous section, the focus now shifts to the formidable engineering challenges of deploying these validated, high-fidelity generative models within real-world industrial ecosystems. While Section 7.3 ensures that synthetic behaviors align with human reality, practical utility ultimately hinges on overcoming an engineering trilemma: balancing the demand for high-quality, personalized generative outputs with strict latency Service Level Agreements (SLAs) and manageable computational costs. Unlike traditional discriminative recommenders that rely on efficient matrix factorization or shallow neural networks, generative recommendation systems involve autoregressive decoding processes that are inherently computationally intensive and memory-bound. In industrial settings, where systems must serve billions of users with millisecond-level response times, the sheer scale of model parameters and the dynamic nature of user queries necessitate a holistic re-architecture of inference infrastructure. Without such optimizations, the high-fidelity models developed in simulation would remain impractical for live deployment, rendering the advances in causal alignment and domain adaptation theoretically sound but operationally inert.

A primary bottleneck in scaling LLM-based recommenders is the memory bandwidth constraint during the token generation phase. The autoregressive nature of these models requires frequent access to Key-Value (KV) caches, which grow linearly with sequence length, often becoming the limiting factor rather than compute throughput. To mitigate this, recent advancements have introduced techniques to reduce the KV cache footprint without sacrificing recommendation quality. For instance, the Keyformer system leverages the observation that attention weights are heavily concentrated on a small subset of “key” tokens, allowing for the pruning of less significant cache entries [446]. This approach has been shown to reduce inference latency by over 2x and improve token generation throughput significantly, a critical optimization for long-context user behavior modeling in recommendation scenarios. Similarly, Dynamic Context Pruning offers a learnable mechanism to drop uninformative tokens during generation, effectively pruning up to 80% of the context while maintaining model expressiveness, thereby addressing the quadratic cost scaling of attention mechanisms in long user history sequences [447].

Beyond memory optimization, reducing the computational load per request is essential for cost-efficiency. Speculative decoding has emerged as a powerful strategy to accelerate inference by employing smaller, faster “draft” models to propose token sequences that are then verified by the larger target model. The Minions framework enhances this by utilizing a collective and adaptive speculative generation mechanism, where multiple small models jointly speculate outputs via a majority-voted system [448]. This approach dynamically determines the optimal speculation length, decoupling decoding from verification to maximize throughput. Furthermore, algorithms like SkipDecode enable token-level early exit strategies that are compatible with batch inferencing and KV caching, bypassing lower layers for easier tokens and dedicating compute resources to complex reasoning steps, yielding speedups of 2x to 5x [449]. These methods are particularly relevant for recommendation tasks where the complexity of generating the next item varies significantly based on user intent clarity.

Cost management in industrial deployment also involves strategic resource allocation and model selection. The heterogeneity of user queries suggests that a one-size-fits-all model deployment is inefficient. The FrugalGPT framework proposes an LLM cascade strategy, learning to route simpler queries to cheaper, smaller models while reserving expensive, high-capacity models for complex reasoning tasks [450]. This can reduce costs by up to 98% while matching the performance of the best individual model. Complementing this, the Hybrid LLM approach utilizes a router to assign queries based on predicted difficulty, dynamically trading quality for cost to meet specific scenario requirements [451]. In the context of recommendation ranking, EcoRank addresses the high cost of LLM-based re-ranking by jointly optimizing budget allocation across different prompt strategies and API providers, ensuring maximal performance within fixed financial constraints [452].

Hardware-software co-design is another critical frontier for achieving scalability. Traditional serving architectures often fail to handle the terabyte-scale embedding tables combined with massive transformer parameters found in modern recommenders. Systems like RecPipe decompose recommendation models into multi-stage pipelines, mapping them onto heterogeneous platforms to exploit parallelism [453]. This is further enhanced by custom accelerators like RPAccel, which implement dual static and dynamic embedding caches and reconfigurable systolic arrays to jointly optimize quality and tail-latency [453]. For memory-constrained environments, such as edge devices or cost-sensitive cloud instances, techniques like storing model parameters in flash memory and loading them on-demand (as seen in LLM in a flash) allow for running models twice the size of available DRAM [454]. Additionally, low-precision arithmetic and quantization are indispensable; studies on low-precision hardware architectures demonstrate that with careful adaptation of compute kernels and toolchains, recommendation models can be deployed with up to 5x complexity on specialized hardware without accuracy loss [455].

Finally, managing uncertainty in inference latency is vital for maintaining consistent user experiences under high-traffic workloads. The RT-LM ecosystem quantifies how input uncertainties affect output length and latency, integrating this knowledge into a system-level scheduler for uncertainty-aware prioritization and dynamic consolidation [456]. This ensures that real-time response demands are met even when facing the intrinsic variability of language generation. Moreover, distributed inference strategies, as characterized in studies on capacity-driven scale-out, reveal that dividing massive models across multiple servers can support terabyte-scale parameters with marginal latency overhead, provided that embedding table distribution and sparsity are managed effectively [457]. By synthesizing these algorithmic innovations, hardware optimizations, and strategic routing mechanisms, industrial systems can overcome the barriers to deploying generative recommendation models at scale. However, as these systems become more complex and distributed, the ability to

fairly compare different architectural choices and reproduce experimental results becomes increasingly difficult, necessitating the standardized evaluation protocols and reproducibility frameworks discussed in the following section.

7.5 Standardized Evaluation Protocols and Reproducibility

7.5 Standardized Evaluation Protocols and Reproducibility

While the previous section detailed the engineering strategies required to make generative recommendation systems scalable and cost-efficient in industrial settings, the rapid evolution of these systems has concurrently created an urgent need for standardized evaluation protocols and reproducible research frameworks. As the field transitions from traditional discriminative ranking to complex text-to-text generation paradigms, the absence of unified benchmarks threatens to fragment progress and obscure genuine advancements. The current landscape is often characterized by ad-hoc experimental settings, private data splits, and inconsistent hyper-parameter configurations, which significantly hinder the ability to reproduce results and fairly compare disparate models. To address these systemic issues and ensure that scalability gains do not come at the expense of scientific rigor, the research community must adopt standards that guarantee transparency, accountability, and validity in the development of next-generation recommenders.

A primary step toward resolving the reproducibility crisis is the establishment of open benchmarking initiatives that integrate all components of the research pipeline. The BARS initiative exemplifies this approach by proposing a standardized benchmarking pipeline that meticulously documents datasets, source code, hyper-parameter settings, running logs, and evaluation results [458]. By moving away from isolated, non-reproducible studies, such frameworks reduce redundant efforts and drive more solid research outcomes. Similarly, in the specific domain of security and robustness, the RecAD library offers a unified benchmark for recommender attack and defense, integrating diverse datasets and standard source codes to ensure fair performance comparisons [459]. These efforts highlight that reproducibility is not merely about sharing code but involves a holistic standardization of the entire experimental lifecycle, from data preprocessing to final metric calculation.

The challenge of standardization is further compounded by the variability in data preprocessing and evaluation protocols, which can drastically alter model performance metrics. Research has shown that heavily preprocessed datasets, such as MovieLens, can lead to difficulties in comparing results across different papers if the specific preprocessing steps are not explicitly transparent [460]. Therefore, standardized protocols must mandate the detailed reporting of data filtration, splitting strategies, and feature engineering processes. Furthermore, the integrity of reported results must be verifiable. Tools like `mlscorecheck` have been developed to numerically identify inconsistencies between reported performance scores and experimental setups, serving as a critical mechanism for validating claims and detecting potential statistical flaws in machine learning research [461]. Such automated validation tools should become an integral part of the submission and review process for generative recommendation studies.

In the context of generative AI and LLMs, the issue of openness extends beyond code to include the completeness of model artifacts. The Model Openness Framework (MOF) provides a ranked classification system to rate models based on their completeness and adherence to open science principles, combating the practice of “openwashing” where models are claimed to be open but lack essential components for reproduction [462]. For LLM-based recommenders specifically, platforms like OpenP5 are crucial as they provide extensible data processing, task-centric optimization, and

standardized evaluations across multiple datasets and indexing methods [463]. These platforms facilitate the fair comparison of encoder-decoder and decoder-only architectures, ensuring that improvements in generative recommendation are attributable to architectural innovations rather than opaque tuning practices.

Moreover, the evaluation of generative systems must account for the ethical dimensions of fairness and bias, which require specialized metrics and testing frameworks. The proliferation of fairness metrics has created a paradox where practitioners struggle to select the appropriate measures for their specific contexts [464]. Standardized protocols must therefore include guidance on selecting and interpreting fairness metrics relative to specific stakeholder distributions. Frameworks like FairRec and CFaiRLLM demonstrate the necessity of systematic testing approaches to uncover hidden biases in deep and LLM-based recommenders, respectively [465][237]. Additionally, the auditing of synthetic data and generative models requires holistic frameworks that assess trustworthiness, privacy, and bias simultaneously, as seen in recent proposals for controllable trust trade-offs [466].

Finally, the role of human participants in evaluating these systems cannot be overlooked. As generative models increasingly simulate user interactions and generate content for human consumption, ethical guidelines for research involving human participants must be strictly followed to ensure transparency and informed consent [467]. The CANGARU guidelines further emphasize the need for consistent reporting standards regarding the use of generative AI in scholarly research to prevent a “Babel Tower Effect” of conflicting rules [468]. By adhering to such comprehensive ethical and technical standards, the community can foster a more trustworthy ecosystem. These rigorous evaluation protocols form the essential foundation upon which broader sustainability and societal impact assessments—discussed in the following section—can be reliably constructed, ensuring that the leap toward Artificial General Recommenders is built on science that is not only reproducible but also ethically sound.

7.6 Sustainability and Ethical Impact Assessment

7.6 Sustainability and Ethical Impact Assessment

Building upon the rigorous evaluation protocols and reproducibility standards established in the preceding section, the assessment of generative recommendation systems must extend beyond technical validity to encompass their broader sustainability and ethical implications. As these systems evolve from discriminative ranking models to massive foundation models capable of text-to-item generation, the sheer scale of Large Language Models (LLMs) introduces profound environmental and societal challenges. Consequently, a holistic evaluation framework is no longer optional but imperative, requiring the quantification of carbon footprints, energy demands, and water consumption alongside traditional accuracy metrics like precision or recall. The computational intensity inherent in training and inferring generative recommenders contributes significantly to global greenhouse gas emissions, necessitating that environmental accountability become a core component of next-generation recommendation research.

The environmental footprint of generative AI is multifaceted, comprising both operational emissions from electricity usage during model training and inference, and embodied carbon resulting from hardware manufacturing. Research indicates that the carbon footprint associated with LLMs is a significant concern, covering emissions from training, inference, experimentation, and storage processes [469]. For generative recommendation systems, which often demand real-time inference across billions of user interactions, the cumulative energy consumption can be staggering. To address this, evaluation frameworks must incorporate metrics that track energy usage and CO₂

equivalents, translating abstract computational costs into human-understandable benchmarks such as automobile miles driven or household energy consumption [470]. Tools like EcoDiag have demonstrated the feasibility of estimating the carbon footprint of digital fleets by inventorying services and equipment, providing a methodology that can be adapted to assess the specific environmental load of recommendation pipelines [471].

A critical dimension of this sustainability evaluation involves navigating the trade-off between recommendation quality and environmental efficiency. The paradigm of “Green AI” advocates for reducing energy consumption without sacrificing system performance. In the context of news recommendation, benchmarks like GreenRec have been introduced to assess the trade-off between accuracy and efficiency, revealing that efficient paradigms can achieve competitive accuracy while delivering massive improvements in sustainability metrics [472]. Similarly, in industrial anomaly detection, studies have utilized multi-objective optimization to highlight the Pareto optimality between model performance and environmental impact, emphasizing that superior outcomes often come with commensurate increases in resource consumption [473]. For generative recommenders, this implies that evaluation protocols must not only measure the relevance of generated items but also the carbon cost per recommendation, encouraging the development of models that are both effective and ecologically responsible.

Beyond mere measurement, ethical impact assessment requires addressing the issue of environmental inequity. The environmental footprint of AI is not distributed uniformly; it can be disproportionately higher in certain regions due to variations in local energy grids and cooling requirements. Emerging research proposes equity-aware geographical load balancing to explicitly address these disparities, ensuring that the burden of AI’s carbon and water footprints does not fall unfairly on disadvantaged regions [474]. Furthermore, the water consumption associated with cooling data centers for large-scale model training has become a pressing ethical concern, prompting calls for public action and regulatory frameworks to promote sustainable generative AI [475]. Evaluation frameworks must therefore include regional impact assessments, considering the local energy mix and water stress levels where inference servers are deployed.

To operationalize these assessments, the community is moving toward standardized reporting and certification mechanisms, mirroring the push for reproducibility discussed previously. Proposals for a “Social and Environmental Certificate for AI Systems” suggest a LEED-esque framework that combines compute-efficient machine learning, federated learning, and data sovereignty to certify products based on their eco-social responsibility [476]. Such certifications would rely on transparency mechanisms, such as disclosing the GHG footprint of AI systems, which is increasingly being mandated by regulations like the EU AI Act [477]. Additionally, the concept of “Carbon Utility” has been introduced to quantify the service quality brought by user activity per unit of carbon emissions, offering a normalized metric for comparing the sustainability of different generative applications [478].

Finally, ethical assessment must consider the broader societal implications of deploying resource-intensive recommendation systems. This includes evaluating whether the marginal gains in user engagement justify the substantial environmental costs. Frameworks like the ESG Digital and Green Index (DGI) offer dashboards for assessing a company’s performance in achieving sustainability targets, monitoring the efficient use of natural resources related to AI technologies [479]. Moreover, the integration of self-reflection mechanisms in recommendation systems can foster profound sustainable consumption decisions, shifting the focus from persuasive selling to empowering users to make environmentally conscious choices [480]. By incorporating these diverse metrics—ranging from granular carbon accounting to high-level societal impact indices—evaluation protocols for gen-

erative recommendation systems can ensure that technological advancement proceeds in harmony with global sustainability goals and ethical imperatives, laying the groundwork for a responsible future in generative AI.

8 Open Challenges, Future Directions, and the Path to Artificial General Recommenders

8.1 Causal Reasoning and De-biasing in Generative Frameworks

8.1 Causal Reasoning and De-biasing in Generative Frameworks

As Large Language Models (LLMs) increasingly underpin generative recommendation systems, a critical limitation emerges: the tendency of these models to conflate statistical correlation with genuine causation. While LLMs excel at pattern recognition within massive corpora, standard training objectives based on next-token prediction inherently optimize for associative likelihood rather than causal mechanisms. This reliance on spurious correlations poses severe risks in generative frameworks, where the model does not merely rank items but actively constructs user narratives and generates synthetic interaction data. Without explicit causal reasoning, generative recommenders risk amplifying historical biases, reinforcing filter bubbles, and failing to generalize to scenarios where past correlations no longer hold. Addressing this requires a paradigm shift from purely associational learning to frameworks that integrate causal structure discovery, counterfactual reasoning, and invariant representation learning.

The fundamental challenge lies in the fact that observational user feedback is often confounded by hidden variables such as item popularity, display position, and the recommender system’s own prior interventions. Traditional discriminative models often suffer from Simpson’s paradox, where aggregated data suggests a preference that contradicts individual causal effects. In the context of generative recommendation, this issue is exacerbated because the model may hallucinate preferences based on these spurious patterns. To mitigate this, researchers have proposed integrating causal graphs directly into the learning process. For instance, the concept of Causal Collaborative Filtering (CCF) provides a unified view where traditional algorithms are seen as special cases under simplified causal assumptions, advocating for conditional interventions to estimate true user-item causal preferences rather than mere correlations [481]. Furthermore, frameworks like Dynamic Causal Collaborative Filtering (∂ CCF) explicitly model feedback loops as cyclic processes within causal graphs, utilizing back-door adjustment and counterfactual reasoning to prevent the system from entering echo chambers where user interests become increasingly narrowed [482].

A promising direction for future generative frameworks involves leveraging the inherent knowledge of LLMs to assist in causal structure learning, thereby bridging the gap between data-driven discovery and knowledge-based reasoning. While LLMs trained on heterogeneous data may implicitly learn some invariant causal features due to the coupling effects of data diversity and overparameterization [483], relying solely on this emergence is insufficient for high-stakes applications. Recent advances suggest using LLMs as “causal architects” that can propose causal graphs based on textual descriptions of variables, which are then refined using statistical data. The Iterative LLM Supervised Causal Structure Learning (ILS-CSL) framework demonstrates how LLMs can provide robust structural constraints to guide causal discovery, overcoming issues of data sparsity and unreliable pairwise analyses [484]. Similarly, the COAT framework utilizes LLMs as factor proposers to extract hidden causal variables from unstructured text, iteratively refining these proposals through feedback from rigorous causal learning algorithms [485]. Integrating such hybrid approaches into generative recommenders could allow the system to distinguish between a user’s genuine interest

in an item and their interaction driven by external confounders like trending topics or aggressive marketing.

De-biasing in generative frameworks also demands sophisticated counterfactual reasoning capabilities. Generative models must be able to answer “what-if” questions, such as predicting how a user’s behavior would change if a specific biasing factor (e.g., item popularity) were removed. Methods like Counterfactual Maximum Likelihood Estimation (CMLE) offer a pathway by optimizing models on interventional distributions rather than observational ones, effectively reducing spurious correlations caused by observed confounders [486]. In sequential recommendation settings, where popularity bias is rampant, disentangled counterfactual reasoning has shown success in separating popularity representations from genuine user interest, leading to unbiased predictions [487]. For generative agents, embedding such counterfactual constraints ensures that the synthesized user behaviors or recommended sequences reflect invariant preferences rather than transient statistical artifacts.

Moreover, breaking the feedback loops inherent in interactive recommendation ecosystems is paramount for long-term system health. When a generative model recommends items based on biased data, it influences user behavior, which in turn generates new biased data, creating a self-reinforcing cycle. The Causal Adjustment for Feedback Loops (CAFL) algorithm proposes a solution by reasoning about intervention distributions, proving that systems can break these loops if they correctly adjust for the recommender’s own predictions [488]. Future generative frameworks should incorporate similar causal adjustments, potentially using reinforcement learning agents that actively perform interventions to test and refine their causal models, as seen in the CORE framework which learns to reconstruct causal graphs through informative interventions [489].

Finally, achieving robust de-biasing requires learning invariant causal representations that generalize across different environments and domains. The Latent Causal Invariant Model (LaCIM) illustrates the importance of separating output-causative factors from spuriously correlated variables to ensure precise inference and out-of-distribution generalization [490]. By regularizing learning procedures with mutual information measures based on hypothetical causal graphs, models can achieve reduced sample complexity and better robustness against adversarial attacks and distribution shifts [491]. Establishing these causal foundations is a prerequisite for the next critical challenge: ensuring that these invariant mechanisms hold when models encounter the pervasive distribution shifts of real-world deployment. As we move towards Artificial General Recommenders, the integration of these causal principles—structure learning, counterfactual inference, and invariant representation—will transform LLM-based recommenders from sophisticated correlational engines into trustworthy agents capable of understanding the “why” behind user actions, ensuring fairness, robustness, and genuine personalization in an increasingly complex digital ecosystem.

8.2 Cross-Domain Generalization and Out-of-Distribution Robustness

8.2 Cross-Domain Generalization and Out-of-Distribution Robustness

Building upon the causal foundations established in the previous subsection, where we discussed disentangling invariant mechanisms from spurious correlations to ensure fairness and robustness, this section addresses the practical manifestation of these principles: cross-domain generalization and out-of-distribution (OOD) robustness. While causal reasoning provides the theoretical framework for identifying “why” a user prefers an item, real-world deployment demands that these causal insights hold when user behaviors, item catalogs, and contextual environments undergo non-stationary shifts. The critical hurdle for Large Language Models (LLMs) in generative recom-

mendation is not merely learning from static data, but maintaining performance when trained on source domains and encountered with unseen target domains. This subsection explores advanced methodologies for disentangling domain-shared causal factors from domain-specific nuisance variables, examining how LLMs can serve as domain-agnostic bridges to transfer knowledge effectively across disparate recommendation contexts.

A fundamental limitation of traditional discriminative models, which rely heavily on associative learning, is their tendency to latch onto spurious correlations specific to the training environment. When faced with distribution shifts, these statistical dependencies collapse, leading to severe performance degradation. To mitigate this, recent research emphasizes the necessity of disentangling latent factors into invariant and variant components, a direct application of the causal invariant representation learning discussed earlier. For instance, the concept of “domain-specific risk minimization” has been proposed to explicitly model the “adaptivity gap” between source and target domains, suggesting that ensembling multiple classifiers or performing online model steering can significantly reduce this gap [492]. Similarly, approaches like “Additive Disentanglement of Domain Features with Remix Loss” challenge the conventional wisdom of solely extracting domain-invariant features. Instead, they argue that valuable information resides in domain-variant characteristics and propose architectures that incorporate both invariant and variant features through additive disentanglement strategies, thereby enhancing generalization capacity [493].

In the specific context of cross-domain recommendation, the challenge is compounded by the need to precisely identify what should and should not be transferred. Naive alignment of marginal distributions often leads to negative transfer, particularly when domains are weakly correlated. To address this, hierarchical subspace disentanglement methods have been developed to explore the joint identifiability of cross-domain distributions. These approaches organize user representations into generic shallow subspaces and domain-oriented deep subspaces, utilizing causality-based data generation graphs to distinguish between consistent cross-domain factors and unique domain-specific components [494]. This aligns with the broader machine learning principle that ideal representations must separate causal factors, which drive classification judgments invariantly across domains, from non-causal factors that are category-independent [495]. By enforcing properties such as separation, joint independence, and causal sufficiency, models can simulate invariant causal mechanisms that remain robust even when the underlying data distribution shifts, effectively operationalizing the causal structures introduced in Section 8.1.

The role of LLMs in this landscape is transformative, acting as powerful domain-agnostic bridges due to their pre-training on vast, diverse corpora. Unlike task-specific models that require extensive retraining for every new domain, LLMs possess a form of “universal” semantic understanding that can be adapted to new recommendation scenarios with minimal supervision. However, the adaptability of these foundation models is not infinite; recent investigations into models like GPT-4V reveal that while they exhibit strong zero-shot capabilities, their robustness can vary significantly across different degrees of distribution shift, particularly in specialized domains like healthcare or molecular science [496]. This suggests that while LLMs provide a strong prior, effective cross-domain generalization still requires careful alignment strategies to handle the specific nuances of the target domain, bridging the gap between general semantic knowledge and domain-specific user behavior.

To further enhance robustness beyond simple feature alignment, researchers are exploring techniques that explicitly account for risk variance and uncertainty. Methods such as “Risk Extrapolation” propose reducing the variance of risks across training domains to minimize sensitivity to extreme distributional shifts, effectively recovering causal mechanisms even when input distributions

contain both causal and anti-causal elements [497]. Additionally, the integration of uncertainty quantification through frameworks like “Quantile Risk Minimization” allows models to learn predictors that perform well with high probability rather than just on average, offering a more reliable safety net for OOD scenarios [498]. In graph-based recommendation systems, which are prone to shifts in both node attributes and topology, hierarchical semantic environment generation has shown promise. By explicitly extracting variant subgraphs and regenerating global environments in a hierarchical manner, these models can better capture the complex heterogeneity of real-world data, such as in drug discovery tasks [499].

Moreover, the interplay between domain generalization and fairness adds another layer of complexity. In evolving domains, sensitive attributes may shift in their relationship with user preferences, necessitating frameworks that simultaneously address distribution shifts and counterfactual fairness. Approaches like “Counterfactual Fairness-Aware Domain Generalization with Sequential Autoencoder” demonstrate that separating environmental information from sensitive attributes can preserve fairness while maintaining accuracy across continuous domain shifts [500]. Finally, the concept of “transferability” itself is being rigorously quantified, with new metrics proposed to distinguish between features that are merely invariant and those that are genuinely transferable to unseen targets [501]. As the field moves toward Artificial General Recommenders, the ability to disentangle domain-shared factors and leverage LLMs as flexible, domain-agnostic intermediaries will be paramount. These capabilities form the essential backbone for the next critical challenge: achieving such robustness and generalization even in the face of severe data scarcity, a topic we explore in the following subsection through few-shot learning and self-supervised paradigms.

8.3 Data Efficiency, Few-Shot Learning, and Self-Supervised Paradigms

8.3 Data Efficiency, Few-Shot Learning, and Self-Supervised Paradigms

While the previous subsection highlighted the critical need for robustness against distribution shifts, the practical realization of such resilient systems is often constrained by the scarcity of high-quality labeled interaction data. The pursuit of Artificial General Recommenders (AGR) faces a fundamental bottleneck: emerging domains, cold-start scenarios, and privacy-sensitive environments rarely offer the massive datasets required by traditional paradigms to learn effective user and item representations. Consequently, achieving the OOD generalization discussed earlier necessitates a shift toward data efficiency. This subsection explores the synergistic integration of few-shot learning, self-supervised auxiliary tasks, and advanced prompt-tuning mechanisms. These approaches aim to distill generalizable knowledge from source domains and adapt it to target tasks with minimal supervision, serving as the essential bridge between the theoretical robustness of causal disentanglement and the scalable, foundational architectures discussed in the following section.

A primary avenue for overcoming data scarcity lies in adapting contrastive self-supervised learning to the few-shot regime. Although contrastive methods have proven successful in learning robust representations from unlabeled data, they traditionally rely on large batches and extensive training sets, creating a paradox when target data is limited. Recent advancements demonstrate that domain adaptation techniques can significantly enhance contrastive learning under these constraints. For instance, methods like Few-Max address the poor generalization of contrastive models in few-shot regimes by proposing domain adaptation strategies that prevent overfitting when fine-tuning on small target samples [502]. Similarly, the exploitation of low-confidence samples in the target domain offers a novel pathway to better capture target data structures, as ignoring these samples often leads to biased transferability [503]. By constructing positive and negative pairs specifically

from low-confidence instances, models can learn more discriminative features that bridge the domain gap without requiring extensive labeled data, thereby reinforcing the invariant mechanisms required for robust generalization.

In the context of graph-based recommendation, which is central to modeling complex user-item structural dependencies, the challenge of data sparsity is even more pronounced. Self-supervised graph co-training has emerged as a powerful solution, particularly for session-based recommendations where interaction history is short. By generating multiple views of session graphs that preserve internal and external connectivities, models can supervise each other through contrastive learning, effectively augmenting data without the information loss associated with traditional dropout strategies [504]. Furthermore, the integration of contrastive learning into spatio-temporal graph forecasting reveals that specific augmentation strategies and rule-based negative filtering can alleviate data scarcity, providing a blueprint for adapting similar techniques to dynamic user behavior modeling [505]. The development of automated graph contrastive learning frameworks that learn priors in a continuous parameter space, rather than relying on hand-crafted augmentations, further reduces the need for domain expertise and enhances generalizability across large-scale graphs [506].

Prompt tuning represents another transformative paradigm for achieving data efficiency, aligning closely with the LLM-centric vision of foundational models. In the graph domain, Contrastive Graph Prompt-tuning has shown remarkable efficacy in cross-domain recommendation. The Personalised Graph Prompt-based Recommendation (PGPRec) framework leverages personalized graph prompts rooted in user history and employs contrastive learning to produce pre-trained embeddings with superior generalizability. This approach not only reduces the number of tuned parameters by up to 74% but also yields significant performance improvements in cold-start situations [507]. Extending this concept, Subgraph-level Universal Prompt Tuning (SUPT) assigns prompt features at the subgraph level, preserving universal capabilities while requiring extremely fewer parameters than full fine-tuning, demonstrating superior performance in both full-shot and few-shot scenarios [508]. The synergy between prompt tuning and domain adaptation is further exemplified by methods like OPTIMA, which regularizes decision boundaries to be smooth across domains, significantly enhancing the sample efficiency of prompt tuning in few-shot settings [509].

Moreover, the combination of self-training and self-supervised learning offers a robust framework for few-shot adaptation in sequential and dialogue-based recommendation tasks. The CSS framework, which combines self-training iterations with contrastive self-supervised learning using dropout augmentation, achieves competitive performance in few-shot dialogue state tracking by effectively utilizing unlabeled data [510]. Similarly, task augmentation techniques that synthesize auxiliary data from unlabeled texts, combined with self-training, have been shown to substantially improve sample efficiency, enabling models to achieve performance comparable to full-supervision with only a handful of examples [511]. In visual and multimodal recommendation contexts, counterfactual prompt learning enhances generalization by identifying minimal non-spurious feature changes, allowing models to learn robust representations from limited examples [512].

Looking forward, the path to data-efficient Artificial General Recommenders requires the development of unified frameworks that seamlessly integrate these disparate techniques, paving the way for the trillion-parameter scalable architectures discussed next. Future systems must be capable of automatically selecting between contrastive pre-training, prompt-based adaptation, and self-supervised augmentation based on the specific constraints of the target domain. The success of methods like GenCo, which repurposes contrastive learning as a pre-training method and uses an auxiliary generator to enrich limited labeled data in feature space, highlights the potential of hybrid approaches in remote sensing and suggests analogous applications in recommendation [513].

Ultimately, achieving true data efficiency will depend on the ability of recommender systems to leverage the latent structure of unlabeled data and the semantic richness of pre-trained foundation models to generalize from just a few interactions. This capability is the prerequisite for realizing the “One Model for All” vision, where systems possess the agility to adapt instantly to any user or domain, setting the stage for the foundational scaling laws explored in the subsequent section.

8.4 Towards Foundational Recommendation Models and Scalable Architectures

8.4 Towards Foundational Recommendation Models and Scalable Architectures

Building upon the data-efficient paradigms and few-shot adaptation strategies discussed in the previous section, the field is now poised for a paradigmatic shift from task-specific, siloed architectures toward unified, trillion-parameter foundational models. While techniques like contrastive learning and prompt tuning enable robustness with limited data, the ultimate realization of Artificial General Recommenders (AGR) demands the capacity to absorb vast amounts of heterogeneous information within a single, scalable framework. This evolution mirrors the trajectory of Large Language Models (LLMs) in natural language processing, where the “One Model for All” architecture has demonstrated unprecedented capabilities in generalization and zero-shot reasoning. The core vision for foundational recommendation models lies in their ability to unify diverse tasks—ranging from sequential prediction and rating estimation to review generation and explainability—thereby transforming recommendation from a discriminative ranking problem into a generative text-to-text task. This unification is not merely an architectural convenience but a critical necessity to fully leverage the semantic richness of language models for a holistic understanding of user intent and item characteristics, effectively scaling the benefits of the data-efficient methods previously outlined.

A critical driver of this transition is the empirical validation of scaling laws within the recommendation domain. Historically, deep learning recommendation models (DLRMs) struggled to exhibit the consistent power-law improvements seen in LLMs as model size and compute increased. However, recent breakthroughs have begun to dismantle this barrier. Research indicates that when reformulated as sequential transduction tasks, generative recommenders can achieve quality improvements that scale as a power law with training compute across three orders of magnitude, reaching scales comparable to GPT-3 and LLaMa-2 [40]. This finding is pivotal, as it suggests that the path to superior recommendation performance is inextricably linked to increasing model capacity and computational resources, provided the architecture is designed to support such scaling. Furthermore, studies on large sequential recommendation models have confirmed that scaling laws hold even in data-constrained scenarios, where enlarging model size significantly boosts performance on challenging tasks like cold-start mitigation and long-term preference modeling [514], thus reinforcing the synergy between scale and the data efficiency goals established earlier.

Despite these promising trends, realizing trillion-parameter foundational models requires overcoming significant architectural and engineering hurdles that go beyond the parameter-efficient tuning discussed in Section 8.3. Traditional DLRMs are often bottlenecked by the inefficiency of their up-scaling mechanisms, particularly regarding embedding layers which can constitute over 99% of the total model parameters. To address this, novel architectures such as Wukong have been proposed, utilizing stacked factorization machines and synergistic upscaling strategies to establish robust scaling laws that extend beyond 100 Gflop, effectively bridging the gap to LLM training scales [515]. Similarly, the Persia system demonstrates the feasibility of training models up to 100 trillion parameters through a hybrid training algorithm that decouples the synchronization mechanisms for memory-intensive embedding layers and computation-intensive dense neural networks [516]. These

systems underscore the necessity of co-designing optimization algorithms and distributed system infrastructures to support the heterogeneous computational demands of massive recommendation models, laying the physical groundwork for the generative capabilities explored next.

The concept of a foundational model also implies a shift toward transferable, domain-agnostic architectures that serve as the backbone for infinite item spaces. The P5 (Pretrain, Personalized Prompt, and Predict) paradigm exemplifies this direction by converting all recommendation data—interactions, metadata, and reviews—into natural language sequences, allowing a single model to learn multiple tasks via a shared language modeling objective [5]. This approach facilitates zero-shot and few-shot learning, reducing the need for extensive task-specific fine-tuning and acting as a bridge to the generative creation paradigm. Extending this further, multimodal foundation models like VIP5 integrate visual, textual, and personalization modalities into a shared architecture, leveraging parameter-efficient training methods such as adapter-based fine-tuning to handle diverse input types without retraining the entire backbone [517]. Such multimodal integration is essential for capturing the full spectrum of user preferences and item attributes, providing the rich representational ground required for the high-fidelity content generation discussed in the following section.

However, the road to scalable foundational models is not without its theoretical and practical complexities. Issues such as “embedding collapse,” where enlarged models fail to improve due to embeddings residing in low-dimensional subspaces, pose significant risks to scalability [518]. Addressing this requires innovative designs, such as multi-embedding strategies with set-specific interaction modules, to ensure that increased capacity translates to genuine representational gains. Moreover, the interpretation of scaling laws must be nuanced; recent critiques argue that aggregate scaling metrics may obscure performance disparities across different demographic communities, suggesting that models may not improve uniformly for all users as datasets grow [519]. Additionally, the phenomenon of “model collapse” due to the inclusion of synthetic data in training corpora presents a future challenge, potentially altering scaling trajectories and leading to performance degeneration if not carefully managed [520]. These challenges highlight the delicate balance required between raw scale and the quality of learned representations.

Looking forward, the development of foundational recommendation models will likely rely on advanced scaling techniques borrowed from the LLM domain, such as Mixture-of-Experts (MoE) architectures and sparse routing mechanisms. These approaches allow for the conditional activation of parameter subsets, enabling models to scale to hundreds of billions or trillions of parameters while maintaining computational efficiency during inference [521]. The integration of such scalable architectures with robust causal reasoning and ecosystem-aware mechanism design will be crucial. Ultimately, these foundational models provide the necessary infrastructure to transition from static retrieval to dynamic creation. By establishing a universal, scalable backbone capable of processing infinite contexts, these systems set the stage for the “infinite item spaces” explored in the subsequent section, where the recommender evolves from a selector of existing items into an autonomous generator of personalized content.

8.5 Multimodal Generative Systems and Infinite Item Spaces

8.5 Multimodal Generative Systems and Infinite Item Spaces

Building upon the scalable architectures and foundational models discussed in the preceding section, the next frontier in recommendation is the transcendence of finite item corpora. Traditional recommender systems have been fundamentally constrained by the necessity to retrieve optimal subsets from a pre-existing, static catalog of items such as movies, products, or articles. However,

the convergence of large language models (LLMs) and multimodal generative AI is precipitating a radical shift toward “infinite item spaces.” In this paradigm, the system does not merely retrieve content but actively generates personalized items tailored to specific user preferences. This transition represents a move from discriminative ranking to a generative creation paradigm, fundamentally altering the definition of recommendation itself. By leveraging the parameter efficiency and universal representation capabilities of foundational models, the boundary between the recommender and the content creator dissolves, enabling the synthesis of novel artifacts that have never existed before yet perfectly align with latent user desires.

The core challenge in this domain lies in effectively navigating the infinite possibility space of generative models to satisfy diverse information needs. Early explorations into this concept demonstrated that deep generative models, such as Variational Autoencoders (VAEs), could be leveraged to propose novel items for groups of users by optimizing a latent space that incorporates both rating performance and item reconstruction [522]. This foundational work established that generating plausible, unobserved items is feasible, yet it was limited by the relatively low fidelity of early generative architectures and a lack of fine-grained control over specific item attributes. The advent of diffusion models and transformer-based architectures, often underpinned by the scaling laws of foundational models, has since expanded this horizon, allowing for the generation of high-fidelity multimodal content ranging from images to complex 3D assets.

A critical component of realizing infinite item spaces is the ability to disentangle and control specific attributes within the generative process. Personalization in an infinite space requires more than just matching a general style; it demands the precise manipulation of semantic factors such as material, layout, and emotional valence. Recent advancements have introduced frameworks like ProSpect, which utilizes a prompt spectrum space to represent images as collections of inverted textual token embeddings corresponding to different generation stages, thereby enabling attribute-aware personalization without extensive fine-tuning [523]. Similarly, research into disentangled latent spaces has shown that users can be provided with “knobs” to control specific item aspects, allowing for dynamic adjustments to recommendations based on real-time feedback [524]. These methods illustrate how the latent space of generative models can be structured to support granular user control, transforming the recommendation interaction from a passive selection process into an active co-creation dialogue.

Furthermore, the evolution toward infinite item spaces necessitates robust multimodal fusion capabilities, extending the multimodal integration strategies introduced by foundational models like VIP5. Modern applications require the seamless integration of text, vision, and 3D geometry to create holistic user experiences. For instance, frameworks like MaterialSeg3D have addressed the challenge of inferring underlying materials from 2D semantic priors to generate relightable 3D assets, overcoming the limitations of baked-in lighting in traditional 2D generative models [525]. In the realm of human-centric content, compositional models have been developed to synthesize natural interactions between virtual humans and 3D environments, enabling the creation of complex scenes controlled by high-level semantic specifications [526]. These developments underscore the necessity for recommenders to understand and manipulate rich, multimodal representations rather than relying solely on static ID embeddings.

The concept of “open-endedness” is central to the vision of infinite item spaces and serves as a bridge toward the Artificial General Recommenders (AGR) envisioned in future discussions. Unlike closed systems that operate within fixed distributions, truly generative recommenders must possess the capacity for transformational creativity, producing outputs that lie outside the training distribution while remaining meaningful. Research into open-ended evolution in artificial systems suggests that

achieving this requires mechanisms for continuous novelty search and the ability to reshape goal representations dynamically [527]. Agents augmented with LLMs, such as the LMA3 architecture, demonstrate how language can serve as a cognitive tool to imagine and pursue self-generated, abstract goals, facilitating the exploration of vast skill and content spaces [528]. In the context of recommendation, this implies a future where the system can autonomously propose entirely new categories of content or hybrid items—such as the “ganimals” created by interpolating GAN categories—that users did not explicitly request but find highly engaging [529].

However, moving toward infinite item spaces introduces significant challenges regarding evaluation and alignment, complexities that will further intersect with the ecosystem dynamics discussed in the following section. Traditional metrics based on recall or precision are ill-suited for assessing generated content that has no ground truth. New evaluation paradigms must account for creativity, diversity, and user satisfaction with novel artifacts. Studies utilizing group-based subset scanning have begun to characterize creative processes by detecting anomalous node activations in generative models, offering a potential pathway for quantifying the “creativity” of generated recommendations [530]. Moreover, the risk of generating irrelevant or low-quality content necessitates robust feedback loops. Frameworks like FABRIC have shown that incorporating iterative human feedback directly into the diffusion process can significantly enhance output quality and alignment with user preferences [531].

Looking ahead, the trajectory points toward “Generative Recommendation” paradigm where AI generators act as both editors and creators, repurposing existing items or synthesizing new ones based on natural language instructions [532]. This evolution will likely lead to systems capable of generating personalized bundle creatives consisting of mixed object types, such as items, slogans, and templates, optimized through contrastive non-autoregressive decoding [533]. As these systems mature, they will transition from being tools that assist human creativity to autonomous agents capable of sustaining open-ended exploration of infinite item spaces. This shift fundamentally redefines the relationship between users and the digital content they consume, setting the stage for the broader ecosystem modeling and long-horizon optimization required to manage the economic and social implications of such powerful generative capabilities. The ultimate goal is to create ecosystems where the item space is not a constraint but a canvas, endlessly expandable and infinitely customizable to the unique contours of individual human intent.

8.6 Ecosystem Modeling, Mechanism Design, and Long-Horizon Optimization

8.6 Ecosystem Modeling, Mechanism Design, and Long-Horizon Optimization

While the preceding section envisioned a future where recommendation systems transcend finite catalogs to generate infinite, personalized item spaces, the realization of such boundless creativity introduces profound systemic complexities. The ability to synthesize novel artifacts on demand fundamentally alters the dynamics between users, creators, and platforms, moving beyond the static constraints of traditional retrieval. However, managing these generative capabilities requires more than just technical proficiency in synthesis; it demands a rigorous shift from myopic optimization of immediate engagement metrics to holistic ecosystem modeling. As we approach the era of Artificial General Recommenders (AGR), the focus must expand from localized preference matching to mechanism design and long-horizon optimization, ensuring that the “infinite canvas” of generative recommendation remains sustainable, fair, and aligned with the collective health of the digital environment.

Modern recommender systems operate at the nexus of complex, interdependent ecosystems where

the behaviors of users, content providers, advertisers, and now, generative agents, are tightly coupled. In a landscape where items can be generated rather than merely retrieved, optimizing solely for individual user utility in a short-sighted manner risks severe long-term consequences, including the degradation of content diversity and the destabilization of the creator economy. If generative models are trained only to maximize immediate clicks, they may inadvertently encourage a race to the bottom in content quality or homogenize the synthetic output to fit narrow preference profiles. To address this, researchers argue that explicitly modeling the incentives and strategic behaviors of all actors is strictly necessary to maximize value for the entire system [534]. This necessitates a transition from simple predictive modeling to frameworks that utilize reinforcement learning for long-horizon optimization, apply social choice methods to manage conflicting stakeholder utilities, and employ mechanism design to mitigate information asymmetry amidst strategic interactions.

A primary challenge in this expanded ecosystem is the strategic response of content providers to algorithmic signals, a dynamic that becomes even more critical when the “content” includes AI-generated assets. When producers—or generative agents acting on their behalf—anticipate how algorithms select content, they may alter production strategies to game the system, potentially leading to supply-side equilibria dominated by low-effort, high-exposure synthetic content [535]. In such environments, competition for visibility can distort content distribution, creating dynamic shifts that prioritize algorithmic favor over social welfare. Theoretical insights suggest that while relevance-driven matching performs reasonably well under specific conditions, the fraction of user welfare loss due to unchecked creator competition remains a significant concern that must be bounded [536]. Furthermore, myopic training based on historical choice data fails to ensure the long-term sustainability of the platform, as it cannot account for the feedback loops created when generated content influences future user preferences and provider strategies [537].

To mitigate these risks, mechanism design offers powerful tools for aligning individual incentives with collective goals, a necessity in markets flooded with generative content. In two-sided markets, myopic agents naturally prefer exploitation over exploration, hindering the system’s ability to discover high-quality novel items or diverse creative styles. Novel recommender systems have been proposed that respect agents’ incentives through randomized and adaptive policies, achieving asymptotically optimal performance while satisfying ex-post fairness criteria [538]. Similarly, in markets with capacity constraints, interactive frameworks utilizing pricing mechanisms derived from Walrasian equilibria can facilitate optimal allocations, minimizing instability and ensuring user acceptance even as the volume of available items scales infinitely [539]. These approaches demonstrate that incorporating economic principles into recommendation logic is vital for maintaining market stability and efficiency in a generative age.

Long-horizon optimization serves as another cornerstone of ecosystem-aware recommendation, particularly when dealing with the cumulative effects of generated content on user psychology and platform health. Many current systems fail to account for future consequences, such as user fatigue from excessive notifications or the long-term impact of ad placements on retention. For instance, optimizing push notification decisions using model-based reinforcement learning to maximize long-term value (LTV) has proven effective in reducing notification volume while maintaining engagement, contrasting sharply with myopic systems that would endlessly spam users [540]. Similarly, joint optimization frameworks that balance immediate advertising revenue with long-term user experience have shown that separating these objectives leads to suboptimal overall performance [541]. The ability to decompose future impacts at the item level, even within request-level formulations, allows for more nuanced policy optimization that better utilizes item characteristics for long-term gains [542].

Furthermore, fostering a healthy ecosystem requires active measures to nurture content creation and ensure fair exposure, especially as the line between human and machine-generated content blurs. Feedback shaping approaches have been deployed to predict how consumer feedback incentivizes creators, allowing platforms to reshape feedback distributions to encourage higher quality content production without compromising consumer experience [543]. Additionally, content prompting policies can reduce information asymmetry by suggesting novel content topics to providers, thereby inducing an equilibrium that optimizes user social welfare while respecting provider incentives [544]. Multi-objective learning frameworks are also emerging to handle the conflicting goals of different stakeholders, such as balancing user utility, platform revenue, and creator fairness through data-centric optimization strategies [545].

Ultimately, the path to Artificial General Recommenders necessitates a fundamental rethinking of the objective function, transforming the system from a passive ranker into an active steward of the ecosystem. Instead of maximizing a single metric, future systems must operate as mediators in a multi-agent game, balancing the diverse and often competing interests of all participants. This involves estimating and penalizing induced preference shifts to avoid manipulative behaviors [546] and designing interventions that achieve high social welfare with minimal disruption [547]. By integrating causal inference to break feedback loops [488] and adopting user-oriented exploration policies that account for varying activity levels [548], the next generation of recommenders can evolve from myopic optimizers into responsible governors of a vibrant, sustainable, and equitable digital ecosystem. This robust ecosystem modeling provides the essential groundwork for the realization of AGRs, ensuring that the autonomous, conversational agents discussed in the following section operate within a framework that prioritizes long-term societal value alongside individual satisfaction.

8.7 The Vision of Artificial General Recommenders (AGR)

8.7 The Vision of Artificial General Recommenders (AGR)

Building upon the foundation of ecosystem modeling and long-horizon optimization discussed in the previous section, the trajectory of recommendation system research is rapidly converging toward a transformative vision: the Artificial General Recommender (AGR). While traditional systems operate as narrow, task-specific engines optimized for static ranking metrics, an AGR represents a paradigm shift toward autonomous agents possessing both universality and conversationality. This emerging class of systems aims to transcend the limitations of domain-specific silos, offering a unified interface capable of engaging in natural, multi-turn dialogues while generating high-quality recommendations across diverse verticals, from e-commerce and entertainment to healthcare and finance. The core essence of an AGR lies in its ability to mimic human-like cognitive processes, integrating deep semantic understanding with dynamic preference elicitation to serve as a truly personalized decision-support companion that acts as a steward of the complex digital ecosystem.

Foundational to the concept of AGR is the dual requirement of conversationality and universality. As articulated in early experimental frameworks, an AGR must be able to engage users in fluid, context-aware dialogues that go beyond simple query-response patterns, effectively functioning as a general-purpose assistant [549]. This necessitates a move away from the disjointed architectures of the past, where recommendation modules and dialogue managers were trained separately, often leading to semantic misalignment and disjointed user experiences. Instead, the AGR vision advocates for a unified “Pretrain, Personalized Prompt, and Predict” paradigm, where all recommendation tasks—whether sequential prediction, review generation, or explanation synthesis—are framed as

natural language sequences [5]. By leveraging the universal reasoning capabilities of Large Language Models (LLMs), AGRs can transfer knowledge learned in one domain to another, enabling zero-shot or few-shot adaptation to new environments without extensive retraining, thereby facilitating the cross-domain flexibility required for holistic ecosystem management.

However, realizing the full potential of AGRs requires overcoming significant gaps in current technology, particularly regarding the fidelity of user simulation and the robustness of agent planning. A critical component in the development and evaluation of AGRs is the ability to simulate realistic human behavior at scale, which is essential for testing the long-horizon strategies and mechanism designs proposed earlier. Recent advances have introduced generative agents equipped with memory, profile, and action modules that can emulate complex human decision-making processes, including emotional responses and social traits [426]. These simulators are indispensable for stress-testing AGRs in scenarios that are difficult to replicate with offline datasets, such as long-horizon interactions and the emergence of filter bubbles. Furthermore, the integration of self-inspiring planning algorithms allows these agents to iteratively refine their strategies by considering all previously explored states, thereby enhancing their ability to provide coherent and contextually relevant recommendations in zero-shot settings [41].

Despite these advancements, several challenges remain before AGRs can be widely deployed. One major hurdle is the “recommendability identification” problem—the ability of the system to discern not just what to recommend, but whether a recommendation is appropriate at a specific moment in the conversation. Current models often assume a constant demand for suggestions, which can lead to user disruption; future AGRs must develop the situational awareness to know when to withhold recommendations and simply converse [550]. Additionally, the transition from single-session interactions to multi-subsession continuity is vital. Real-world user journeys are rarely linear; users often return with vague interests or fragmented memories of past interactions. A true AGR must possess long-term memory mechanisms to activate dormant interests and maintain coherence across multiple conversation sessions, bridging the gap between immediate satisfaction and long-term user value [551].

Another pivotal gap lies in the alignment of AGRs with the broader ecosystem dynamics and ethical principles outlined in Section 8.6. The vision of AGR extends beyond maximizing individual user utility to encompassing the well-being of content providers and the sustainability of the information ecosystem. Future systems must be designed as multi-stakeholder agents that balance user satisfaction with provider incentives, preventing the collapse of content diversity and ensuring fair exposure [552]. Moreover, as AGRs become more autonomous, the need for rigorous evaluation protocols that capture both system-centric efficiency and user-centric experience becomes paramount. Emerging frameworks propose comprehensive metrics that assess fluency, empathy, and the ability to handle unexpected conversational turns, moving beyond traditional accuracy metrics to evaluate the holistic quality of the interaction [244].

Ultimately, the path to Artificial General Recommenders involves synthesizing these disparate capabilities into a cohesive architecture that respects human autonomy while delivering profound personalization. It requires a shift from viewing recommendation as a retrieval problem to seeing it as a collaborative, generative process where the system acts as a trusted advisor. By addressing the current limitations in cross-domain generalization, long-term planning, and ethical alignment, the research community can pave the way for AGRs that not only predict user preferences but also actively help users discover new horizons, fulfilling the promise of AI that adapts to humans rather than forcing humans to adapt to machines [553]. The realization of this vision marks the dawn of a new era where recommender systems evolve from passive filters into active, intelligent partners

in the human journey of discovery, effectively operationalizing the theoretical ideals of ecosystem sustainability and long-term optimization.

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